EXERCISES COURSE

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1. Week 3 (9.19)

Problem 1.1 (1.3)**.** *For* $\forall\, n \in \mathbb{N}$*, let* $A_n = \left[-1 + \frac{1}{2n}, 1 - \frac{1}{n}\right]$*, prove that* $\bigcup\limits_{n=1}^{\infty} A_n = (-1, 1)$*.*

Proof. Firstly, by definition we have that

$$\forall\, x \in \bigcup_{n=1}^{\infty} A_n, \quad \exists\, n_0 \in \mathbb{N}, \quad s.t.\ x \in A_{n_0} \subset (-1, 1).$$

Hence

$$x \in (-1, 1),$$

i.e.

$$\bigcup_{n=1}^{\infty} A_n \subset (-1, 1). \tag{1.1}$$

On the other hand, $\forall\, x \in (-1, 1)$, set

$$N = \left[\max\left\{\frac{1}{2(x+1)}, \frac{1}{1-x}\right\}\right] + 1,$$

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then there is $x \in A_N$, thus $x \in \bigcup_{n=1}^{\infty} A_n$, which means

$$(-1,1) \subset \bigcup_{n=1}^{\infty} A_n. \tag{1.2}$$

Combining (1.1) and (1.2) yields

$$\bigcup_{n=1}^{\infty} A_n = (-1,1).$$

□

Problem 1.2 (1.5)**.** *Prove that* $A = \left\{ n \sin \frac{n\pi}{2}; n \in \mathbb{Z} \right\}$ *is unbounded.*

Proof. **Definition:** A set X is unbounded if and only if $\forall\, M > 0$, there exists a $x \in X$, such that $|x| > M$. Then $\forall\, M > 0$, let

$$N = 2([M] + 1) + 1,$$

hence

$$\left| N \sin \frac{N\pi}{2} \right| = 2([M] + 1) + 1 > M,$$

which implies that A is unbounded. □

Problem 1.3 (1.23)**.** *Suppose that* $f(x)$ *is defined on* E*, where* $|E| \geq 3$*. Prove that* $f(x)$ *is strictly monotonic on* E *iff for* $\forall\, x_1, x_2, x_3 \in E$*, if* $x_1 < x_2 < x_3$*, then there must be*

$$(f(x_1) - f(x_2))(f(x_2) - f(x_3)) > 0. \tag{1.3}$$

Proof. " only if ": Obviously.

" if ": Suppose that $f(x)$ is not a strictly monotonic function, hence there exist x_1, x_2, x_3, $x_4 \in E$ with $x_1 < x_2$ and $x_3 < x_4$ such that

$$f(x_1) \leq f(x_2) \quad \text{and} \quad f(x_3) \geq f(x_4).$$

If $x_1 < x_2 < x_3 < x_4$, we know that at least one of

$$(f(x_1) - f(x_2))(f(x_2) - f(x_3)) \quad \text{and} \quad (f(x_2) - f(x_3))(f(x_3) - f(x_4))$$

is non-positive, contradicting with (1.3).

If $x_1 < x_3 < x_2 < x_4$, we know that at least one of

$$(f(x_1) - f(x_2))(f(x_2) - f(x_4)) \quad \text{and} \quad (f(x_1) - f(x_3))(f(x_3) - f(x_4))$$

is non-positive, contradicting with (1.3).

Similar for other cases. Then we obtain contradictions for all cases, wihch means that $f(x)$ is strictly monotonic on E. □

Problem 1.4 (1.25)**.** *Prove that* $\sin(x^2 + x)$ *is not a periodic function.*

Proof. Suppose $T > 0$ is the period of $f(x) := \sin(x^2 + x)$, we have that

$$f(T) = f(-T) = f(0) = 0.$$

Hence there are $k_1, k_2 \in \mathbb{Z}$ such that

$$T^2 + T = k_1\pi \quad \text{and} \quad T^2 - T = k_2\pi,$$

which yields

$$T = \frac{k_1 - k_2}{2}\pi.$$

Then

$$\frac{(k_1 - k_2)^2}{4}\pi + \frac{k_1 - k_2}{2} = k_1,$$

i.e.

$$\pi = \frac{2(k_1 + k_2)}{(k_1 - k_2)^2} \in \mathbb{Q},$$

contradiction. □

Problem 1.5 (1.26)**.** *Suppose that $f(x)$ is defined on $(0, +\infty)$, $x_1, x_2 > 0$. Prove*

(1) *If $\frac{f(x)}{x}$ is decreasing, then $f(x_1 + x_2) \leq f(x_1) + f(x_2)$;*
(2) *If $\frac{f(x)}{x}$ is increasing, then $f(x_1 + x_2) \geq f(x_1) + f(x_2)$.*

Proof. (1) Since $\frac{f(x)}{x}$ is decreasing, then $f(x_1 + x_2) \leq f(x_1) + f(x_2)$, we have

$$\frac{f(x_1 + x_2)}{x_1 + x_2} \leq \frac{f(x_1)}{x_1} \quad \text{and} \quad \frac{f(x_1 + x_2)}{x_1 + x_2} \leq \frac{f(x_2)}{x_2}.$$

Hence

$$f(x_1 + x_2) = \frac{x_1}{x_1 + x_2} f(x_1 + x_2) + \frac{x_2}{x_1 + x_2} f(x_1 + x_2) \leq f(x_1) + f(x_2).$$

Similar for (2). □

Problem 1.6 (1.27)**.** *Suppose $f(x)$ is defined on $(-\infty, +\infty)$, and $f(f(x)) \equiv x$.*

(1) *Is $f(x)$ unique? If not, please give an example;*
(2) *If $f(x)$ is strictly increasing, is it unique? Why?*

Solution. (1) $f(x)$ is not unique. For example $f_1(x) = x$, $f_2(x) = -x$, $f_3(x) = \frac{1}{x}, x \neq 0; f_3(0) = 0$.

(2) We prove that $f(x) = x$. Suppose not, if there is a $x_0 \in \mathbb{R}$, such that $f(x_0) \neq x_0$. Without loss of generality, we assume that $f(x_0) > x_0$. Then $f(f(x_0)) > f(x_0)$ by f is strictly increasing. Hence

$$x_0 = f(f(x_0)) > f(x_0) > x_0,$$

contradiction. □

Problem 1.7. *Prove that $f(x) = \sin x + \sin \sqrt{2}x$, $x \in \mathbb{R}$ is not a periodic function.*

Proof. Note that

$$f''(x) = -\sin x - 2\sin\sqrt{2}x.$$

Hence we have

$$\begin{aligned} f(x) + f''(x) &= -\sin\sqrt{2}x, \\ 2f(x) + f''(x) &= \sin x. \end{aligned}$$

If $T > 0$ is the period of $f(x)$, then T is also the period of $f(x)+f''(x)$ and $2f(x)+f''(x)$. Hence

$$T = \frac{2k\pi}{\sqrt{2}} = 2m\pi, \quad \text{for some } k, m \in \mathbb{N},$$

which implies

$$\sqrt{2} = \frac{k}{m} \in \mathbb{Q},$$

contradiction. □

Problem 1.8. *Suppose $f(x)$ is an increasing function defined on closed interval $[a, b]$, and f satisfies $f(a) \geq a$, $f(b) \leq b$. Prove that there exists $x_0 \in [a, b]$, such that $f(x_0) = x_0$.*

Proof. Set

$$A = \{x \in [a, b] : f(x) \geq x\}.$$

Obviously, A is not empty since $f(a) \geq a$, and b is an upper bound of A. Hence $\sup A$ exists. We denote $x_0 := \sup A$. For $\forall\, x \in A$, there is $x \leq x_0$. Since f is increasing, we have $x \leq f(x) \leq f(x_0)$, i.e. $f(x_0)$ is an upper bound of A, thus $x_0 \leq f(x_0)$. If $x_0 < f(x_0)$, then $f(x_0) \leq f(f(x_0))$, which means $f(x_0) \in A$, contradiction. Hence $f(x_0) = x_0$. □

Problem 1.9. *Is there a function, whose period are all rational numbers but none of irrational number?*

Solution. Dirichlet function, i.e.

$$D(x) = \begin{cases} 1, & x \in \mathbb{Q}, \\ 0, & x \in \mathbb{R}\backslash\mathbb{Q}. \end{cases}$$

□

2. Week 4 (9.26)

Problem 2.1 (2.3(5))**.** *Using definition to prove*

$$\lim_{n\to\infty} n^3 q^n = 0\,(|q| < 1).$$

Proof. It suffices to prove $\lim\limits_{n\to\infty} n^3|q|^n = 0$, and it is obviously valid when $|q| = 0$, hence we assume that $|q| \neq 0$ in the following. Let

$$\frac{1}{|q|} = 1 + \alpha.$$

By binomial theorem, we have

$$\frac{1}{|q|^n} = (1+\alpha)^n = \sum_{k=0}^{n} C_n^k \alpha^k \geq \frac{n(n-1)(n-2)(n-3)}{4!}\alpha^4$$

provided with $n \geq 4$. Hence for $n \geq 4$,

$$n^3|q|^n \leq \frac{24n^2}{(n-1)(n-2)(n-3)\alpha^4} < \frac{72}{\alpha^4}\frac{1}{n-3}.$$

Then $\forall\, \varepsilon > 0$, choosing $N = \left[\frac{72}{\alpha^4\varepsilon}\right] + 4$, for any $n > N$, there is $n^3|q|^n < \varepsilon$, i.e. $\lim\limits_{n\to\infty} n^3|q|^n = 0$. □

Problem 2.2 (2.4)**.** *Suppose for all $n \in \mathbb{N}$, there is $x_n \leq a \leq y_n$ and $\lim\limits_{n\to\infty}(y_n - x_n) = 0$. Prove*

$$\lim_{n\to\infty} x_n = a = \lim_{n\to\infty} y_n.$$

Proof. Since $x_n \leq a \leq y_n$, we know that

$$0 \leq a - x_n \leq y_n - x_n.$$

By Sandwich Theorem, we have

$$\lim_{n\to\infty} x_n = a.$$

Similar for y_n. □

Problem 2.3. *Suppose $\{a_n\}$ is monotonic increasing, $\{b_n\}$ is monotonic decreasing, and $\lim\limits_{n\to\infty}(b_n - a_n) = 0$. Prove that $\lim\limits_{n\to\infty} a_n$, $\lim\limits_{n\to\infty} b_n$ exist, and they are equal.*

Proof. Assume $\lim\limits_{n\to\infty} a_n$ doesn't exist, since $\{a_n\}$ is monotonic increasing, we know there is $\lim\limits_{n\to\infty} a_n = +\infty$. By $\{b_n\}$ is monotonic decreasing, and $\lim\limits_{n\to\infty}(b_n - a_n) = 0$, there must be $\lim\limits_{n\to\infty} b_n = -\infty$. Hence, for $\forall\, M > 0$, there exists a $N \in \mathbb{N}$, such that for $\forall\, n > N$, there are

$$a_n > M \quad \text{and} \quad b_n < -M.$$

Then

$$b_n - a_n < -2M,$$

which contradicts with $\lim_{n\to\infty}(b_n - a_n) = 0$. Thus, $\lim_{n\to\infty} a_n$ exists. It's easy to see that $\lim_{n\to\infty} a_n = \lim_{n\to\infty} b_n$. $\square$

Problem 2.4 (2.6(3))**.** *$\{F_n\}$ is the Fibonacci sequence, defined by*

$$F_0 = F_1 = 1, \quad F_{n+1} = F_n + F_{n-1}.$$

Prove $\lim_{n\to\infty} F_n = +\infty$ by definition.

Proof. **First way:** We can calculate the general terms of F_n by 'eigenvalue method' (See Problem 2.10 for details). Then we know

$$F_n = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^{n+1} - \left(\frac{1-\sqrt{5}}{2}\right)^{n+1}\right].$$

Since $\frac{1+\sqrt{5}}{2} > 1$ and $\frac{\sqrt{5}-1}{2} < 1$, we have $\lim_{n\to\infty} F_n = +\infty$.

Second way: We can prove by induction that $F_n \geq n$ for $n \geq 1$. Since $\lim_{n\to\infty} n = +\infty$, we know $\lim_{n\to\infty} F_n = +\infty$. $\square$

Problem 2.5 (2.10(6))**.** *Calculate*

$$\lim_{n\to\infty} \sqrt[3]{n}(\sqrt[3]{n+1} - \sqrt[3]{n}).$$

Solution.

$$\lim_{n\to\infty} \sqrt[3]{n}(\sqrt[3]{n+1} - \sqrt[3]{n}) = \lim_{n\to\infty} \frac{\sqrt[3]{n}}{(\sqrt[3]{n+1})^2 + \sqrt[3]{n+1}\sqrt[3]{n} + (\sqrt[3]{n})^2} = 0.$$

$\square$

Problem 2.6 (2.14)**.** *Calculate $\lim_{n\to\infty} x_n$.*

(1) $x_n = \dfrac{1\cdot 3\cdot 5\cdots(2n-1)}{2\cdot 4\cdot 6\cdots(2n)}$;

(2) $x_n = \displaystyle\sum_{k=n^2}^{(n+1)^2} \frac{1}{\sqrt{k}}$;

(3) $x_n = \sqrt[n]{n\ln n}$.

Solution. (1)Note that

$$\frac{q}{p} < \frac{q+1}{p+1}, \quad \text{for } 0 < q < p.$$

Then

$$x_n = \frac{1\cdot 3\cdot 5\cdots(2n-1)}{2\cdot 4\cdot 6\cdots(2n)}$$

$$
\begin{aligned}
&= \frac{1}{2}\cdot\frac{3}{4}\cdot\frac{5}{6}\cdots\frac{2n-1}{2n} \\
&< \frac{2}{3}\cdot\frac{4}{5}\cdot\frac{6}{7}\cdots\frac{2n-2}{2n-1}\cdot\frac{2n}{2n+1} \\
&= \frac{2}{1}\cdot\frac{4}{3}\cdot\frac{6}{5}\cdots\frac{2n}{2n-1}\cdot\frac{1}{2n+1} \\
&= \frac{1}{x_n}\frac{1}{2n+1},
\end{aligned}
$$

which gives us

$$x_n < \frac{1}{\sqrt{2n+1}}.$$

Hence $\lim_{n\to\infty} x_n = 0$.

(2) Note that $(n+1)^2 - n^2 + 1 = 2(n+1)$, we have

$$2 = \frac{2(n+1)}{n+1} \le \sum_{k=n^2}^{(n+1)^2} \frac{1}{\sqrt{k}} \le \frac{2(n+1)}{n} \to 2, \quad \text{as } n \to \infty.$$

Hence by Sandwich Theorem, $\lim_{n\to\infty} x_n = 2$.

(3) When $n > 3$, there is $n < n\ln n < n^2$. Then

$$1 \leftarrow \sqrt[n]{n} < \sqrt[n]{n\ln n} < \sqrt[n]{n^2} \to 1, \quad \text{as } n \to \infty.$$

Hence by Sandwich Theorem, $\lim_{n\to\infty} x_n = 1$. □

Problem 2.7 (2.17)**.** *Sequence $\{q_n\}$ satisfies*

$$0 < q_n < 1, \quad (1-q_n)q_{n+1} > \frac{1}{4}, \quad \forall\, n \in \mathbb{N}. \tag{2.1}$$

Prove that $\{q_n\}$ is monotonic increasing and $\lim_{n\to\infty} q_n = \frac{1}{2}$.

Proof. By (2.1) and the mean value inequality, we have

$$\frac{q_{n+1}}{q_n} > \frac{1}{4q_n(1-q_n)} \ge \frac{1}{4\left(\frac{q_n+1-q_n}{2}\right)^2} = 1,$$

i.e. $q_{n+1} > q_n$. Hence $\{q_n\}$ is monotonic increasing. Then by the monotone bounded convergence theorem, we know $\lim_{n\to\infty} q_n$ exists. Denote $q := \lim_{n\to\infty} q_n$, by (2.1),

$$(1-q)q \ge \frac{1}{4},$$

i.e.

$$\left(q - \frac{1}{2}\right)^2 \le 0,$$

which means $q = \frac{1}{2}$. □

Problem 2.8 (2.19)**.** *Suppose that* $0 < a_1 < b_1$*, let*

$$a_{n+1} = \sqrt{a_n b_n}, \quad b_{n+1} = \frac{1}{2}(a_n + b_n) \quad (n = 1, 2, \cdots).$$

Prove $\lim_{n\to\infty} a_n$ *and* $\lim_{n\to\infty} b_n$ *exist, and* $\lim_{n\to\infty} a_n = \lim_{n\to\infty} b_n$*.*

Proof. By the mean value inequality, we have

$$a_{n+1} = \sqrt{a_n b_n} \le \frac{1}{2}(a_n + b_n) = b_{n+1} \quad \text{for } n \ge 1.$$

Hence

$$a_{n+1} = \sqrt{a_n b_n} \ge a_n \quad \text{and} \quad b_{n+1} = \frac{1}{2}(a_n + b_n) \le b_n \quad \text{for } n \ge 2.$$

Then

$$a_2 \le \cdots \le a_n \le b_n \le \cdots \le b_2 \quad \text{for all } n \ge 2.$$

By the monotone bounded convergence theorem, we know $\lim_{n\to\infty} a_n$ and $\lim_{n\to\infty} b_n$ exist, and it's easy to see that $\lim_{n\to\infty} a_n = \lim_{n\to\infty} b_n$. □

Problem 2.9. *Suppose that* $a_1 > b_1 > 0$*, let*

$$a_{n+1} = \frac{a_n + b_n}{2}, \quad b_{n+1} = \frac{2a_n b_n}{a_n + b_n} \quad (n = 1, 2, \cdots).$$

Prove $\lim_{n\to\infty} a_n$ *and* $\lim_{n\to\infty} b_n$ *exist, and* $\lim_{n\to\infty} a_n = \lim_{n\to\infty} b_n = \sqrt{a_1 b_1}$*.*

Proof. The existence is similar to Problem 2.8. Thus, we only need to prove $\lim_{n\to\infty} a_n = \lim_{n\to\infty} b_n = \sqrt{a_1 b_1}$, and if we notice that

$$a_{n+1} b_{n+1} = a_n b_n = \cdots = a_1 b_1,$$

and

$$a_{n+1} \ge \sqrt{a_n b_n} \ge b_{n+1},$$

it's easy to obtain the conclusion. □

Problem 2.10. *Suppose that* $a_1 = \alpha$*,* $b_1 = \beta$*. Let*

$$a_{n+1} = \frac{a_n + b_n}{2}, \quad b_{n+1} = \frac{a_{n+1} + b_n}{2} \quad (n = 1, 2, \cdots).$$

Prove $\lim_{n\to\infty} a_n$ *and* $\lim_{n\to\infty} b_n$ *exist,* $\lim_{n\to\infty} a_n = \lim_{n\to\infty} b_n$*, and find the limitation.*

Proof. By $a_{n+1} = (a_n + b_n)/2$, we have $b_n = 2a_{n+1} - a_n$. Hence

$$2a_{n+2} - a_{n+1} = \frac{a_{n+1} + 2a_{n+1} - a_n}{2} = \frac{3}{2}a_{n+1} - \frac{1}{2}a_n,$$

which is

$$a_{n+2} = \frac{5}{4}a_{n+1} - \frac{1}{4}a_n. \tag{2.2}$$

Then the characteristic equation of (2.2) is

$$x^2 - \frac{5}{4}x + \frac{1}{4} = 0.$$

We find that $x = 1$ and $x = \frac{1}{4}$ are the solution. Hence the general form of a_n is

$$a_n = A + B\left(\frac{1}{4}\right)^n.$$

By $a_1 = \alpha$, $a_2 = (a_1 + b_1)/2 = (\alpha + \beta)/2$, we have

$$a_n = \frac{1}{3}\alpha + \frac{2}{3}\beta + \frac{2}{3}(\alpha - \beta)\left(\frac{1}{4}\right)^{n-1}.$$

Hence $\lim\limits_{n\to\infty} a_n$ and $\lim\limits_{n\to\infty} b_n$ exist, and $\lim\limits_{n\to\infty} a_n = \lim\limits_{n\to\infty} b_n = \frac{1}{3}\alpha + \frac{2}{3}\beta$. □

Problem 2.11 (2.20)**.** *Calculate following limitations.*

(2) $\lim\limits_{n\to\infty}\left(1 + \frac{1}{n^2}\right)^n$;

(3) $\lim\limits_{n\to\infty}\left(1 + \frac{1}{n}\right)^{n^2}$.

Solution. (2)We first note that

$$\left(1 + \frac{1}{n^2}\right)^{n^2} < e.$$

(This can be proved by definition of e.) Then we have

$$1 < \left(1 + \frac{1}{n^2}\right)^n < e^{\frac{1}{n}} \to 1 \quad \text{as } n \to \infty.$$

Hence by Sandwich Theorem

$$\lim_{n\to\infty}\left(1 + \frac{1}{n^2}\right)^n = 1.$$

(3)**First way:** By binomial theorem, we know

$$\left(1 + \frac{1}{n}\right)^{n^2} = 1 + n^2\frac{1}{n} + \cdots \geq n.$$

Hence

$$\lim_{n\to\infty}\left(1 + \frac{1}{n}\right)^{n^2} = +\infty.$$

Second way: By the definition of e, we know there exists a $N \in \mathbb{N}$, such that for $n > N$, there is

$$\frac{e}{2} < \left(1 + \frac{1}{n}\right)^n < e.$$

(Via choosing $\varepsilon = \frac{e}{2}$.) Hence we have

$$\left(1+\frac{1}{n}\right)^{n^2} > \left(\frac{e}{2}\right)^n \to +\infty \quad \text{as } n \to \infty,$$

i.e.

$$\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^{n^2} = +\infty.$$

□

Problem 2.12. *Using* $\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n = e$ *to prove* $\lim_{n\to\infty}\sum_{k=0}^{n}\frac{1}{k!} = e$.

Proof. By binomial theorem, we have

$$\begin{aligned}
\left(1+\frac{1}{n}\right)^n &= \sum_{k=0}^{n} C_n^k \frac{1}{n^k} = \sum_{k=0}^{n} \frac{n!}{k!(n-k)!}\frac{1}{n^k} \\
&= \sum_{k=0}^{n} \frac{n(n-1)\cdots(n-k+1)}{k!}\frac{1}{n^k} \\
&= \sum_{k=0}^{n} \frac{1}{k!}\left(1-\frac{1}{n}\right)\cdots\left(1-\frac{k-1}{n}\right) \\
&\le \sum_{k=0}^{n} \frac{1}{k!}.
\end{aligned}$$

From the above calculation, we know for $\forall\, m > n$, there is

$$\left(1+\frac{1}{m}\right)^m \ge \sum_{k=0}^{n} \frac{1}{k!}\left(1-\frac{1}{m}\right)\cdots\left(1-\frac{k-1}{m}\right).$$

Let $m \to +\infty$, there is

$$e \ge \sum_{k=0}^{n} \frac{1}{k!} \ge \left(1+\frac{1}{n}\right)^n.$$

Hence

$$\lim_{n\to\infty}\sum_{k=0}^{n}\frac{1}{k!} = e.$$

□

Problem 2.13. *Calculate the limitation of*

$$\prod_{k=2}^{n}\left(1-\frac{1}{k^2}\right).$$

Solution.

$$\lim_{n\to\infty}\prod_{k=2}^{n}\left(1-\frac{1}{k^2}\right)=\lim_{n\to\infty}\frac{1}{2}\cdot\frac{3}{2}\cdot\frac{2}{3}\cdot\frac{4}{3}\cdots\frac{n-1}{n}\cdot\frac{n+1}{n}=\frac{1}{2}.$$

□

Problem 2.14. *Suppose* $\lim_{n\to\infty} a_n = a$. *Prove*

$$\lim_{n\to\infty}\frac{p_1a_n+p_2a_{n-1}+\cdots+p_na_1}{p_1+p_2+\cdots+p_n}=a,$$

where $p_k>0$ *and* $\lim_{n\to\infty}\dfrac{p_n}{p_1+p_2+\cdots+p_n}=0$.

Proof. Without loss of generality, we can assume $a=0$. (Otherwise, we can consider a_n-a instead.) Since $\lim_{n\to\infty} a_n=0$, we know for $\forall\,\varepsilon>0$, there exists $N_1\in\mathbb{N}$, such that for all $n>N_1$, there is $|a_n|<\varepsilon/2$. Then

$$\left|\frac{p_1a_n+\cdots+p_{n-N_1}a_{N_1+1}}{p_1+p_2+\cdots+p_n}\right|\le\frac{p_1+\cdots+p_{n-N_1}}{p_1+p_2+\cdots+p_n}\cdot\frac{\varepsilon}{2}<\frac{\varepsilon}{2}.$$

Since a_n is convergence, we know $\{a_n\}$ is bounded. We assume $|a_n|\le M$ for some $M>0$. By $\lim_{n\to\infty}\dfrac{p_n}{p_1+p_2+\cdots+p_n}=0$, we have

$$\begin{aligned}\left|\frac{p_{n-N_1+1}a_{N_1}+\cdots+p_na_1}{p_1+p_2+\cdots+p_n}\right|&\le\left(\frac{p_{n-N_1+1}+\cdots+p_n}{p_1+p_2+\cdots+p_n}\right)M\\&<M\sum_{k=1}^{N_1}\frac{p_{n-N_1+k}}{p_1+p_2+\cdots+p_{n-N_1+k}}\\&\to 0\quad\text{as } n\to\infty.\end{aligned}$$

Hence there exists a $N_2\in\mathbb{N}$ such that for all $n>N_2$, there is

$$\left|\frac{p_{n-N_1+1}a_{N_1}+\cdots+p_na_1}{p_1+p_2+\cdots+p_n}\right|<\frac{\varepsilon}{2}.$$

Then, set $N=\max\{N_1,N_2\}$, we know for all $n>N$, there is

$$\begin{aligned}\left|\frac{p_1a_n+p_2a_{n-1}+\cdots+p_na_1}{p_1+p_2+\cdots+p_n}\right|&\le\left|\frac{p_1a_n+\cdots+p_{n-N_1}a_{N_1+1}}{p_1+p_2+\cdots+p_n}\right|+\left|\frac{p_{n-N_1+1}a_{N_1}+\cdots+p_na_1}{p_1+p_2+\cdots+p_n}\right|\\&<\frac{\varepsilon}{2}+\frac{\varepsilon}{2}=\varepsilon,\end{aligned}$$

i.e.

$$\lim_{n\to\infty}\frac{p_1a_n+p_2a_{n-1}+\cdots+p_na_1}{p_1+p_2+\cdots+p_n}=0.$$

□

Problem 2.15. *Define sequence* $\{a_n\}$ *by* $a_{n+1}=2a_n-a_n^2$, *where* a_0 *is given. Discuss the convergence and divergence of* $\{a_n\}$ *regards the choice of* a_0.

Solution. Firstly, we have

$$1 - a_{n+1} = 1 - 2a_n + a_n^2 = (1 - a_n)^2.$$

By induction, we have

$$1 - a_{n+1} = (1 - a_n)^2 = \cdots = (1 - a_0)^{2^{n+1}},$$

i.e.

$$a_n = 1 - (1 - a_0)^{2^n} \quad \text{for } n \geq 1.$$

Hence,

When $|1 - a_0| < 1$, i.e. $0 < a_0 < 2$, $\{a_n\}$ is convergent, and $\lim\limits_{n\to\infty} a_n = 1$.

When $|1 - a_0| = 1$, i.e. $a_0 = 0$ or $a_0 = 2$, $\{a_n\}$ is convergent, and $\lim\limits_{n\to\infty} a_n = 0$.

When $|1 - a_0| > 1$, i.e. $a_0 < 0$ or $a_0 > 2$, $\{a_n\}$ is divergent, $\lim\limits_{n\to\infty} a_n = -\infty$. $\square$

3. Week 6 (10.10)

Problem 3.1 (2.21)**.** *Suppose that* $\{b_n\}$ *is strictly increasing and* $\lim_{n\to\infty} b_n = +\infty$*. Prove that if*

$$\lim_{n\to\infty} \frac{a_n - a_{n-1}}{b_n - b_{n-1}} = A, \tag{3.1}$$

where A *is finite or* $\pm\infty$*. Then*

$$\lim_{n\to\infty} \frac{a_n}{b_n} = A.$$

Proof. We first prove the case for A is finite. By (3.1), we know $\forall\, \varepsilon > 0$, there exists a $N_1 \in \mathbb{N}$, such that for $\forall\, n > N_1$, there is

$$\left| \frac{a_n - a_{n-1}}{b_n - b_{n-1}} - A \right| < \varepsilon.$$

Since $b_n > b_{n-1}$ for all $n \in \mathbb{N}$, we have

$$(A - \varepsilon)(b_n - b_{n-1}) < a_n - a_{n-1} < (A + \varepsilon)(b_n - b_{n-1}).$$

For given N_1, summing all those inequalities, we obtain

$$(A - \varepsilon)(b_n - b_{N_1}) < a_n - a_{N_1} < (A + \varepsilon)(b_n - b_{N_1}),$$

i.e.

$$\left| \frac{a_n - a_{N_1}}{b_n - b_{N_1}} - A \right| < \varepsilon.$$

Note the identity

$$\frac{a_n}{b_n} - A = \left(1 - \frac{b_{N_1}}{b_n}\right) \cdot \left(\frac{a_n - a_{N_1}}{b_n - b_{N_1}} - A\right) + \frac{a_{N_1} - Ab_{N_1}}{b_n},$$

and combining $\lim_{n\to\infty} b_n = +\infty$, we know there exists a $N_2 \in \mathbb{N}$, such that for $n > N_2$, there is

$$0 < 1 - \frac{b_{N_1}}{b_n} < 2 \quad \text{and} \quad \left| \frac{a_{N_1} - Ab_{N_1}}{b_n} \right| < \varepsilon.$$

Choosing $N = \max\{N_1, N_2\}$, then for $\forall\, n > N$, we have

$$\left| \frac{a_n}{b_n} - A \right| < 3\varepsilon,$$

i.e.

$$\lim_{n\to\infty} \frac{a_n}{b_n} = A.$$

Next, we prove the case for $A = +\infty$. By (3.1), we know $\forall\, M > 0$, there exists a $N_1 \in \mathbb{N}$, such that for $\forall\, n > N_1$, there is

$$\frac{a_n - a_{n-1}}{b_n - b_{n-1}} > 3M.$$

Since $b_n > b_{n-1}$ for all $n \in \mathbb{N}$, we have

$$a_n - a_{n-1} > 3M(b_n - b_{n-1}).$$

For given N_1, summing all those inequalities, we obtain

$$a_n - a_{N_1} > 3M(b_n - b_{N_1}),$$

i.e.

$$\frac{a_n - a_{N_1}}{b_n - b_{N_1}} > 3M.$$

Note the identity

$$\frac{a_n}{b_n} = \left(1 - \frac{b_{N_1}}{b_n}\right) \cdot \left(\frac{a_n - a_{N_1}}{b_n - b_{N_1}}\right) + \frac{a_{N_1}}{b_n},$$

and combining $\lim_{n\to\infty} b_n = +\infty$, we know there exists a $N_2 \in \mathbb{N}$, such that for $n > N_2$, there is

$$\frac{1}{2} < 1 - \frac{b_{N_1}}{b_n} \quad \text{and} \quad \left|\frac{a_{N_1}}{b_n}\right| < \frac{1}{2}M.$$

Choosing $N = \max\{N_1, N_2\}$, then for $\forall\, n > N$, we have

$$\frac{a_n}{b_n} > \frac{3}{2}M - \frac{1}{2}M = M,$$

i.e.

$$\lim_{n\to\infty} \frac{a_n}{b_n} = +\infty.$$

Similar for $A = -\infty$. □

Problem 3.2 ($\frac{0}{0}$ type of Stolz theorem)**.** *Suppose that* $\lim_{n\to\infty} a_n = 0$, $\lim_{n\to\infty} b_n = 0$, *and* $\{b_n\}$ *is strictly decreasing. Prove that if*

$$\lim_{n\to\infty} \frac{a_n - a_{n-1}}{b_n - b_{n-1}} = A, \tag{3.2}$$

where A is finite or $\pm\infty$. Then

$$\lim_{n\to\infty} \frac{a_n}{b_n} = A.$$

Proof. We only prove the case for A is finite. By (3.2), we know $\forall\, \varepsilon > 0$, there exists a $N \in \mathbb{N}$, such that for $\forall\, n > N_1$, there is

$$\left|\frac{a_n - a_{n-1}}{b_n - b_{n-1}} - A\right| < \varepsilon.$$

Since $b_n > b_{n+1}$ for all $n \in \mathbb{N}$, we have

$$(A - \varepsilon)(b_n - b_{n+1}) < a_n - a_{n+1} < (A + \varepsilon)(b_n - b_{n+1}).$$

For any $m > n$, summing all those inequalities, we obtain

$$(A - \varepsilon)(b_n - b_m) < a_n - a_m < (A + \varepsilon)(b_n - b_m),$$

i.e.

$$\left|\frac{a_n - a_m}{b_n - b_m} - A\right| < \varepsilon.$$

Let $m \to \infty$, and combining $\lim\limits_{n\to\infty} a_n = 0$, $\lim\limits_{n\to\infty} b_n = 0$, we know for $\forall\, n > N$, there is

$$\left|\frac{a_n}{b_n} - A\right| \le \varepsilon,$$

i.e.

$$\lim_{n\to\infty} \frac{a_n}{b_n} = A.$$

□

Remark 3.3. *If $A = \infty$ in Problem 3.1 and Problem 3.2, the conclusion is not correct. For example $a_n = (-1)^n n$, $b_n = n$ in Problem 3.1 and $a_n = (-1)^n \frac{1}{n}$, $b_n = \frac{1}{n}$ Problem 3.2.*

Problem 3.4. *Using Stolz theorem to prove*

(1) *If $\lim\limits_{n\to\infty} a_n = a$, then $\lim\limits_{n\to\infty} \frac{1}{n}\sum\limits_{k=1}^{n} a_k$ exists and $\lim\limits_{n\to\infty} \frac{1}{n}\sum\limits_{k=1}^{n} a_k = a$.*

(2) *If $\lim\limits_{n\to\infty} \frac{1}{n}\sum\limits_{k=1}^{n} a_k = a$, and $\lim\limits_{n\to\infty} n(a_n - a_{n-1}) = 0$, then $\lim\limits_{n\to\infty} a_n$ exists and $\lim\limits_{n\to\infty} a_n = a$.*

Proof. (1) By Stolz theorem,

$$\lim_{n\to\infty} \frac{1}{n}\sum_{k=1}^{n} a_k = \lim_{n\to\infty} \frac{\sum\limits_{k=1}^{n} a_k - \sum\limits_{k=1}^{n-1} a_k}{n-(n-1)} = \lim_{n\to\infty} a_n = a.$$

(2) Assume $a_0 = 0$. Let $A_n := a_n - a_{n-1}$, then $a_n = \sum\limits_{k=1}^{n} A_k$. The conditions become

$$a = \lim_{n\to\infty} \frac{1}{n}\sum_{k=1}^{n} a_k = \lim_{n\to\infty} \frac{nA_1 + (n-1)A_2 + \cdots + A_n}{n},$$

and

$$0 = \lim_{n\to\infty} n(a_n - a_{n-1}) = \lim_{n\to\infty} nA_n.$$

Hence

$$\begin{aligned}
\lim_{n\to\infty} a_n &= \lim_{n\to\infty} \frac{na_n}{n} = \lim_{n\to\infty} \frac{n\sum\limits_{k=1}^{n} A_k}{n} \\
&= \lim_{n\to\infty} \frac{nA_1 + nA_2 + \cdots + nA_n}{n} \\
&= \lim_{n\to\infty} \left(\frac{nA_1 + (n-1)A_2 + \cdots + A_n}{n} + \frac{A_2 + \cdots + (n-1)A_n}{n}\right)
\end{aligned}$$

$$
\begin{aligned}
&= \lim_{n\to\infty} \frac{nA_1 + (n-1)A_2 + \cdots + A_n}{n} + \lim_{n\to\infty} \frac{A_2 + \cdots + (n-1)A_n}{n} \\
&= a + \lim_{n\to\infty} \frac{A_2 + \cdots + (n-1)A_n}{n} \\
&= a + \lim_{n\to\infty} \frac{(n-1)A_n}{n-(n-1)} \qquad \text{(Stolz theorem)} \\
&= a + \lim_{n\to\infty} \frac{n-1}{n} \cdot \lim_{n\to\infty} nA_n \\
&= a.
\end{aligned}
$$

□

Problem 3.5. *Suppose that* $x_{n+1} = x_n(1-x_n)$, $n = 1, 2, \cdots$, $0 < x_1 < 1$. *Prove that* $\lim_{n\to\infty} x_n = 0$ *and* $\lim_{n\to\infty} nx_n = 1$.

Proof. Firstly. note that

$$x_{n+1} - x_n = -x_n^2 \leq 0.$$

Then $\{x_n\}$ is monotonic decreasing. Since $0 < x_1 < 1$, we can prove $0 < x_n < 1$ by induction. Hence by monotone bounded convergence theorem, we know $\lim_{n\to\infty} x_n$ exists and $\lim_{n\to\infty} x_n = 0$. And by Stolz theorem, we have

$$
\begin{aligned}
\lim_{n\to\infty} nx_n &= \lim_{n\to\infty} \frac{n}{\frac{1}{x_n}} = \lim_{n\to\infty} \frac{n-(n-1)}{\frac{1}{x_n} - \frac{1}{x_{n-1}}} \\
&= \lim_{n\to\infty} \frac{x_{n+1}x_n}{x_n - x_{n+1}} = \lim_{n\to\infty} \frac{x_{n+1}}{x_n} \\
&= \lim_{n\to\infty} (1 - x_n) = 1.
\end{aligned}
$$

□

Remark 3.6. *If we replace* $x_{n+1} = x_n(1-x_n)$ *by* $x_{n+1} = \ln(1+x_n)$, *we can have a similar conclusion.*

Problem 3.7 (2.23)**.** *Suppose* f *is defined on* (a,b), *and for* $\forall\, \xi \in (a,b)$, *there exists a* $\delta > 0$, *such that for* $x \in (\xi - \delta, \xi + \delta) \cap (a,b)$,

(1) *If* $x < \xi$, *there is* $f(x) < f(\xi)$;
(2) *If* $x > \xi$, *there is* $f(x) > f(\xi)$.

Prove that f *is strictly increasing in* (a,b).

Proof. For $\forall\, x_1, x_2 \in (a,b)$ with $x_1 < x_2$, we want to show $f(x_1) < f(x_2)$. By assumption, $\forall\, \xi \in [x_1, x_2]$, there exists a $\delta = \delta(\xi)$ (If necessary, we can shrink δ so that $U(\xi,\delta) \subset (a,b)$), such that for $x \in U(\xi,\delta)$,

(1) If $x < \xi$, there is $f(x) < f(\xi)$;
(2) If $x > \xi$, there is $f(x) > f(\xi)$.

Then we know

$$[x_1, x_2] \subset \bigcup_{\xi \in [x_1, x_2]} U(\xi, \delta(\xi))$$

is an open covering. By Heine–Borel theorem, there exists a finite subcovering, i.e. there are $\xi_1, \cdots, \xi_n \in [x_1, x_2]$ and $\delta_1, \cdots, \delta_n > 0$, such that

$$[x_1, x_2] \subset \bigcup_{i=1}^{n} U(\xi_i, \delta_i).$$

Without loss of generality, we assume that

$$\xi_1 < \xi_2 < \cdots < \xi_n \quad \text{and} \quad U(\xi_i, \delta_i) \cap U(\xi_{i+1}, \delta_{i+1}) \neq \emptyset, \ i = 1, 2, \cdots, n-1.$$

Hence, we have $f(x_1) \leq f(\xi_1) < f(\xi_2) < \cdots < f(\xi_n) \leq f(x_2)$, i.e. $f(x_1) < f(x_2)$ valids for any $x_1 < x_2$. Thus, f is strictly increasing in (a, b). □

Problem 3.8 (2.25)**.** *Using supremum and infimum principle to prove accumulation point principle.*

Proof. **First way:** Suppose S is a bounded set consists of an infinite number of elements. By supremum and infimum principle, we know that $\sup S$ and $\inf S$ exist. If one of them is not a isolated point of S, it's obviously a accumulation point. Now, we assume none of them is the accumulation point of S. Set

$$E := \{x \in \mathbb{R} \,|\text{There are only a finite number of elments in } S \text{ that are less than } x\}.$$

Then E is nonempty and has an upper bound. Let $\eta := \sup E$, we prove η is a accumulation point of S. Indeed, by the construction of E, we know that $\forall\, \varepsilon > 0$, there must be $\eta + \varepsilon \notin E$, i.e. there are an infinite number of elements in S that are less that $\eta + \varepsilon$. Since there eixsts a $x_0 \in E$ such that $\eta - \varepsilon < x_0$, we know that there are only a finite number of elments in S that are less than $\eta - \varepsilon$. Then $(\eta - \varepsilon, \eta + \varepsilon)$ contains an infinite number of elements in S, which means that η is a accumulation point of S.

Second way: We first claim that any sequence in $\mathbb{R}$ has at least a monotonic subsequence (either increasing or decreasing). Indeed, if there is no increasing subsequence in $\{x_n\}$, we know there exists a $n_1 > 0$, such that $\forall\, n > n_1$, there is $x_n < x_{n_1}$. Similarly, there is no increasing subsequence in $\{x_n\}_{n > n_1}$, we know there exists a $n_2 > n_1$, such that $\forall\, n > n_2$, there is $x_n < x_{n_2} < x_{n_1}$. Proceeding like this, we can find a strictly decreasing subsequence $\{x_{n_k}\}$. Then it's easy to see that accumulation point principle is valid. □

Problem 3.9 (2.34)**.** *Prove that if $x_n > 0$ and*

$$\overline{\lim_{n\to\infty}}\, x_n \cdot \overline{\lim_{n\to\infty}}\, \frac{1}{x_n} = 1,$$

then the sequence $\{x_n\}$ is convergent.

Proof. Suppose there is a subsequence $\{x_{n_k}\} \subset \{x_n\}$, such that

$$\lim_{k\to\infty} x_{n_k} = \underline{\lim_{n\to\infty}}\, x_n.$$

By the definition of limit superior and $x_n > 0$, we know

$$\frac{1}{x_{n_k}} \leq \sup_{n \geq n_k} \frac{1}{x_n}.$$

Hence

$$\varliminf_{n\to\infty} x_n = \lim_{k\to\infty} x_{n_k} \geq \lim_{k\to\infty} \frac{1}{\sup\limits_{n\geq n_k} \frac{1}{x_n}} = \frac{1}{\varlimsup\limits_{n\to\infty} \frac{1}{x_n}} = \varlimsup_{n\to\infty} x_n,$$

i.e.

$$\varliminf_{n\to\infty} x_n = \varlimsup_{n\to\infty} x_n.$$

Hence the sequence $\{x_n\}$ is convergent. □

Problem 3.10 (2.35)**.** *Suppose that $\{x_n\}$ is bounded, and*

$$\lim_{n\to\infty} (x_{n+1} - x_n) = 0. \tag{3.3}$$

Denote $l = \varliminf\limits_{n\to\infty} x_n$ and $L = \varlimsup\limits_{n\to\infty} x_n$. Prove that any number in $[l, L]$ is the limitation of a subsequence of $\{x_n\}$.

Proof. **First way:** By definition, l, L are both accumulation points of $\{x_n\}$. Then we assume $l < L$ and $a \in (l, L)$, and we will prove a is a accumulation point of $\{x_n\}$ in the following. We first prove the claim that for any given $\varepsilon > 0$ and $N \in \mathbb{N}$, there must exist a $\bar{n} > N$ such that

$$|x_{\bar{n}} - a| < \varepsilon.$$

By (3.3), we know there exists a $N' \in \mathbb{N}$, such that for $\forall\, n > N'$, there is

$$|x_{n+1} - x_n| < \varepsilon.$$

Let $N_0 := \max\{N, N'\}$. We know there must exist at least two points $x_{n'}$, $x_{n''}$ in $\{x_n\}_{n\geq N_0}$ such that $x_{n'} < a$, $x_{n''} > a$ (otherwise, if there is no points that are less than a, we must have $\varliminf\limits_{n\to\infty} x_n \geq a$, contradicts with $l < a$; if there is no points that are large than a, we must have $\varlimsup\limits_{n\to\infty} x_n \leq a$, contradicts with $a < L$). Without loss of generality, we assume $n' < n''$. Let $\bar{n}$ be the maximal integer which satisfies $n' \leq n \leq n''$ and $x_n \leq a$. Clearly, $\bar{n} \leq n'' - 1$ and $x_{\bar{n}} \leq a$, $x_{\bar{n}+1} > a$. Hence $\bar{n} > N$, and

$$|x_{\bar{n}} - a| \leq x_{\bar{n}+1} - x_{\bar{n}} < \varepsilon.$$

The claim is proved.

Now choosing $\varepsilon_1 = 1$, $N_1 = 1$, there exists a x_{n_1} $(n_1 > 1)$ such that $|x_{n_1} - a| < 1$. Then choosing $\varepsilon_2 = \frac{1}{2}$, $N_2 = n_1$, there exists a x_{n_2} $(n_2 > n_1)$ such that $|x_{n_2} - a| < \frac{1}{2}$. Next, choosing $\varepsilon_3 = \frac{1}{3}$, $N_3 = n_2$, there exists a x_{n_3} $(n_3 > n_2)$ such that $|x_{n_3} - a| < \frac{1}{3}$. Proceeding like this, we obtain a subsequence $\{x_{n_k}\} \subset \{x_n\}$ satisfies

$$|x_{n_k} - a| < \frac{1}{k}.$$

Hence $\lim\limits_{k\to\infty} x_{n_k} = a$, i.e. a is a accumulation point of $\{x_n\}$.

Second way: By definition, l, L are both accumulation points of $\{x_n\}$. Then we assume $l < L$ and $a \in (l, L)$, and we will prove a is a accumulation point of $\{x_n\}$ in the following. Let $\delta := \min\{L - a, a - l\}$. For $\forall\, i \in \mathbb{N}$, there exists a $k_i \in \mathbb{N}$, when $n \geq k_i$, we have $|x_{n+1} - x_n| < \frac{\delta}{2^i}$.

By definition of l, L, we can choose $l_1 \geq k_1$, such that $x_{l_1} > L - \frac{\delta}{2}$, and choose $m_1 > l_1$, such that $x_{m_1} < l + \frac{\delta}{2}$. Again, choosing $l_2 > \max\{k_2, m_1\}$, such that $x_{l_2} > L - \frac{\delta}{2^2}$, and choose $m_2 > l_2$, such that $x_{m_2} < l + \frac{\delta}{2^2}$. Proceeding like this, we can choose $l_i > \max\{k_i, m_{i-1}\}$, such that $x_{l_i} > L - \frac{\delta}{2^i}$, and choose $m_i > l_i$, such that $x_{m_1} < l + \frac{\delta}{2^i}$. Hence, we know there exists n_i, $l_i < n_i < m_i$ $(i \in \mathbb{N})$, such that

$$x_{n_i} < l + \delta \leq a, \quad x_{n_i - 1} \geq a \quad (i \in \mathbb{N}).$$

Then

$$|x_{n_i} - a| < |x_{n_i} - x_{n_i - 1}| \quad (i \in \mathbb{N}).$$

By $n_i > l_i > k_i$, we know $n_i - 1 \geq k_i$. Hence

$$|x_{n_i} - a| < |x_{n_i} - x_{n_i - 1}| < \frac{\delta}{2^i}.$$

Hence $\lim\limits_{i \to \infty} x_{n_i} = a$, i.e. a is a accumulation point of $\{x_n\}$. □

Problem 3.11 (2.36)**.** *Suppose $\{x_n\}$ and $\{y_n\}$ satisfy*

$$x_{n+1} = y_n + q x_n \, (0 < q < 1), \quad n = 1, 2, \cdots.$$

Prove $\{y_n\}$ converges iff $\{x_n\}$ converges.

Proof. "if": Note that $y_n = x_{n+1} - q x_n$, it's easy to show $\{y_n\}$ converges if $\{x_n\}$ is convergent.

"only if": We first prove that $\{x_n\}$ is bounded. Since $\{y_n\}$ is convergent, we know it is bounded. Assume that $|y_n| \leq M$ for some $M > 0$. Then

$$\begin{aligned}
|x_{n+1}| &= |y_n + q x_n| = |y_n + q(y_{n-1} + q x_{n-1})| \\
&= \left|y_n + q y_{n-1} + q^2 x_{n-1}\right| \\
&= \left|y_n + q y_{n-1} + \cdots + q^{n-1} y_1 + q^n x_1\right| \\
&\leq M\left(1 + q + \cdots + q^{n-1}\right) + q^n |x_1| \\
&< \frac{M}{1-q} + |x_1|.
\end{aligned}$$

Hence the upper limit and lower limit of $\{x_n\}$ exist. We have from $x_{n+1} = y_n + q x_n$ that

$$\overline{\lim_{n\to\infty}}\, x_n = \overline{\lim_{n\to\infty}}\,(y_{n-1} + q x_{n-1}) = \lim_{n\to\infty} y_n + q \overline{\lim_{n\to\infty}}\, x_{n-1} \leq \lim_{n\to\infty} y_n + q \overline{\lim_{n\to\infty}}\, x_n,$$

which yields

$$\overline{\lim_{n\to\infty}}\, x_n \leq (1-q)^{-1} \lim_{n\to\infty} y_n.$$

Similarly, we have

$$\underline{\lim_{n\to\infty}}\, x_n \geq (1-q)^{-1} \lim_{n\to\infty} y_n.$$

Hence

$$\overline{\lim_{n\to\infty}}\, x_n = \underline{\lim_{n\to\infty}}\, x_n = (1-q)^{-1} \lim_{n\to\infty} y_n,$$

i.e. $\{x_n\}$ converges. $\square$

Problem 3.12.

(1) *(2.37). Suppose $\{x_n\}$ satisfies for $\forall\, n, m \in \mathbb{N}$, there is*

$$0 \le x_{n+m} \le x_n + x_m.$$

Prove that $\left\{\dfrac{x_n}{n}\right\}$ has a limitation.

(2) *Suppose $\{x_n\}$ satisfies for $\forall\, n, m \in \mathbb{N}$, there is*

$$0 \le x_{n+m} \le x_n \cdot x_m.$$

Prove that $\left\{x_n^{\frac{1}{n}}\right\}$ has a limitation.

Proof. (1) By

$$x_n \le x_{n-1} + x_1 \le x_{n-2} + 2x_1 \le \cdots \le nx_1,$$

we have

$$0 \le \frac{x_n}{n} \le x_1.$$

Hence $\left\{\dfrac{x_n}{n}\right\}$ is bounded. Denote $\underline{\lim_{n\to\infty}}\, \dfrac{x_n}{n} = l$, then $0 \le l \le x_1$. By definition, for $\forall\, \varepsilon > 0$, there exists a $N \in \mathbb{N}$, such that

$$\frac{x_N}{N} < l + \varepsilon.$$

For any $n > N$, we choose $q \in \mathbb{N}$ and $0 \le r < N$, such that $n = qN + r$. Then

$$x_n = x_{qN+r} \le x_{qN} + x_r \le qx_N + rx_1 \le qx_N + Nx_1.$$

Hence

$$\frac{x_n}{n} \le \frac{qx_N}{n} + \frac{Nx_1}{n} \le \frac{x_N}{N} + \frac{Nx_1}{n} < l + \varepsilon + \frac{Nx_1}{n}.$$

Thus

$$\overline{\lim_{n\to\infty}}\, \frac{x_n}{n} \le l + \varepsilon.$$

Let $\varepsilon \to 0$, we have

$$\overline{\lim_{n\to\infty}}\, \frac{x_n}{n} \le l = \underline{\lim_{n\to\infty}}\, \frac{x_n}{n}.$$

Hence

$$\overline{\lim_{n\to\infty}}\, \frac{x_n}{n} = \underline{\lim_{n\to\infty}}\, \frac{x_n}{n},$$

i.e. $\lim\limits_{n\to\infty} \dfrac{x_n}{n}$ exists.

(2) The proof is similar to (1), we leave it to readers. $\square$

Problem 3.13 (3.8)**.** *Calculate the following limitations.*

(6) $\lim\limits_{t\to 1}(1-t)\tan\dfrac{\pi t}{2}$;

(8) $\lim\limits_{x\to\frac{\pi}{4}}\dfrac{\tan x-1}{x-\frac{\pi}{4}}$;

(9) $\lim\limits_{x\to 0}\dfrac{\cos(n\arccos x)}{x}$ $\quad$ (n *is odd*).

Solution. (6)

$$\lim_{t\to 1}(1-t)\tan\frac{\pi t}{2}=\lim_{t\to 1}\frac{1-t}{\cos\frac{\pi t}{2}}=\lim_{t\to 1}\frac{1-t}{\sin\frac{\pi}{2}(1-t)}=\frac{2}{\pi}.$$

(8)

$$\lim_{x\to\frac{\pi}{4}}\frac{\tan x-1}{x-\frac{\pi}{4}}=\lim_{x\to\frac{\pi}{4}}\frac{1}{\cos x}\cdot\lim_{x\to\frac{\pi}{4}}\frac{\sin x-\cos x}{x-\frac{\pi}{4}}=2\lim_{x\to\frac{\pi}{4}}\frac{\sin(x-\frac{\pi}{4})}{x-\frac{\pi}{4}}=2.$$

(9)

$$\begin{aligned}\lim_{x\to 0}\frac{\cos(n\arccos x)}{x}&=\lim_{x\to 0}\frac{(-1)^{\frac{n-1}{2}}\sin(\frac{n\pi}{2}-n\arccos x)}{x}\\&=\lim_{x\to 0}\frac{(-1)^{\frac{n-1}{2}}\sin\big(n(\frac{\pi}{2}-\arccos x)\big)}{x}\\&=\lim_{x\to 0}\frac{(-1)^{\frac{n-1}{2}}\sin\big(n(\arcsin x)\big)}{x}\\&=(-1)^{\frac{n-1}{2}}n,\end{aligned}$$

where we have used $\arcsin x+\arccos x=\dfrac{\pi}{2}$. $\square$

Problem 3.14 (3.9(4))**.** *Calculate the following limitation.*

$$\lim_{x\to\infty}\left(\cos\frac{a}{x}\right)^{x^2}\qquad(a\neq 0).$$

Solution. Note that

$$\begin{aligned}\lim_{x\to\infty}\left(\cos\frac{a}{x}\right)^{x^2}&=\lim_{x\to 0}(\cos ax)^{\frac{1}{x^2}}=\lim_{x\to 0}(1+\cos ax-1)^{\frac{1}{x^2}}\\&=\lim_{x\to 0}\left(1-2\sin^2\frac{ax}{2}\right)^{\frac{1}{x^2}}\\&=\lim_{x\to 0}\left(\left(1-2\sin^2\frac{ax}{2}\right)^{\frac{1}{-2\sin^2\frac{ax}{2}}}\right)^{\frac{-2\sin^2\frac{ax}{2}}{x^2}}.\end{aligned}$$

Since

$$\lim_{x\to 0}\left(1-2\sin^2\frac{ax}{2}\right)^{\frac{1}{-2\sin^2\frac{ax}{2}}}=e,$$

we know that for $\forall\, \varepsilon > 0$, there exists a $\delta > 0$, such that $\forall\, x : |x| < \delta$, there is

$$e - \varepsilon < \left(1 - 2\sin^2 \frac{ax}{2}\right)^{\frac{1}{-2\sin^2 \frac{ax}{2}}} < e + \varepsilon.$$

Hence, we have for $|x| < \delta$ that

$$(e+\varepsilon)^{\frac{-2\sin^2 \frac{ax}{2}}{x^2}} < \left(\left(1 - 2\sin^2 \frac{ax}{2}\right)^{\frac{1}{-2\sin^2 \frac{ax}{2}}}\right)^{\frac{-2\sin^2 \frac{ax}{2}}{x^2}} < (e-\varepsilon)^{\frac{-2\sin^2 \frac{ax}{2}}{x^2}}.$$

Then

$$\begin{aligned}
(e+\varepsilon)^{\frac{-a^2}{2}} &\leq \varliminf_{x\to 0} \left(\left(1 - 2\sin^2 \frac{ax}{2}\right)^{\frac{1}{-2\sin^2 \frac{ax}{2}}}\right)^{\frac{-2\sin^2 \frac{ax}{2}}{x^2}} \\
&\leq \varlimsup_{x\to 0} \left(\left(1 - 2\sin^2 \frac{ax}{2}\right)^{\frac{1}{-2\sin^2 \frac{ax}{2}}}\right)^{\frac{-2\sin^2 \frac{ax}{2}}{x^2}} \\
&\leq (e-\varepsilon)^{\frac{-a^2}{2}}.
\end{aligned}$$

Since ε is arbitrary, we have

$$\varliminf_{x\to 0} \left(\left(1 - 2\sin^2 \frac{ax}{2}\right)^{\frac{1}{-2\sin^2 \frac{ax}{2}}}\right)^{\frac{-2\sin^2 \frac{ax}{2}}{x^2}} = \varlimsup_{x\to 0} \left(\left(1 - 2\sin^2 \frac{ax}{2}\right)^{\frac{1}{-2\sin^2 \frac{ax}{2}}}\right)^{\frac{-2\sin^2 \frac{ax}{2}}{x^2}} = e^{\frac{-a^2}{2}},$$

i.e.

$$\lim_{x\to 0} \left(\left(1 - 2\sin^2 \frac{ax}{2}\right)^{\frac{1}{-2\sin^2 \frac{ax}{2}}}\right)^{\frac{-2\sin^2 \frac{ax}{2}}{x^2}} = e^{\frac{-a^2}{2}}.$$

$\square$

4. Week 7 (10.17)

Problem 4.1.

(1) *Suppose* $\{x_n\}$ *is a positive sequence. Prove that*

$$\overline{\lim_{n\to\infty}}\, n\left(\frac{1+x_{n+1}}{x_n}-1\right)\geq 1.$$

(2) *Suppose* $\{x_n\}$ *is a positive sequence. Prove that*

$$\overline{\lim_{n\to\infty}}\left(\frac{x_1+x_{n+1}}{x_n}\right)^n\geq e.$$

Proof. (1) Proof by contradiction. Assume that

$$\overline{\lim_{n\to\infty}}\, n\left(\frac{1+x_{n+1}}{x_n}-1\right)<1.$$

Hence there exists a $N\in\mathbb{N}$, such that $\forall\, n\geq N$, there is

$$n\left(\frac{1+x_{n+1}}{x_n}-1\right)<1.$$

Then we have

$$\frac{1}{n+1}<\frac{x_n}{n}-\frac{x_{n+1}}{n+1},\quad n\geq N.$$

Summing all inequalities from N to n yields

$$\frac{1}{N+1}+\frac{1}{N+2}+\cdots+\frac{1}{n+1}<\frac{x_N}{N}-\frac{x_{n+1}}{n+1}\leq\frac{x_N}{N}.$$

However, we already konw $\lim\limits_{n\to\infty}\left(\frac{1}{N+1}+\frac{1}{N+2}+\cdots+\frac{1}{n+1}\right)=+\infty$, which makes a contradiction.

(2) Proof by contradiction. Assume that

$$\overline{\lim_{n\to\infty}}\left(\frac{x_1+x_{n+1}}{x_n}\right)^n<e.$$

Hence there exists a $N\in\mathbb{N}$, such that $\forall\, n\geq N$, there is

$$\left(\frac{x_1+x_{n+1}}{x_n}\right)^n<e<\left(1+\frac{1}{n-1}\right)^n.$$

Then we have

$$\frac{1}{n}<\frac{x_n}{n-1}-\frac{x_{n+1}}{n},\quad n\geq N.$$

Summing all inequalities from N to n yields

$$\frac{1}{N}+\frac{1}{N+1}+\cdots+\frac{1}{n}<\frac{x_N}{N-1}-\frac{x_{n+1}}{n}\leq\frac{x_N}{N-1}.$$

However, we already konw $\lim\limits_{n\to\infty}\left(\frac{1}{N}+\frac{1}{N+1}+\cdots+\frac{1}{n}\right)=+\infty$, which makes a contradiction. $\square$

Remark 4.2. *Both constants in Problem 4.1 are optimal. Indeed, we can choose* $x_n = n\ln n$.

Exercise 4.3 (Leave to readers)**.** *Let* $\{a_n\}_{n=1}^{\infty}$ *be a sequence of positive numbers. Prove that*

$$\varliminf_{n\to\infty}\left(n^2\left(4a_n(1-a_{n-1})-1\right)\right)\le\frac{1}{4}.$$

Hint: It suffices to prove the conclusion for $a_n\in(0,1)$. Assume by contradiction that $\varliminf\limits_{n\to\infty}\left(n^2\left(4a_n(1-a_{n-1})-1\right)\right)>\dfrac{1}{4}$. Then for $\varliminf\limits_{n\to\infty}\left(n^2\left(4a_n(1-a_{n-1})-1\right)\right)>l>\dfrac{1}{4}$, there exists a $N\in\mathbb{N}$, such that $\forall\, n>N$(without loss of generality, we can assume that $N=1$), there is

$$n^2\left(4a_n(1-a_{n-1})-1\right)>l.$$

Firstly, to prove that $\{a_n\}$ is monotonic increasing and $\lim\limits_{n\to\infty}a_n=\frac{1}{2}$ (by the monotone bounded convergence theorem). Secondly, let $a_n:=\frac{1}{2}-b_n$, where $b_n\ge 0$ and $\lim\limits_{n\to\infty}b_n=0$. Note that

$$b_{n-1}-b_n-2b_nb_{n-1}>\frac{l}{2n^2}.$$

Next, to prove that $\{nb_n\}$ is monotonic decreasing for large enough n. Indeed, there is

$$\begin{aligned}
nb_n-(n-1)b_{n-1}&<\frac{nb_{n-1}-\frac{l}{2n}}{1+2b_{n-1}}-(n-1)b_{n-1}\\
&=\frac{nb_{n-1}-\frac{l}{2n}-(n-1)b_{n-1}-2(n-1)b_{n-1}^2}{1+2b_{n-1}}\\
&=\frac{-2(n-1)\left(b_{n-1}-\frac{1}{4(n-1)}\right)^2+\frac{1}{8(n-1)}-\frac{l}{2n}}{1+2b_{n-1}}\\
&=\frac{-2(n-1)\left(b_{n-1}-\frac{1}{4(n-1)}\right)^2+\frac{(1-4l)n+4l}{8n(n-1)}}{1+2b_{n-1}}\\
&\le 0,
\end{aligned}$$

provided with $n\ge\frac{4l}{4l-1}$. Hence $\lim\limits_{n\to\infty}nb_n$ exists. Denote that $\lim\limits_{n\to\infty}nb_n=A$. By Stolz theorem (for lower limit), we have

$$\begin{aligned}
A=\lim_{n\to\infty}nb_n&=\varliminf_{n\to\infty}\frac{b_n}{\frac{1}{n}}\\
&\ge\varliminf_{n\to\infty}\frac{b_n-b_{n-1}}{\frac{1}{n}-\frac{1}{n-1}}
\end{aligned}$$

$$
\begin{aligned}
&\geq \lim_{n\to\infty} n(n-1)\left(\frac{l}{2n^2}+2b_nb_{n-1}\right)\\
&=\frac{l}{2}+2\lim_{n\to\infty} n(n-1)b_nb_{n-1}\\
&=\frac{l}{2}+2A^2,
\end{aligned}
$$

i.e.

$$2A^2-A+\frac{l}{2}\leq 0.$$

However, $\Delta = 1^2-4\times 2\times \frac{l}{2} = 1-4l<0$, which makes a contradiction.

From the above proof, we can see that $\frac{1}{4}$ is the optimal constant. Actually, we can choose $a_n=\frac{1}{2}-\frac{1}{4n}$. □

Problem 4.4. *Suppose that $x_n>0$. Prove that $\overline{\lim\limits_{n\to\infty}}\sqrt[n]{x_n}\leq 1$ iff $\lim\limits_{n\to\infty}\frac{x_n}{l^n}=0,\ \forall\, l>1$.*

Proof. "if": If $\lim\limits_{n\to\infty}\frac{x_n}{l^n}=0$, we know there exists a $N\in\mathbb{N}$ such that $\forall\, n>N$, there is $x_n<\frac{1}{2}l^n$. Hence $\sqrt[n]{x_n}<\left(\frac{1}{2}\right)^{\frac{1}{n}}l$, which means $\overline{\lim\limits_{n\to\infty}}\sqrt[n]{x_n}\leq l$. Since $l>1$ is arbitrary, we know $\overline{\lim\limits_{n\to\infty}}\sqrt[n]{x_n}\leq 1$.

"only if": For $\forall\, l>1$, by $\overline{\lim\limits_{n\to\infty}}\sqrt[n]{x_n}\leq 1$ we have there exists a $N\in\mathbb{N}$, such that $\forall\, n>N$, there is

$$\sqrt[n]{x_n}<1+\frac{l-1}{2}=\frac{l+1}{2}.$$

Then

$$\frac{x_n}{l^n}<\left(\frac{l+1}{2l}\right)^n\to 0\quad \text{as } n\to\infty,$$

since $\frac{l+1}{2l}<1$. Thus, $\lim\limits_{n\to\infty}\frac{x_n}{l^n}=0,\ \forall\, l>1$. □

Problem 4.5. *Suppose that $x_n>0$. If $\overline{\lim\limits_{n\to\infty}}\sqrt[n]{x_n}=1$, prove that $\overline{\lim\limits_{n\to\infty}}\sqrt[n]{\sum\limits_{k=1}^{n}x_k}=1$.*

Proof. It is clear that $\overline{\lim\limits_{n\to\infty}}\sqrt[n]{\sum\limits_{k=1}^{n}x_k}\geq\overline{\lim\limits_{n\to\infty}}\sqrt[n]{x_n}=1$. Hence, we only need to prove the inverse inequality. Since $\overline{\lim\limits_{n\to\infty}}\sqrt[n]{x_n}=1$, we know for any $l>1$, there exists a $N\in\mathbb{N}$, such that $\forall\, n>N$, there is $\sqrt[n]{x_n}<l$. Then

$$\sum_{k=1}^{n}x_k\leq\sum_{k=1}^{N}x_k+\sum_{k=N+1}^{n}l^k$$

$$
\begin{aligned}
&\leq \sum_{k=1}^{N} x_k + \sum_{k=1}^{n} l^k \\
&< \sum_{k=1}^{N} x_k + \frac{l^{n+1}}{l-1}.
\end{aligned}
$$

Hence

$$
\overline{\lim_{n\to\infty}} \sqrt[n]{\sum_{k=1}^{n} x_k} \leq \overline{\lim_{n\to\infty}} \sqrt[n]{\sum_{k=1}^{N} x_k + \frac{l^{n+1}}{l-1}} = l.
$$

Since $l > 1$ is arbitrary, we know $\overline{\lim\limits_{n\to\infty}} \sqrt[n]{\sum\limits_{k=1}^{n} x_k} \leq 1$. Then we are done. $\square$

Problem 4.6. *Suppose that* $x_n > 0$, $\lim\limits_{n\to\infty} \frac{x_n}{n} = 0$ *and* $\overline{\lim\limits_{n\to\infty}} \frac{1}{n} \sum\limits_{k=1}^{n} x_k < +\infty$. *Prove that* $\overline{\lim\limits_{n\to\infty}} \frac{1}{n^2} \sum\limits_{k=1}^{n} x_k^2 = 0$.

Proof. Since $\lim\limits_{n\to\infty} \frac{x_n}{n} = 0$ and $L := \overline{\lim\limits_{n\to\infty}} \frac{1}{n} \sum\limits_{k=1}^{n} x_k < +\infty$, we know $\forall\, \varepsilon > 0$, there exists a $N \in \mathbb{N}$, such that $\forall\, n > N$, there is

$$
\frac{x_n}{n} < \varepsilon \quad \text{and} \quad \frac{1}{n} \sum_{k=1}^{n} x_k < L + 1.
$$

Then for $\forall\, n > N$,

$$
\begin{aligned}
\frac{1}{n^2} \sum_{k=1}^{n} x_k^2 &= \frac{1}{n^2} \sum_{k=1}^{N} x_k^2 + \frac{1}{n^2} \sum_{k=N+1}^{n} x_k^2 \\
&< \frac{1}{n^2} \sum_{k=1}^{N} x_k^2 + \frac{1}{n^2} \sum_{k=N+1}^{n} x_k \cdot k\varepsilon \\
&\leq \frac{1}{n^2} \sum_{k=1}^{N} x_k^2 + \left(\frac{1}{n} \sum_{k=N+1}^{n} x_k \right) \varepsilon \\
&< \frac{1}{n^2} \sum_{k=1}^{N} x_k^2 + (L+1)\varepsilon.
\end{aligned}
$$

Hence

$$
\overline{\lim_{n\to\infty}} \frac{1}{n^2} \sum_{k=1}^{n} x_k^2 \leq \overline{\lim_{n\to\infty}} \left(\frac{1}{n^2} \sum_{k=1}^{N} x_k^2 + (L+1)\varepsilon \right) = (L+1)\varepsilon.
$$

Sicne $\varepsilon > 0$ is arbitrary, we know $\overline{\lim\limits_{n\to\infty}} \frac{1}{n^2}\sum\limits_{k=1}^{n} x_k^2 = 0$, i.e. $\lim\limits_{n\to\infty} \frac{1}{n^2}\sum\limits_{k=1}^{n} x_k^2 = 0$. $\square$

Problem 4.7. *Calculate the following limitations.*

(1) $\lim\limits_{x\to 0} \dfrac{\sqrt[m]{1+\alpha x} - \sqrt[n]{1+\beta x}}{x}$, *where* $m, n \in \mathbb{N}$, $\alpha, \beta \in \mathbb{R}$ *are constants;*

(2) $\lim\limits_{n\to\infty} \sum\limits_{k=1}^{n} \sin\left(\dfrac{2k-1}{n^2}x\right)$, *where* $x \in \mathbb{R}$ *is a constant;*

(3) $\lim\limits_{x\to+\infty} \sin\sqrt{x+1} - \sin\sqrt{x}$;

(4) $\lim\limits_{x\to 0} (x+e^x)^{\frac{1}{x}}$;

(5) $\lim\limits_{x\to+\infty} \dfrac{[xf(x)]}{x}$, *where* $\lim\limits_{x\to+\infty} f(x) = 1$;

(6) $\lim\limits_{x\to+\infty} \left(\dfrac{1}{p}\sum\limits_{k=1}^{p} a_k^x\right)^{\frac{1}{x}}$, *where* $a_1, \cdots, a_p$ $(p \geq 2)$ *are positive.*

(7) $\lim\limits_{x\to 0+} \left(\dfrac{1}{p}\sum\limits_{k=1}^{p} a_k^x\right)^{\frac{1}{x}}$, *where* $a_1, \cdots, a_p$ $(p \geq 2)$ *are positive.*

Solution. (1) By replacement with equivalent infinitesimal, we have

$$\begin{aligned}\lim_{x\to 0} \frac{\sqrt[m]{1+\alpha x} - \sqrt[n]{1+\beta x}}{x} &= \lim_{x\to 0} \frac{\sqrt[m]{1+\alpha x} - 1}{x} + \lim_{x\to 0} \frac{1 - \sqrt[n]{1+\beta x}}{x} \\ &= \frac{\alpha}{m} - \frac{\beta}{n}.\end{aligned}$$

(2) It's clear that the limitation is 0 when $x = 0$. Next, we assume that $x \neq 0$. By the Prosthaphaeresis formula, we have

$$\begin{aligned}\lim_{n\to\infty} \sum_{k=1}^{n} \sin\left(\frac{2k-1}{n^2}x\right) &= \lim_{n\to\infty} \frac{1}{2\sin\frac{x}{n^2}} \sum_{k=1}^{n} 2\sin\frac{x}{n^2}\sin\left(\frac{2k-1}{n^2}x\right) \\ &= \lim_{n\to\infty} \frac{1}{2\sin\frac{x}{n^2}} \sum_{k=1}^{n} \left[\cos\left(\frac{2k-2}{n^2}x\right) - \cos\left(\frac{2k}{n^2}x\right)\right] \\ &= \lim_{n\to\infty} \frac{1-\cos\left(\frac{2}{n}x\right)}{2\sin\frac{x}{n^2}} \\ &= x.\end{aligned}$$

(3) By the Prosthaphaeresis formula, we have

$$\lim_{x\to+\infty} \sin\sqrt{x+1} - \sin\sqrt{x} = \lim_{x\to+\infty} 2\cos\left(\frac{\sqrt{x+1}+\sqrt{x}}{2}\right)\sin\left(\frac{\sqrt{x+1}-\sqrt{x}}{2}\right)$$

$$
\begin{aligned}
&= \lim_{x\to+\infty} 2\cos\left(\frac{\sqrt{x+1}+\sqrt{x}}{2}\right)\sin\left(\frac{1}{2(\sqrt{x+1}+\sqrt{x})}\right)\\
&= 0.
\end{aligned}
$$

(4) By replacement with equivalent infinitesimal, we have

$$
\begin{aligned}
\lim_{x\to 0}(x+e^x)^{\frac{1}{x}} &= \lim_{x\to 0} e^{\frac{\ln(x+e^x)}{x}}\\
&= \lim_{x\to 0} e^{\frac{\ln(1+x+e^x-1)}{x}}\\
&= \lim_{x\to 0} e^{\frac{x+e^x-1}{x}}\\
&= e^2.
\end{aligned}
$$

(5) Note that

$$
f(x)-\frac{1}{x} < \frac{[xf(x)]}{x} \le f(x).
$$

It's easy to obtain $\lim\limits_{x\to+\infty}\dfrac{[xf(x)]}{x}=1$ by the Sandwich Theorem.

(6) Note that

$$
\frac{\max\{a_1,\cdots,a_p\}}{p^{\frac{1}{x}}} \le \left(\frac{1}{p}\sum_{k=1}^{p} a_k^x\right)^{\frac{1}{x}} \le \max\{a_1,\cdots,a_p\}.
$$

By the Sandwich Theorem, we have

$$
\lim_{x\to+\infty}\left(\frac{1}{p}\sum_{k=1}^{p} a_k^x\right)^{\frac{1}{x}} = \max\{a_1,\cdots,a_p\}.
$$

(7) Note that

$$
\begin{aligned}
\frac{1}{x}\ln\left(\frac{1}{p}\sum_{k=1}^{p} a_k^x\right) &= \frac{1}{x}\ln\left(1+\frac{1}{p}\sum_{k=1}^{p}(a_k^x-1)\right)\\
&\sim \frac{1}{p}\sum_{k=1}^{p}\frac{a_k^x-1}{x} \quad \text{as } x\to 0+\\
&\to \frac{1}{p}\sum_{k=1}^{p}\ln a_k \quad \text{as } x\to 0+.
\end{aligned}
$$

Hence

$$
\lim_{x\to 0+}\left(\frac{1}{p}\sum_{k=1}^{p} a_k^x\right)^{\frac{1}{x}} = e^{\frac{1}{p}\sum\limits_{k=1}^{p}\ln a_k} = \sqrt[p]{a_1a_2\cdots a_p}.
$$

□

Problem 4.8. *Calculate* $\lim\limits_{n\to\infty}\prod\limits_{k=1}^{n}\cos\dfrac{x}{2^k}$, *and prove the Viète formula*

$$\frac{\pi}{2}=\frac{1}{\sqrt{\frac{1}{2}}\cdot\sqrt{\frac{1}{2}+\frac{1}{2}\sqrt{\frac{1}{2}}}\cdot\sqrt{\frac{1}{2}+\frac{1}{2}\sqrt{\frac{1}{2}+\frac{1}{2}\sqrt{\frac{1}{2}}}}\cdots}.$$

Proof. It's clear that $\lim\limits_{n\to\infty}\prod\limits_{k=1}^{n}\cos\dfrac{x}{2^k}=1$ when $x=0$. Next, we assume that $x\neq 0$. By $\sin 2x=2\sin x\cos x$, we have

$$\lim_{n\to\infty}\prod_{k=1}^{n}\cos\frac{x}{2^k}=\lim_{n\to\infty}\frac{\cos\frac{x}{2}\cos\frac{x}{2^2}\cdots\cos\frac{x}{2^n}\sin\frac{x}{2^n}}{\sin\frac{x}{2^n}}=\lim_{n\to\infty}\frac{\sin x}{2^n\sin\frac{x}{2^n}}=\frac{\sin x}{x}.$$

Note that $\cos 2\theta=2\cos^2\theta-1$, we know that $\cos\theta=\sqrt{\dfrac{1}{2}+\dfrac{1}{2}\cos 2\theta}$. Choosing $x=\frac{\pi}{2}$, we know that $\cos\frac{x}{2}=\sqrt{\frac{1}{2}}$, $\cos\frac{x}{4}=\sqrt{\frac{1}{2}+\frac{1}{2}\sqrt{\frac{1}{2}}}$. Hence

$$\frac{\sin x}{x}=\sqrt{\frac{1}{2}}\cdot\sqrt{\frac{1}{2}+\frac{1}{2}\sqrt{\frac{1}{2}}}\cdot\sqrt{\frac{1}{2}+\frac{1}{2}\sqrt{\frac{1}{2}+\frac{1}{2}\sqrt{\frac{1}{2}}}}\cdots,$$

i.e.

$$\frac{1}{\sqrt{\frac{1}{2}}\cdot\sqrt{\frac{1}{2}+\frac{1}{2}\sqrt{\frac{1}{2}}}\cdot\sqrt{\frac{1}{2}+\frac{1}{2}\sqrt{\frac{1}{2}+\frac{1}{2}\sqrt{\frac{1}{2}}}}\cdots}=\frac{\frac{\pi}{2}}{\sin\frac{\pi}{2}}=\frac{\pi}{2}.$$

□

Problem 4.9. *Suppose that* $f(x)$, $g(x)$ *are periodic function defined on* $\mathbb{R}$, *and satisfy* $\lim\limits_{x\to+\infty}(f(x)-g(x))=0$. *Prove that* $f(x)=g(x)$, $\forall\, x\in\mathbb{R}$.

Proof. Denote T_1 as the period of $f(x)$ and T_2 as the period of $g(x)$. We first note that by $\lim\limits_{x\to+\infty}(f(x)-g(x))=0$, there is

$$f(x)-g(x+nT_1)=f(x+nT_1)-g(x+nT_1)\to 0\quad\text{as } n\to\infty.$$

Similarly, we have

$$g(x)=\lim_{n\to\infty}f(x+nT_2),\quad\forall\, x\in\mathbb{R}.$$

Hence we have

$$\begin{aligned}f(x)-g(x)&=\lim_{n\to\infty}(g(x+nT_1)-f(x+nT_2))\\&=\lim_{n\to\infty}(g(x+nT_1+nT_2)-f(x+nT_2+nT_1))\\&=0,\quad\forall\, x\in\mathbb{R}.\end{aligned}$$

$\square$

Problem 4.10. *Suppose that $f(x)$ does not have an upper bound in (a,b). Prove that there exists a sequence $\{x_n\} \subset (a,b)$ such that $\lim\limits_{n\to\infty} f(x_n) = +\infty$.*

Proof. Since $f(x)$ has no upper bound, we know that for any $n \in \mathbb{N}$, there exists a $x_n \in (a,b)$ such that $f(x_n) > n$. Then we have that $\lim\limits_{n\to\infty} f(x_n) = +\infty$. $\square$

Problem 4.11. *Suppose $f(x)$ is defined on $(0,1)$, and $\lim\limits_{x\to 0} f(x) = 0$, $\lim\limits_{x\to 0} \dfrac{f(x) - f\left(\frac{x}{2}\right)}{x} = 0$. Prove that $\lim\limits_{x\to 0} \dfrac{f(x)}{x} = 0$.*

Proof. By $\lim\limits_{x\to 0} \dfrac{f(x) - f\left(\frac{x}{2}\right)}{x} = 0$, we know that $\forall\, \varepsilon > 0$, there exists a $\delta > 0$, such that $\forall\, x : 0 < x < \delta$, there is

$$\frac{\left|f(x) - f\left(\frac{x}{2}\right)\right|}{x} < \varepsilon,$$

i.e.

$$\left|f(x) - f\left(\frac{x}{2}\right)\right| < \varepsilon x.$$

Then for $0 < x < \delta$, there is

$$\begin{aligned}
|f(x)| &= \left|f(x) - f\left(\frac{x}{2}\right) + f\left(\frac{x}{2}\right) - f\left(\frac{x}{2^2}\right) + \cdots + f\left(\frac{x}{2^n}\right)\right| \\
&\le \left|f(x) - f\left(\frac{x}{2}\right)\right| + \left|f\left(\frac{x}{2}\right) - f\left(\frac{x}{2^2}\right)\right| + \cdots + \left|f\left(\frac{x}{2^n}\right)\right| \\
&< \varepsilon\left(x + \frac{x}{2} + \cdots + \frac{x}{2^n}\right) + \left|f\left(\frac{x}{2^n}\right)\right| \\
&< 2\varepsilon x + \left|f\left(\frac{x}{2^n}\right)\right|.
\end{aligned}$$

Since $\lim\limits_{x\to 0} f(x) = 0$, we have by letting $n \to \infty$ that

$$|f(x)| \le 2\varepsilon x, \quad 0 < x < \delta.$$

Thus,

$$\lim_{x\to 0} \frac{f(x)}{x} = 0.$$

$\square$

Problem 4.12. *Suppose $f(x) > 0$ is an increasing function defined on $(0,+\infty)$ satisfies $\lim\limits_{x\to+\infty} \dfrac{f(2x)}{f(x)} = 1$. Prove that $\lim\limits_{x\to+\infty} \dfrac{f(ax)}{f(x)} = 1$, $\forall\, a \in (0,+\infty)$.*

Proof. It suffices to prove $\lim\limits_{x\to+\infty}\dfrac{f(ax)}{f(x)}=1$ for all $a>1$. Indeed, for any $0<a<1$, we have

$$\lim_{x\to+\infty}\frac{f(ax)}{f(x)}=\lim_{x\to+\infty}\frac{f(ax)}{f(\frac{1}{a}ax)}=\frac{1}{\lim\limits_{ax\to+\infty}\frac{f(\frac{1}{a}ax)}{f(ax)}}=1.$$

For $a>1$, we know there exist some $N\in\mathbb{N}$ such that $2^N\le a<2^{N+1}$. Since $f(x)$ is increasing, we know

$$\frac{f(2^Nx)}{f(x)}\le\frac{f(ax)}{f(x)}\le\frac{f(2^{N+1}x)}{f(x)}.$$

Then the problem is reduced to prove $\lim\limits_{x\to+\infty}\dfrac{f(2^Kx)}{f(x)}=1$ for all $K\in\mathbb{N}$, and it is clear since

$$\lim_{x\to+\infty}\frac{f(2^Kx)}{f(x)}=\lim_{x\to+\infty}\frac{f(2^Kx)}{f(2^{K-1}x)}\cdot\lim_{x\to+\infty}\frac{f(2^{K-1}x)}{f(2^{K-2}x)}\cdots\lim_{x\to+\infty}\frac{f(2x)}{f(x)}=1.$$

□

Problem 4.13. *Suppose $f:(0,+\infty)\to\mathbb{R}$ satisfies $\forall\, a>0$, f is bounded on $(0,a)$. Prove that if $\lim\limits_{x\to+\infty}[f(x+1)-f(x)]=l$, then $\lim\limits_{x\to+\infty}\dfrac{f(x)}{x}=l$.*

Proof. Without loss of generality, we can assume that $l=0$ (otherwise, we replace $f(x)$ by $f(x)-lx$). Since $\lim\limits_{x\to+\infty}[f(x+1)-f(x)]=0$, we know $\forall\,\varepsilon>0$, there exists a $X>0$, such that $\forall\, x>X$, there is

$$-\varepsilon<f(x+1)-f(x)<\varepsilon.$$

Summing all inequalities, we have

$$-\left([x-X]+1\right)\varepsilon<f(x+1)-f\left(x-[x-X]\right)<\left([x-X]+1\right)\varepsilon.$$

Note that $\lim\limits_{x\to+\infty}\dfrac{f\left(x-[x-X]\right)}{x+1}=0$ since f is bounded on $(0,X+1)$. We have

$$-\varepsilon\le\varliminf_{x\to+\infty}\frac{f(x+1)}{x+1}\le\varlimsup_{x\to+\infty}\frac{f(x+1)}{x+1}\le\varepsilon.$$

Since ε is arbitrary, we obtain $\lim\limits_{x\to+\infty}\dfrac{f(x)}{x}=0$. □

Problem 4.14. *Suppose $f,g:(a,+\infty)\to\mathbb{R}$ satisfy $\forall\, b>a$, f,g are bounded on (a,b); g is strictly increasing, and $\lim\limits_{x\to+\infty}g(x)=+\infty$. Prove that if $\lim\limits_{x\to+\infty}\dfrac{f(x+1)-f(x)}{g(x+1)-g(x)}=l$, then $\lim\limits_{x\to+\infty}\dfrac{f(x)}{g(x)}=l$.*

Proof. The proof is very similar to Problem 4.13, we omit details here and leave it to readers. □

Remark 4.15. *(1) Problem 4.14 is called the function version of Stolz theorem.*
(2) We can change the constant "1" to any positive constant "T".
(3) l can be chosen as $\pm\infty$, but it's not correct for ∞.

Problem 4.16. *Suppose that f is defined on $\mathbb{R}$, and f is bounded in some neighborhood of $x = 0$. If there exist $a > 1$, $b > 1$, such that $f(ax) = bf(x)$, prove that $f(x)$ is continuous at $x = 0$.*

Proof. First, by $f(ax) = bf(x)$, it's easy to know $f(0) = 0$. Assume that there exist $M > 0$, $\delta_0 > 0$, such that $\forall\, x : |x| < \delta_0$, there is $|f(x)| \leq M$. For $\forall\, \varepsilon > 0$, choosing $\delta = \delta_0/a^{N+1}$, where N satisfies $M/b^N < \varepsilon$. Then for $\forall\, x : |x| < \delta$, there is

$$|f(x)| = \frac{1}{b^N}|f(a^N x)| \leq \frac{M}{b^N} < \varepsilon.$$

Hence

$$\lim_{x\to 0} f(x) = 0 = f(0),$$

i.e. $f(x)$ is continuous at $x = 0$. □

Problem 4.17. *Suppose that $f(x) \in C[0, +\infty)$ is bounded, and $\lim\limits_{x\to+\infty} f(x)$ does not exist. Prove there exists $t \in \mathbb{R}$ such that $f(x) = t$ has an infinite number of solutions.*

Proof. Since $f(x) \in C[0, +\infty)$ is bounded, and $\lim\limits_{x\to+\infty} f(x)$ does not exist, we know

$$-\infty < \varliminf_{x\to+\infty} f(x) < \varlimsup_{x\to+\infty} f(x) < +\infty.$$

Denote $l := \varliminf\limits_{x\to+\infty} f(x)$ and $\varlimsup\limits_{x\to+\infty} f(x)$. Let $t = \frac{l+L}{2}$, we will show $f(x) = t$ has an infinite number of solutions. Indeed, by the definition of upper limit and lower limit, we know that for $X_1 = 1$, there exist $x_1, y_1 > X_1$, such that $f(x_1) < t$, $f(y_1) > t$. Hence there exists z_1 between x_1 and y_1 such that $f(z_1) = t$ by $f(x)$ is continuous. For $X_2 = \max\{2, x_1, y_1\}$, there exist $x_2, y_2 > X_2$, such that $f(x_2) < t$, $f(y_2) > t$. Hence there exists z_2 between x_2 and y_2 such that $f(z_2) = t$ by $f(x)$ is continuous. Proceeding like this, we can find an infinite number of z_n such that $f(z_n) = t$, i.e. $f(x) = t$ has an infinite number of solutions. □

Problem 4.18. *Suppose that $f(x)$ is uniformly continuous on $[0, +\infty)$, and $\forall\, h > 0$, $\lim\limits_{n\to\infty} f(nh)$ exists. Prove that $\lim\limits_{x\to+\infty} f(x)$ exists.*

Proof. **First method: Find the limitation** Pick $h = 1$, assume that the sequence $\{f(n)\}_{n=1}^{\infty}$ converges to L. Then for each $m \in \mathbb{N}$, let $h = 1/m$. Then $\{f(n/m\}_{n=1}^{\infty}$ contains the subsequence $\{f(n)\}_{n=1}^{\infty}$. Thus $\{f(n/m\}_{n=1}^{\infty}$ converges to L for all m. Now we show that $\lim\limits_{x\to\infty} f(x) = L$.

Let $\varepsilon > 0$. Since f is uniformly continuous, there is $\delta > 0$ so that if $x, y \in \mathbb{R}_+$ and $|x - y| < \delta$, then

$$|f(x) - f(y)| < \frac{\varepsilon}{2}.$$

Now let $m \in \mathbb{N}$ so that $1/m < \delta$. Since $\{f(n/m\}_{n=1}^{\infty}$ converges to L, there is $N \in \mathbb{N}$ so that

$$\left| f\left(\frac{n}{m}\right) - L \right| < \frac{\varepsilon}{2}$$

for all $n \geq N$. Let $M = N/m$. Then if $x \geq M$, there is $n \geq N$ so that $|x - n/m| < \delta$ (we used $1/m < \delta$ here). Then $|f(x) - f(n/m)| < \varepsilon/2$ and thus

$$|f(x) - L| \leq \left| f(x) - f\left(\frac{n}{m}\right) \right| + \left| f\left(\frac{n}{m}\right) - L \right| < \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary we conclude $\lim\limits_{x\to\infty} f(x) = L$.

Second method: Cauchy principle Let $\varepsilon > 0$. Since f is uniformly continuous, there is $\delta > 0$ so that if $x, y \in \mathbb{R}_+$ and $|x - y| < \delta$, then

$$|f(x) - f(y)| < \frac{\varepsilon}{3}.$$

For ε, δ ginven as above, since $\lim\limits_{n\to\infty} f(n\delta)$ exists, we know there exists $N \in \mathbb{N}$, such that $\forall\, m, n > N$, there is

$$|f(m\delta) - f(n\delta)| < \frac{\varepsilon}{3}.$$

Then, let $X = (N+1)\delta$, we know that $\forall\, x_1, x_2 > X$, there are

$$\left[\frac{x_i}{\delta}\right] > \frac{x_i}{\delta} - 1 > N \quad (i = 1, 2)$$

and

$$\left| x_i - \left[\frac{x_i}{\delta}\right]\delta \right| = \delta \left| \frac{x_i}{\delta} - \left[\frac{x_i}{\delta}\right] \right| < \delta \quad (i = 1, 2).$$

Hence

$$\begin{aligned}
|f(x_1) - f(x_2)| \leq & \left| f(x_1) - f\left(\left[\frac{x_1}{\delta}\right]\delta\right) \right| \\
& + \left| f\left(\left[\frac{x_1}{\delta}\right]\delta\right) - f\left(\left[\frac{x_2}{\delta}\right]\delta\right) \right| \\
& + \left| f\left(\left[\frac{x_2}{\delta}\right]\delta\right) - f\left(x_2\right) \right| \\
< & \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} \\
= & \varepsilon.
\end{aligned}$$

□

Remark 4.19. *We can make a weak assumption that $f(x)$ is just continuous on $[0, +\infty)$, but to prove this conclusion is so difficult, we omit the detail here and leave it to someone who interested.*

Problem 4.20. *Suppose that $f(x)$ is uniformly continuous on $[0, +\infty)$, and $\forall\, x > 0$, $\lim\limits_{n\to\infty} f(x + n) = 0$. Prove that $\lim\limits_{x\to+\infty} f(x) = 0$.*

Proof. By $f(x)$ is uniformly continuous on $[0,+\infty)$, we know $\forall\,\varepsilon>0$, there is $\delta>0$, such that $\forall\,x,y\geq 0:|x-y|<\delta$, there is

$$|f(x)-f(y)|<\frac{\varepsilon}{2}.$$

Take $k>\frac{1}{\delta}$, and cut $[0,1]$ uniformly into k pieces. Let $x_i=\frac{i}{k}$ $(i=1,2,\cdots,k)$ be the cut points. Note that $x_i-x_{i-1}=\frac{1}{k}<\delta$. Since for every x_i, there is $\lim\limits_{n\to\infty}f(x_i+n)=0$. We know there exists $N_i>0$, such that $\forall\,n>N_i$, there is $|f(x_i+n)|<\dfrac{\varepsilon}{2}$. Let $N=\max\{N_1,N_2,\cdots,N_k\}$, then $\forall\,n>N$, there is

$$|f(x_i+n)|<\frac{\varepsilon}{2}\quad(i=1,2,\cdots,k).$$

Choose $X=N+1$, for $\forall\,x>X$, there is $[x]>N$. Since $x-[x]\in[0,1)$, we know there exists $i\in\{1,2,\cdots,k\}$, such that $|(x-[x])-x_i|<\delta$, i.e. $|x-(x_i+[x])|<\delta$. Hence, there is

$$\begin{aligned}|f(x)|\leq&|f(x)-f(x_i+[x])|+|f(x_i+[x])|\\<&\frac{\varepsilon}{2}+\frac{\varepsilon}{2}\\=&\varepsilon.\end{aligned}$$

i.e. $\lim\limits_{x\to+\infty}f(x)=0$. □

Remark 4.21. *(1) We can use the same method to prove that if $f(x)$ is uniformly continuous on $[0,+\infty)$, and $\forall\,x>0$, $\lim\limits_{n\to\infty}f(x+n)=A$, then $\lim\limits_{x\to+\infty}f(x)=A$.*
(2) From the proof, we can see that the conclusion is still right for a weak assumption, i.e. $\forall\,x\in[0,1]$, $\lim\limits_{n\to\infty}f(x+n)=0$.
(3) Find a counterexample if we do not assume $f(x)$ is uniformly continuous, but just continuous. Indeed, define

$$f_n(x)=\begin{cases}2nx, & 0\leq x\leq\dfrac{1}{2n},\\ 2-2nx, & \dfrac{1}{2n}<x\leq\dfrac{1}{n},\\ 0, & \textit{otherwise}.\end{cases}$$

Let $f(x)=\sum\limits_{n=1}^{\infty}f_n(x-n)$, then $f(x)$ is a counterexample.

Exercise 4.22 (Leave to readers)**.** *Find x such that $\lim\limits_{m\to\infty}\sqrt{1+\sqrt{x+\sqrt{x^2+\cdots+\sqrt{x^m}}}}=2$.*

Hint: Let me describe a sketch of proof that $x=4$.

A. Observe that if $f(x) = \lim_{n\to\infty}\sqrt{1+\sqrt{x+\sqrt{x^2+\cdots\sqrt{x^n}}}}$, then f is strictly increasing.

B. We shall show that $f(4) = 2$, and hence $x = 4$ is the unique answer.

B_1. Fix $m \in \mathbb{N}$ and show that, for $n = m, m-1, m-2, \cdots$ (induction backwards)

$$2^n < \sqrt{4^n+\sqrt{4^{n+1}+\cdots\sqrt{4^{m-1}+\sqrt{4^m}}}} < 2^n+1,$$

while

$$\sqrt{4^n+\sqrt{4^{n+1}+\cdots\sqrt{4^{m-1}+\sqrt{4^m}+1}}} = 2^n+1.$$

B_2. Next estimate the difference

$$\begin{aligned}
&(2^n+1)-\sqrt{4^n+\sqrt{4^{n+1}+\cdots\sqrt{4^{m-1}+\sqrt{4^m}}}}\\
=&\sqrt{4^n+\sqrt{4^{n+1}+\cdots\sqrt{4^{m-1}+\sqrt{4^m}+1}}}-\sqrt{4^n+\sqrt{4^{n+1}+\cdots\sqrt{4^{m-1}+\sqrt{4^m}}}}\\
=&\frac{\sqrt{4^{n+1}+\cdots\sqrt{4^{m-1}+\sqrt{4^m}+1}}-\sqrt{4^{n+1}+\cdots\sqrt{4^{m-1}+\sqrt{4^m}}}}{\sqrt{4^n+\sqrt{4^{n+1}+\cdots\sqrt{4^{m-1}+\sqrt{4^m}+1}}}+\sqrt{4^n+\sqrt{4^{n+1}+\cdots\sqrt{4^{m-1}+\sqrt{4^m}}}}}\\
<&\frac{\sqrt{4^{n+1}+\cdots\sqrt{4^{m-1}+\sqrt{4^m}+1}}-\sqrt{4^{n+1}+\cdots\sqrt{4^{m-1}+\sqrt{4^m}}}}{2\cdot 2^n}\\
<&\cdots<\frac{(\sqrt{4^m}+1)-\sqrt{4^m}}{2^{m-n}\cdots 2^{n+(n+1)+\cdots+(m-1)}}\\
=&2^{-\frac{(m-n)(n+m+1)}{2}}.
\end{aligned}$$

Thus

$$\lim_{m\to\infty}\sqrt{4^n+\sqrt{4^{n+1}+\cdots\sqrt{4^{m-1}+\sqrt{4^m}}}} = 2^n+1.$$

For $n = 0$ we have

$$\lim_{m\to\infty}\sqrt{1+\sqrt{4+\cdots\sqrt{4^{m-1}+\sqrt{4^m}}}} = 2^0+1 = 2.$$

□

A similar but easier question: Find x in:

$$\sqrt{x^2+\sqrt{4x^2+\sqrt{16x^2+\sqrt{64x^2+\cdots}}}} = 5.$$

Hint: $x+1=\sqrt{x^2+2x+1}=\sqrt{x^2+\sqrt{4x^2+4x+1}}=\cdots$.

Exercise 4.23 (Challenge!)**.** *Assume $f \in C[0,+\infty)$, and for all $a>0$, we have*

$$\lim_{x\to+\infty}(f(x+a)-f(x))=0.$$

Prove that $f(x)$ is uniformly continuous.

Hint: Fix $\varepsilon>0$, we want to find $\delta>0$ such that

$$|x-y|<\delta \Rightarrow |f(x)-f(y)|<\varepsilon. \tag{4.1}$$

For every $N\in\mathbb{N}$, let $E_N := \{a \mid x\geq N \Rightarrow |f(x+a)-f(x)|\leq \varepsilon/4\}$. E_N is closed (by continuity of f) and $\bigcup_{N\in\mathbb{N}} E_N=[0,\infty)$. By Baire Category Theorem, at least one of them, say, E_N contains a closed interval $[b,c]$. For $x,y\geq N+c$, without loss of generality, say $y\geq x$, if $|y-x|<c-b$, there always exists $z\geq N$ such that $[x,y]\subset[z+b,z+c]$. Then $|f(x)-f(y)|\leq|f(x)-f(z)|+|f(y)-f(z)|=|f(z+d)-f(z)|+|f(z+e)-f(z)|\leq\varepsilon/2$ where $d,e\in[b,c]$. For $x,y\leq N+c$, as $[0,N+c]$ is compact, f restricted to $[0,N+c]$ is uniformly continuous, hence there exists $\delta'>0$ satisfing the requirements in (4.1). Let $\delta=\min(c-b,\delta')$, then we are done. $\square$

5. Week 8 (10.24)

Problem 5.1. *Suppose that $f(x)$ is uniformly continuous on $\mathbb{R}$. Prove that there exist $a > 0$, $b > 0$ such that $|f(x)| \le a|x| + b$, $\forall\, x \in \mathbb{R}$.*

Proof. Since $f(x)$ is uniformly continuous on $\mathbb{R}$, we have that $\forall\, \varepsilon > 0$, there exists $\delta > 0$, such that $\forall\, x, y \in \mathbb{R} : |x - y| < \delta$, there is $|f(x) - f(y)| < \varepsilon$. Now, fix ε and δ. For $\forall\, x \in \mathbb{R}$, there exists $n \in \mathbb{Z}$, such that $x = n\delta + x_0$, where $x_0 \in (-\delta, \delta)$. Note that $f(x)$ is bounded on $[-\delta, \delta]$, i.e. $\exists M > 0$, such that $|f(x)| \le M$ $(\forall\, x \in)[-\delta, \delta]$. Hence,

$$\begin{aligned}
|f(x)| &= \left| \sum_{k=1}^{n} [f(k\delta + x_0) - f((k-1)\delta + x_0)] + f(x_0) \right| \\
&\le \sum_{k=1}^{n} |f(k\delta + x_0) - f((k-1)\delta + x_0)| + |f(x_0)| \\
&\le |n|\varepsilon + M \\
&= \frac{\varepsilon}{\delta}|x - x_0| + M \qquad \left(\text{since } \left|\frac{x - x_0}{\delta}\right| = |n|\right) \\
&\le \frac{\varepsilon}{\delta}|x| + \left(M + \frac{\varepsilon}{\delta}|x_0|\right) \\
&\le \frac{\varepsilon}{\delta}|x| + (M + \varepsilon).
\end{aligned}$$

Denote $a = \varepsilon/\delta$, $b = M + \varepsilon$, hence

$$|f(x)| \le a|x| + b \quad (\forall\, x \in (-\infty, +\infty)).$$

□

Problem 5.2 (4.24)**.** *Prove that at any point of the curve*

$$\begin{cases} x = a(\cos t + t\sin t), \\ y = a(\sin t - t\cos t), \end{cases} \quad (a > 0),$$

the distance of the normal line to the origin is equal to a.

Proof. Differentiating respect to t yields

$$\begin{cases} dx = at\cos t\, dt, \\ dy = at\sin t\, dt, \end{cases} \quad (a > 0)$$

Then the normal line at point (x, y), $t \ne k\pi$ is

$$Y - a(\sin t - t\cos t) = -\frac{dx}{dy}(X - a(\cos t + t\sin t)),$$

i.e.

$$Y - a(\sin t - t\cos t) = -\cot t(X - a(\cos t + t\sin t)).$$

Hence

$$\begin{aligned} d &= \frac{|a\cot t(\cos t + t\sin t) + a(\sin t - t\cos t)|}{\sqrt{1+\cot^2 t}} \\ &= |a\cos^2 t + at\sin t\cos t + a\sin^2 t - at\sin t\cos t| \\ &= a. \end{aligned}$$

When $t = k\pi$, we know the normal line is $x = a(-1)^k$, thus $d = a$.

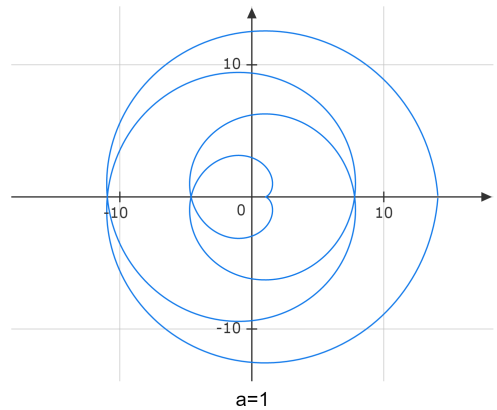

FIGURE 1. Graph of the curve for $a = 1$

□

Problem 5.3. *Calculate the derivative of* $f(x) = x^{\arcsin x}$.

Solution. Note that

$$f(x) = x^{\arcsin x} = e^{\ln x \cdot \arcsin x}.$$

Denote $g(x) = \ln x \cdot \arcsin x$, we have

$$g'(x) = \frac{\arcsin x}{x} + \frac{\ln x}{\sqrt{1-x^2}}.$$

Hence

$$f'(x) = \left(e^{g(x)}\right)' = e^{g(x)}g'(x) = x^{\arcsin x}\left(\frac{\arcsin x}{x} + \frac{\ln x}{\sqrt{1-x^2}}\right).$$

□

Problem 5.4. *Calculate the left right derivative of* $f(x) = \begin{cases} \dfrac{x}{e^{1/x}+1}, & x \neq 0, \\ 0, & x = 0, \end{cases}$ *at* $x = 0$, *and determine whether* $f(x)$ *is differentiable at* $x = 0$.

Solution. Left derivative:

$$f'_-(0) = \lim_{x\to 0-0} \frac{\frac{x}{e^{1/x}+1} - 0}{x - 0} = \lim_{x\to 0-0} \frac{1}{e^{1/x}+1} = 1.$$

Right derivative:

$$f'_+(0) = \lim_{x\to 0+0} \frac{\frac{x}{e^{1/x}+1} - 0}{x - 0} = \lim_{x\to 0+0} \frac{1}{e^{1/x}+1} = 0.$$

Hence $f(x)$ is not differentiable at $x = 0$. The graph of $f(x)$ is as follows:

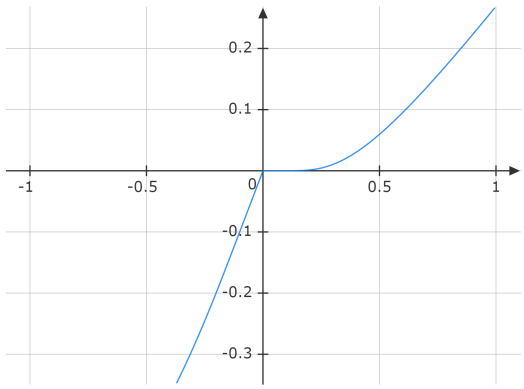

FIGURE 2. Graph of $f(x)$

□

Problem 5.5. *Suppose that $f(x)$ is differentiable on $\mathbb{R}$ and satisfies $f(x) \geq x$, $f(x) \geq 1 - x$, $\forall\, x \in \mathbb{R}$. Prove that $f(\frac{1}{2}) > \frac{1}{2}$.*

Proof. Suppose that $f(\frac{1}{2}) \leq \frac{1}{2}$. Then we have

$$\begin{aligned} f'_-\left(\frac{1}{2}\right) &= \lim_{x\to\frac{1}{2}-0} \frac{f(x) - f(\frac{1}{2})}{x - \frac{1}{2}} \\ &= \lim_{x\to\frac{1}{2}-0} \frac{f(\frac{1}{2}) - f(x)}{\frac{1}{2} - x} \\ &\leq \lim_{x\to\frac{1}{2}-0} \frac{\frac{1}{2} - (1 - x)}{\frac{1}{2} - x} \\ &= -1, \end{aligned}$$

and

$$\begin{aligned} f'_+\left(\frac{1}{2}\right) &= \lim_{x\to\frac{1}{2}+0} \frac{f(x) - f(\frac{1}{2})}{x - \frac{1}{2}} \\ &\geq \lim_{x\to\frac{1}{2}+0} \frac{x - \frac{1}{2}}{x - \frac{1}{2}} \\ &= 1, \end{aligned}$$

contradicts with $f(x)$ is differentiable at $x = \frac{1}{2}$. Hence $f(\frac{1}{2}) > \frac{1}{2}$. □

Remark 5.6. *From the proof of Problem 5.5, we know that it only needs to assume $f(x)$ is differentiable at $x = \frac{1}{2}$.*

Problem 5.7. *Suppose that $f(x)$ is differentiable. Prove that $F(x) = f(x)(1 + |\sin x|)$ is differentiable at $x = 0$ if and only if $f(0) = 0$.*

Proof. Calculating the left derivative of F at $x = 0$ yields

$$\begin{aligned} F'_-(0) &= \lim_{x\to 0-0} \frac{f(x)(1+|\sin x|) - f(0)}{x} \\ &= \lim_{x\to 0-0} \frac{f(x) - f(0)}{x} + \lim_{x\to 0-0} \frac{f(0)|\sin x|}{x} \\ &= f'(0) - f(0). \end{aligned}$$

Similarly, the right left derivative of F at $x = 0$ is

$$\begin{aligned} F'_+(0) &= \lim_{x\to 0+0} \frac{f(x)(1+|\sin x|) - f(0)}{x} \\ &= \lim_{x\to 0+0} \frac{f(x) - f(0)}{x} + \lim_{x\to 0+0} \frac{f(0)|\sin x|}{x} \\ &= f'(0) + f(0). \end{aligned}$$

Hence $F(x)$ is differentiable at $x = 0$ if and only if $F'_-(0) = F'_+(0)$, if and only if $f'(0) - f(0) = f'(0) + f(0)$, if and only if $f(0) = 0$. □

Problem 5.8. *Suppose that $y = y(x)$ is determined by parametric equation $\begin{cases} x = 2t + |t| \\ y = t^2 + 2t|t| \end{cases}$, $t \in \mathbb{R}$. Prove that $y(x)$ is differentiable at $x = 0$, and find $y'(0)$.*

Proof. **First way:** By $x = 2t + |t|$, we have

$$t = \begin{cases} \dfrac{x}{3}, & x \geq 0, \\ x, & x < 0. \end{cases}$$

Hence by $y = t^2 + 2t|t|$, we know

$$y = \begin{cases} \dfrac{1}{3}x^2, & x \geq 0, \\ -x^2, & x < 0. \end{cases}$$

It's easy to see that $y(x)$ is differentiable at $x = 0$, and $y'(0) = 0$.

Second way: A direct differentiating yields

$$\begin{cases} dx = \left(2 + \dfrac{t}{|t|}\right) dt \\ dy = (2t + 4|t|)\, dt \end{cases}$$

Hence

$$\frac{dy}{dx} = \frac{2t + 4|t|}{2 + \frac{t}{|t|}} = 2|t|.$$

It's easy to see that $y(x)$ is differentiable at $x = 0$, and $y'(0) = 0$. $\square$

Problem 5.9. *Suppose that* $x^2y^2+x^2+y^2 = 1$, $(xy > 0)$. *Prove that* $\dfrac{dx}{\sqrt{1-x^4}} + \dfrac{dy}{\sqrt{1-y^4}}$.

Proof. A direct differentiating yields

$$x(1 + y^2)dx + y(1 + x^2)dy = 0.$$

Note by $x^2y^2 + x^2 + y^2 = 1$ that

$$x^2(1 + y^2)^2 = 1 - y^4,$$

and

$$y^2(1 + x^2)^2 = 1 - x^4.$$

Since $xy > 0$, we know $x(1 + y^2)$ and $y(1 + x^2)$ have the same sign. Then

$$\sqrt{1 - y^4}dx + \sqrt{1 - x^4}dy = 0,$$

i.e.

$$\frac{dx}{\sqrt{1-x^4}} + \frac{dy}{\sqrt{1-y^4}}.$$

The graph of the curve is as follows:

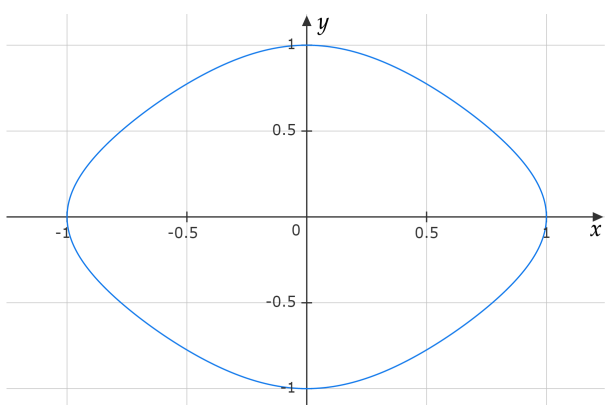

FIGURE 3. Graph of the curve

$\square$

Problem 5.10. *Prove the following identities:*

(1) $\displaystyle\sum_{k=1}^{n} kC_n^k = n2^{n-1}$, $n \in \mathbb{N}_+$;

(2) $\sum_{k=1}^{n} k^2 C_n^k = n(n+1)2^{n-2}$, $n \in \mathbb{N}_+$.

Proof. (1) Note that

$$\sum_{k=1}^{n} kC_n^k x^{k-1} = \left(\sum_{k=1}^{n} C_n^k x^k\right)' = ((1+x)^n - 1)' = n(1+x)^{n-1}.$$

Set $x = 1$, and there is

$$\sum_{k=1}^{n} kC_n^k = n2^{n-1}.$$

(2) **First way:** Note that

$$\begin{aligned}
\sum_{k=1}^{n} k^2 C_n^k x^{k-1} &= \left(\sum_{k=1}^{n} kC_n^k x^k\right)' \\
&= \left(x\sum_{k=1}^{n} kC_n^k x^{k-1}\right)' \\
&= \left(nx(1+x)^{n-1}\right)' \\
&= n(1+x)^{n-1} + n(n-1)x(1+x)^{n-2}.
\end{aligned}$$

Set $x = 1$, and there is

$$\sum_{k=1}^{n} k^2 C_n^k = n(n+1)2^{n-2}.$$

Second way: Firstly, we have

$$k^2 C_n^k = k^2 \frac{n!}{k!(n-k)!} = nkC_{n-1}^{k-1}.$$

Then

$$\begin{aligned}
\sum_{k=1}^{n} k^2 C_n^k x^{k-1} &= n\sum_{k=1}^{n} kC_{n-1}^{k-1} x^{k-1} \\
&= \left(n\sum_{k=1}^{n} C_{n-1}^{k-1} x^k\right)' \\
&= \left(nx(1+x)^{n-1}\right)' \\
&= n(1+x)^{n-1} + n(n-1)x(1+x)^{n-2}.
\end{aligned}$$

Set $x = 1$, and there is

$$\sum_{k=1}^{n} k^2 C_n^k = n(n+1)2^{n-2}.$$

□

Problem 5.11. *Suppose that $f : [a, b] \to [a, b]$ satisfies*

$$|f(x) - f(y)| \le |x - y|, \quad \forall\, x, y \in [a, b].$$

Define $x_{n+1} = \frac{1}{2}(x_n + f(x_n))$ for any given $x_1 \in [a, b]$. Prove that $\lim\limits_{n\to\infty} x_n$ exists.

Proof. Note that

$$\begin{aligned} x_{n+1} - x_n &= \frac{1}{2}(x_n + f(x_n)) - \frac{1}{2}(x_{n-1} + f(x_{n-1})) \\ &= \frac{1}{2}(f(x_n) - f(x_{n-1})) + \frac{1}{2}(x_n - x_{n-1}). \end{aligned}$$

Then we have

$$\begin{aligned} (x_{n+1} - x_n)(x_n - x_{n-1}) &= \frac{1}{2}(f(x_n) - f(x_{n-1}))(x_n - x_{n-1}) + \frac{1}{2}(x_n - x_{n-1})^2 \\ &\ge -\frac{1}{2}|x_n - x_{n-1}|^2 + \frac{1}{2}(x_n - x_{n-1})^2 = 0, \end{aligned}$$

since

$$|f(x_n) - f(x_{n-1})||x_n - x_{n-1}| \le |x_n - x_{n-1}|^2.$$

Hence, $\{x_n\}$ is monotonic. Clearly, $\{x_n\}$ is bounded, it's easy to know that $\lim\limits_{n\to\infty} x_n$ exists. □

Problem 5.12. *Suppose that $f(x)$ is differentiable on $[0, 1]$, and $\{x \in [0, 1] \big| f(x) = 0,\ f'(x) = 0\} = \emptyset$. Prove that f has a finite number of zero pionts in $[0, 1]$.*

Proof. Assume that f has an infinite number of zero points. Let $Z := \{x \in [0, 1] \big| f(x) = 0\}$. Since Z is a bounded set with an infinite number of elements, we know by the Bolzano-Weierstrass theorem that there is a sequence $\{x_n\} \subset Z$ converges, say $\lim\limits_{n\to\infty} x_n = x_0$. By the continuity of f, we know that $f(x_0) = \lim\limits_{n\to\infty} f(x_n) = 0$. Since $f(x)$ is differentiable, we have

$$f'(x_0) = \lim_{n\to\infty} \frac{f(x_n) - f(x_0)}{x_n - x_0} = 0,$$

i.e. $x_0 \in \{x \in [0, 1] \big| f(x) = 0,\ f'(x) = 0\}$, contradiction. □

Problem 5.13.

(1) *Suppose that $f \in C[0, 1]$, $f(0) = f(1)$. Prove that for $0 < \alpha < 1$, if $\frac{1}{\alpha} \in \mathbb{N}$, then there exists $\xi \in [0, 1 - \alpha]$ such that $f(\xi) = f(\xi + \alpha)$;*

(2) *Prove that for $0 < \alpha < 1$, $\frac{1}{\alpha} \notin \mathbb{N}$, there always exists $f \in C[0, 1]$, $f(0) = f(1)$ such that $\forall\, x \in [0, 1 - \alpha]$, there is $f(x) \ne f(x + \alpha)$.*

Proof. (1) Let $g(x) = f(x) - f(x + \alpha)$. Since $\frac{1}{\alpha} \in \mathbb{N}$, we know

$$\sum_{k=0}^{\frac{1}{\alpha}-1} g(k\alpha) = f(0) - f(1) = 0.$$

Then there must be $i, j \in \{0, 1, \cdots, \frac{1}{\alpha} - 1\}$, $i \neq j$, such that $g(i\alpha)g(j\alpha) \leq 0$. Hence there exists $\xi \in [i\alpha, j\alpha] \subset [0, 1-\alpha]$ such that $f(\xi) = f(\xi + \alpha)$.

(2) For $0 < \alpha < 1$, we define

$$f(x) = \sin^2\left(\frac{\pi x}{\alpha}\right) - x\sin^2\left(\frac{\pi}{\alpha}\right).$$

Clearly, f is continuous and $f(0) = 0 = f(1)$. If there is some $x_0 \in [0, 1-\alpha]$, such that $f(x_0) = f(x_0 + \alpha)$, we know there is $\alpha \sin^2\left(\frac{\pi}{\alpha}\right) = 0$. However, since $\frac{1}{\alpha} \notin \mathbb{N}$, we know it's impossible for $\sin^2\left(\frac{\pi}{\alpha}\right) = 0$. Hence, we have that $\forall\, x \in [0, 1-\alpha]$, there is $f(x) \neq f(x + \alpha)$. □

Problem 5.14. *Define $f \in C(\mathbb{R})$ satisfying*

$$f(f(x)) = -x^3 + \sin(x^2 + \ln(1 + |x|)). \tag{5.1}$$

Prove that this equation has no continuous solution.

Proof. Assume by contradiction that there is a continuous function f satisfies (5.1). Then we know that

$$\lim_{x\to+\infty} f(f(x)) = -\infty \quad \text{and} \quad \lim_{x\to-\infty} f(f(x)) = +\infty.$$

We conclude that there must be $\lim\limits_{x\to+\infty} f(x) = -\infty$. Indeed, if there exists a sequence $\{x_n\}$ satisfying $\lim\limits_{n\to\infty} x_n = +\infty$ but $f(x_n)$ is bounded, we know that $f(f(x_n))$ must be bounded since f is continuous, which contradicts with $\lim\limits_{x\to+\infty} f(f(x)) = -\infty$. Hence by the continuity of f, we have that $\lim\limits_{x\to+\infty} f(x) = +\infty$ or $\lim\limits_{x\to+\infty} f(x) = -\infty$. If $\lim\limits_{x\to+\infty} f(x) = +\infty$, then we'll get $\lim\limits_{x\to+\infty} f(f(x)) = +\infty$ which generates contradiction. So we must have $\lim\limits_{x\to+\infty} f(x) = -\infty$. Similarly we must have $\lim\limits_{x\to-\infty} f(x) = +\infty$. But using $\lim\limits_{x\to+\infty} f(x) = -\infty$ and $\lim\limits_{x\to-\infty} f(x) = +\infty$ we have $\lim\limits_{x\to+\infty} f(f(x)) = +\infty$, which also generates contradiction. Hence, (5.1) has no continuous solution. □

Remark 5.15. *We can prove a general conclusion: For any $f \in C(\mathbb{R})$, $\lim\limits_{x\to+\infty} f(f(x)) = -\infty$ and $\lim\limits_{x\to-\infty} f(f(x)) = +\infty$ cannot be simultaneously true.*

Problem 5.16. *Suppose that $g(x)$ is defined on $[0, 1]$, and $g(0) = 1$, $g(1) = 0$. If there exists a continuous function $h(x)$ such that $g(x) + h(x)$ is monotonic increasing on $[0, 1]$, prove that $[0, 1] \subset g([0, 1])$.*

Proof. Denote that $f = g + h$. Since f is monotonic increasing, we have that $f(x-0) \leq f(x) \leq f(x+0)$, $(0 < x < 1)$. Since h is continuous, we know that $g(x-0) \leq g(x) \leq g(x+0)$, $(0 < x < 1)$. For $\forall\, y \in (0, 1)$, we define

$$E_y = \{t \in [0, 1] \big| g(x) > y, \forall\, x \in [0, t]\}.$$

Since $g(0) = 1 > y$, we know that E_y is not empty. Then the supremum of E_y exists. Let $x_0 = \sup E_y$, then $g(x_0 - 0) \geq y \geq g(x_0 + 0)$. Combining above, we obtain that

$g(x_0-0)=g(x_0)=g(x_0+0)=y$, i.e. $y\in g([0,1])$. Hence $(0,1)\subset g([0,1])$. It is clear that $0,1\in g([0,1])$, we have that $[0,1]\subset g([0,1])$. $\square$

Exercise 5.17. *Let $f(x)$ be continuous on $\mathbb{R}$. Suppose that f is periodic with the minimal postive period $\mu>0$, μ is irrational. Show that $\lim\limits_{n\to\infty} f(n)$ does not exist.*

Hint: By Kronecker's Approximation Theorem, we know that the sequence of numbers $\{n\mu-[n\mu]\}$ is dense in the unit interval. Hence we know that for any $x_0\in[0,1]$, there exists a sequence $\{n_j\mu\}$ such that $n_j\mu-[n_j\mu]\to x_0$, $j\to\infty$. Then

$$f([n_j\mu])=f([n_j\mu]-n_j\mu)\to f(-x_0),\quad j\to\infty,$$

which means $\{f(n)\}$ does not converge. $\square$

Exercise 5.18. *Suppose that $x_0=1$, $x_n=x_{n-1}+\cos x_{n-1}$, $(n=1,2,\cdots)$. Prove that $x_n-\dfrac{\pi}{2}=o\left(\dfrac{1}{n^n}\right)$ as $n\to\infty$.*

Hint: Let $y_n=\frac{\pi}{2}-x_n$, $n=0,1,2,\cdots$. Then $y_n=y_{n-1}-\sin y_{n-1}$. By the inequality

$$x-\frac{x^3}{6}<\sin x<x,\quad x\in(0,+\infty),$$

we have

$$0<y_n=y_{n-1}-\sin y_{n-1}<\frac{y_{n-1}^3}{6}<y_{n-1}^3,\quad n\in\mathbb{N}_+.$$

Hence

$$y_n<y_0^{3^n}.$$

Note that $0<y_0<1$. We konw that there is $N\in\mathbb{N}$, such that $\forall\, n>N$,

$$0<y_nn^n<y_0^{3^n}n^n<n^ny_0^{n^2}=\left(\frac{n}{\left(\frac{1}{y_0}\right)^n}\right)^n<\frac{1}{2^n}.$$

Hence, we have

$$y_n=o\left(\frac{1}{n^n}\right),\quad n\to\infty.$$

$\square$

Exercise 5.19. *Suppose that $f\in C[0,1]$, $\lim\limits_{x\to0+0}\dfrac{f(x)-f(0)}{x}=\alpha<\beta=\lim\limits_{x\to1-0}\dfrac{f(x)-f(1)}{x-1}$. Prove that $\forall\,\lambda\in(\alpha,\beta)$, $\exists\, x_1,x_2\in[0,1]$, such that $\lambda=\dfrac{f(x_1)-f(x_2)}{x_1-x_2}$.*

Hint: Let $g(x)=f(x)-\lambda x$. Since

$$\lim_{x\to0+0}\frac{g(x)-g(0)}{x}=\alpha-\lambda<0,$$

we know there exists $\delta_1 > 0$, such that $0 < x < \delta_1$, there is $g(x) < g(0)$. Similarly, since

$$\lim_{x \to 1-0} \frac{g(x) - g(1)}{x - 1} = \beta - \lambda > 0,$$

we know there exists $\delta_2 > 0$, such that $0 < x < \delta_2$, there is $g(x) < g(1)$. Then we know that the minimum of g is achieved in $(0, 1)$. Assume that $g(x_0) = \min_{x \in [0,1]} g(x)$, then $g(x_0) < g(0)$, $g(x_0) < g(1)$. Note that if there exist $x_1, x_2 \in [0, 1]$, $x_1 \neq x_2$, such that $g(x_1) = g(x_2)$, then we have $f(x_1) - \lambda x_1 = f(x_2) - \lambda x_2$, i.e.

$$\frac{f(x_1) - f(x_2)}{x_1 - x_2} = \lambda.$$

Indeed, if $g(0) = g(1)$, we are done. Next, we assume that $g(0) \neq g(1)$. Without loss of generality, we assume that $g(0) < g(1)$, then there is $g(x_0) < g(0) < g(1)$. Since $g(x)$ is continuous on $[x_0, 1]$, we know there exists $\xi \in (x_0, 1)$ such that $g(\xi) = g(0)$, then we are done. □

Exercise 5.20. *Let a, b be two nonzero real numbers and a function $f : \mathbb{R} \longrightarrow [0, \infty)$ satisfying the functional equation*

$$f(x + a + b) + f(x) = f(x + a) + f(x + b). \tag{5.2}$$

(1) *Prove that f is periodic if a/b is rational.*
(2) *If a/b is not rational, could f be nonperiodic?*

Hint: (1) From (5.2), easy induction gives

$$f(x + a + nb) - f(x + nb) = f(x + a) - f(x), \quad \forall\, n \in \mathbb{Z}.$$

From $f(x + a + nb) - f(x + a) = f(x + nb) - f(x)$, easy induction gives

$$f(x + ma + nb) - f(x + ma) = f(x + nb) - f(x), \quad \forall\, m \in \mathbb{Z}.$$

So

$$f(x + ma + nb) + f(x) = f(x + ma) + f(x + nb), \quad \forall\, x \in \mathbb{R} \text{ and } \forall\, m, n \in \mathbb{Z}. \tag{5.3}$$

If $\frac{a}{b}$ is rational, we can choose m, n such that $ma + nb = 0$. Let then $u = |ma| > 0$ and (5.3) becomes

$$2f(x) = f(x + u) + f(x - u),$$

or also

$$f(x + u) - f(x) = f(x) - f(x - u).$$

From there, we easily get

$$f(x + nu) = f(x) + n(f(x + u) - f(x)), \quad \forall\, x \in \mathbb{R},\ \forall\, n \in \mathbb{Z}. \tag{5.4}$$

Then, if $f(x + u) - f(x) \neq 0$, setting $n \to +\infty$ or $n \to -\infty$ (depending on sign), (5.4) implies $f(x + nu) < 0$, contradicts with $f(x) \geq 0$. Hence

$$f(x + u) = f(x), \quad \forall\, x \in \mathbb{R} \text{ and for some } u > 0.$$

(2) Define

$$f(x) = \frac{1}{2}\left(\cos\frac{2\pi x}{a} + \cos\frac{2\pi x}{b}\right) + 1.$$

It's easy to verify that f satisfies (5.3). But f is periodic if and only if a/b is rational. This can be seen by assuming $f(T) = f(0) = 2$, which implies both $\cos\frac{2\pi T}{a}$ and $\cos\frac{2\pi T}{b}$ have to be 1, i.e. T/a and T/b have to be integers. $\square$

Remark 5.21. *From* (5.4) *in the proof of Exercise 5.20, we can see that f is only needed to be bounded above or below.*

Exercise 5.22. *Suppose that $f(x)$ is a uniformly continuous function on $[1, +\infty)$. Prove that $\overline{\lim\limits_{x\to+\infty}} \dfrac{f(x)}{x} < +\infty$.*

Hint: By Problem 5.1, we know that there exist $a, b > 0$ such that $|f(x)| \le ax+b$. Hence

$$\overline{\lim_{x\to+\infty}} \frac{f(x)}{x} \le \overline{\lim_{x\to+\infty}} \frac{|f(x)|}{x} \le \overline{\lim_{x\to+\infty}} \left(a + \frac{b}{x}\right) = a < +\infty.$$

$\square$

6. Week 9 (10.31)

Problem 6.1 (Mid 1)**.** *Calculate limitations.*

(1) $\lim\limits_{n\to\infty}[\sin(\ln(n+1))-\sin(\ln n)]$;

(2) $\lim\limits_{x\to 0}\dfrac{\sqrt[3]{1+x\sin x}-1}{\arctan x^2}$;

(3) $\lim\limits_{x\to 0}(1+2x)^{\frac{(x+1)^2}{x}}$;

(4) $\lim\limits_{n\to\infty}[(n+\ln n)^a-n^a]$, *where* $0<a<1$;

(5) $\lim\limits_{x\to+\infty}\left(\dfrac{2^{1/x}+8^{1/x}}{2}\right)^x$.

Solution. (1)

$$\begin{aligned}\lim_{n\to\infty}|\sin(\ln(n+1))-\sin(\ln n)| &= \lim_{n\to\infty}\left|2\cos\left(\frac{\ln(n+1)+\ln n}{2}\right)\sin\left(\frac{\ln(n+1)-\ln n}{2}\right)\right| \\ &\le \lim_{n\to\infty}(\ln(n+1)-\ln n)=0.\end{aligned}$$

(2)

$$\lim_{x\to 0}\frac{\sqrt[3]{1+x\sin x}-1}{\arctan x^2}=\lim_{x\to 0}\frac{1+\frac{1}{3}x\sin x-1}{x^2}=\frac{1}{3}.$$

(3)

$$\lim_{x\to 0}(1+2x)^{\frac{(x+1)^2}{x}}=\lim_{x\to 0}(1+2x)^{x+\frac{1}{x}+2}=e^2.$$

(4)

$$\begin{aligned}\lim_{n\to\infty}[(n+\ln n)^a-n^a] &= \lim_{n\to\infty}n^a\left[\left(1+\frac{\ln n}{n}\right)^a-1\right] \\ &= \lim_{n\to\infty}\frac{a\ln n}{n^{1-a}}=0.\end{aligned}$$

(5) 4. (Problem 4.7 (7)). □

Problem 6.2 (Mid 2)**.** *Discuss the continuity of the following functions.*

(1) $f(x)=[|\cos x|]$;

(2) $f(x)=\dfrac{1}{1-e^{\frac{x}{1-x}}}$.

Solution. (1) It is easy to see that

$$0\le|\cos x|<1,\quad \forall\, x\in(n\pi,(n+1)\pi),\ n\in\mathbb{Z}.,$$

and

$$|\cos x|=1,\quad x=n\pi,\ n\in\mathbb{Z}.$$

Then f is discontinuous at $x = n\pi$, $\forall\, n \in \mathbb{Z}$, and it's the removable discontinuity.

(2) Note that

$$\lim_{x\to 1-0} \frac{1}{1-e^{\frac{x}{1-x}}} = 0, \qquad \lim_{x\to 1+0} \frac{1}{1-e^{\frac{x}{1-x}}} = 1,$$

and

$$\lim_{x\to 0-0} \frac{1}{1-e^{\frac{x}{1-x}}} = +\infty, \qquad \lim_{x\to 0+0} \frac{1}{1-e^{\frac{x}{1-x}}} = -\infty,$$

we know that $x = 1$ is the jump discontinuity and $x = 0$ is the discontinuity of second kind. The graph of $f(x)$ is as follows: $\square$

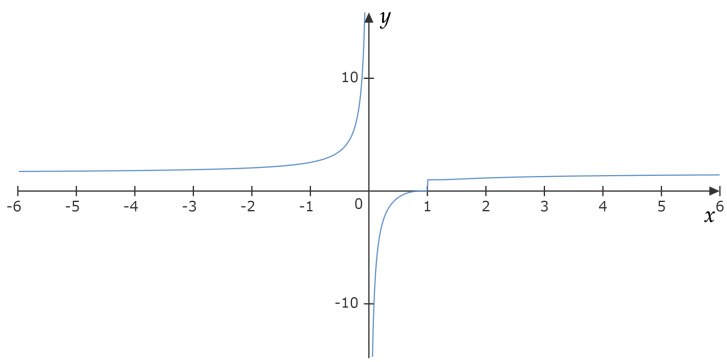

FIGURE 4. Graph of $f(x)$

Problem 6.3 (Mid 3)**.** *Find $n \in \mathbb{N}_+$, such that when $x \to 0$, $e^{x^n} - 1$ is a infinitesimals whose order is lower than $x(\cos\sqrt{x} - 1)(\sqrt[3]{x+1} - 1)$ but higher than $\sqrt{x}\ln(1 + \sqrt[3]{x})$.*

Solution. Note that when $x \to 0$, there are

$$x(\cos\sqrt{x} - 1)(\sqrt[3]{x+1} - 1) \sim -\frac{1}{6}x^3,$$

and

$$\sqrt{x}\ln(1 + \sqrt[3]{x}) \sim x^{\frac{5}{6}}.$$

Hence $n = 1, 2$. $\square$

Problem 6.4 (Mid 5)**.** *Suppose that $f(x)$ satisfying*

$$\lim_{x\to 0} \left(1 + x + \frac{f(x)}{x}\right)^{\frac{1}{x}} = e^3.$$

Prove that $\lim\limits_{x\to 0} \dfrac{f(x)}{x^2}$ exists, and find the limitation.

Proof. By $\lim_{x\to 0}\left(1+x+\frac{f(x)}{x}\right)^{\frac{1}{x}}=e^3$, we know that there must be $\lim_{x\to 0}\left(x+\frac{f(x)}{x}\right)=0$. What's more, we have

$$\lim_{x\to 0}\frac{1}{x}\ln\left(1+x+\frac{f(x)}{x}\right)=3.$$

Hence

$$\begin{aligned}\lim_{x\to 0}\frac{f(x)}{x^2}&=\lim_{x\to 0}\frac{1}{x}\left(x+\frac{f(x)}{x}\right)-1\\&=\lim_{x\to 0}\frac{1}{x}\ln\left(1+x+\frac{f(x)}{x}\right)\cdot\lim_{x\to 0}\frac{x+\frac{f(x)}{x}}{\ln\left(1+x+\frac{f(x)}{x}\right)}-1\\&=3-1=2.\end{aligned}$$

□

Problem 6.5 (Mid 6)**.** *Suppose sequence* $\{x_n\}$ *satisfies* $x_0=0$, $x_{2k}=\frac{x_{2k-1}}{2}$ *and* $x_{2k+1}=x_{2k}+\frac{1}{2}$. *Find the upper and lower limit of* $\{x_n\}$.

Solution. Firstly, we have

$$x_{2k+1}=x_{2k}+\frac{1}{2}=\frac{x_{2k-1}}{2}+\frac{1}{2},$$

which gives us

$$x_{2k+1}=\frac{2^{k+1}-1}{2^{k+1}}\quad\text{and}\quad x_{2k}=\frac{2^k-1}{2^{k+1}}.$$

Hence, we know

$$\varliminf_{n\to\infty}x_n=\frac{1}{2},\qquad\varlimsup_{n\to\infty}x_n=1.$$

□

Problem 6.6 (Mid 7)**.** *Soppose that* $x_0=1$, $x_{n+1}=\frac{1}{x_n^3+4}$. *Prove that* $\{x_n\}$ *converges to the unique positive zero point of equation* $x^4+4x-1=0$.

Proof. Let

$$g(x)=x^4+4x-1.$$

It's easy to see that $g(x)$ is increasing on $[0,+\infty)$. Since $g(0)=-1<0$, $g(1)=4>0$, we know that $g(x)$ has only one zero point in $[0,+\infty)$, say α. In particular, $0<\alpha<1$. Since $g(\alpha)=0$, we have that $\alpha=\frac{1}{\alpha^3+4}$. By induction, we know that $0<x_n<1$, $n\geq 1$. Hence, there is

$$|x_{n+1}-\alpha|=\left|\frac{1}{x_n^3+4}-\frac{1}{\alpha^3+4}\right|$$

$$
\begin{aligned}
&= \frac{|x_n^3 - \alpha^3|}{(x_n^3+4)(\alpha^3+4)} \\
&= \frac{|x_n - \alpha||x_n^2 + \alpha x_n + \alpha^2|}{(x_n^3+4)(\alpha^3+4)} \\
&\le \frac{3}{16}|x_n - \alpha| \le \cdots \\
&\le \left(\frac{3}{16}\right)^n |x_1 - \alpha| \\
&\to 0, \quad \text{as } n \to \infty.
\end{aligned}
$$

□

Problem 6.7 (Mid 10)**.** *Suppose that $f(x)$ is Lipschitz on $[1, +\infty)$, i.e. there exists a constant $C > 0$, such that $\forall\, x, y \in [1, +\infty)$, there is $|f(x) - f(y)| \le C|x - y|$. Prove that $\dfrac{f(x)}{x}$ is uniformly continuous on $[1, +\infty)$.*

Proof. **First way:** By Exercise 5.22, we know that $\dfrac{f(x)}{x}$ is bounded on $[1, +\infty)$, i.e. there exists $M > 0$, such that $\left|\dfrac{f(x)}{x}\right| \le M$. Note that $\forall\, x, y \in [1, +\infty)$, there is

$$
\begin{aligned}
\left|\frac{f(x)}{x} - \frac{f(y)}{y}\right| &= \left|\frac{f(x)}{x} - \frac{f(y)}{x} + \frac{f(y)}{x} - \frac{f(y)}{y}\right| \\
&\le \frac{|f(x) - f(y)|}{x} + \frac{1}{x}\left|\frac{f(y)}{y}\right||x - y| \\
&\le (C + M)|x - y|.
\end{aligned}
$$

i.e. $\dfrac{f(x)}{x}$ is also Lipschitz. Thus, it is uniformly continuous.

Second way: We will show $\dfrac{f(x)}{x}$ is Lipschitz, directly. Indeed,

$$
\begin{aligned}
\left|\frac{f(x)}{x} - \frac{f(y)}{y}\right| &= \left|\frac{f(x)}{x} - \frac{f(y)}{x} + \frac{f(y)}{x} - \frac{f(y)}{y}\right| \\
&\le \frac{|f(x) - f(y)|}{x} + \frac{|f(y)|}{xy}|x - y| \\
&\le \frac{|f(x) - f(y)|}{x} + \frac{|f(y) - f(1)| + |f(1)|}{xy}|x - y| \\
&\le \frac{C|x - y|}{x} + \frac{C(y-1) + |f(1)|}{xy}|x - y| \\
&\le (2C + |f(1)|)|x - y|.
\end{aligned}
$$

□

7. Week 10 (11.7)

Problem 7.1 (4.36)**.**

(4) $y = \sin^3 x$;

(6) $y = \dfrac{x^n}{1-x}$;

(8) $\dfrac{\ln x}{x}$.

Solution. (4) Note that

$$\begin{aligned}\sin^3 x &= \sin x(1-\cos^x) = \sin x - \sin x \cos^2 x \\ &= \frac{1}{2}\sin x - \frac{1}{2}\sin x \cos 2x \\ &= \frac{1}{2}\sin x - \frac{1}{4}\sin 3x + \frac{1}{4}\sin x \\ &= \frac{3}{4}\sin x - \frac{1}{4}\sin 3x.\end{aligned}$$

Hence

$$y^{(n)} = \frac{3}{4}\sin\left(x + \frac{n\pi}{2}\right) - \frac{3^n}{4}\sin\left(3x + \frac{n\pi}{2}\right).$$

(6)**First way:** Note that

$$x^n - 1 = (x-1)(x^{n-1} + x^{n-1} + \cdots + 1).$$

Second way: Note that

$$x^n = (x-1+1)^n = 1 + \sum_{k=1}^{n} C_n^k (x-1)^k.$$

(8) By the Leibniz formula, we have

$$\begin{aligned}y^{(n)} &= \left(\frac{\ln x}{x}\right)^{(n)} \\ &= \ln x \left(\frac{1}{x}\right)^{(n)} + \sum_{k=1}^{n} C_n^k (\ln x)^{(k)} \left(\frac{1}{x}\right)^{(n-k)} \\ &= \frac{(-1)^n n!}{x^{n+1}} \ln x + \sum_{k=1}^{n} \left(\frac{n!}{k!(n-k)!} \frac{(-1)^{k-1}(k-1)!}{x^k} \frac{(-1)^{n-k}(n-k)!}{x^{n-k+1}}\right) \\ &= \frac{(-1)^n n!}{x^{n+1}} \left(\ln x - \sum_{k=1}^{n} \frac{1}{k}\right).\end{aligned}$$

□

Problem 7.2 (4.41(3))**.** *If* $y = x^{n-1}e^{\frac{1}{x}}$*, then* $y^{(n)} = \dfrac{(-1)^n}{x^{n+1}}e^{\frac{1}{x}}$.

Proof. Prove by induction. Let $y_n := x^{n-1}e^{\frac{1}{x}}$. Then $y_1 = e^{\frac{1}{x}}$, we have

$$y_1' = \frac{-1}{x^2}e^{\frac{1}{x}}.$$

Suppose that $y_n^{(n)} = \dfrac{(-1)^n}{x^{n+1}}e^{\frac{1}{x}}$, we calculate $y_{n+1}^{(n+1)}$. By the Leibniz formula, we have

$$\begin{aligned}
y_{n+1}^{(n+1)} &= (y_{n+1}^{(n)})' = ((xy_n)^{(n)}) \\
&= (xy_n^{(n)} + ny_n^{(n-1)})' \\
&= xy_n^{(n+1)} + (n+1)y_n^{(n)} \\
&= \frac{(-1)^{n+1}}{x^{n+2}}e^{\frac{1}{x}} + \frac{(-1)^{n+1}(n+1)}{x^{n+1}}e^{\frac{1}{x}} + \frac{(-1)^n(n+1)}{x^{n+1}}e^{\frac{1}{x}} \\
&= \frac{(-1)^{n+1}}{x^{n+2}}e^{\frac{1}{x}}.
\end{aligned}$$

□

Problem 7.3 (4.44)**.** *Suppose that* $f(x)$ *is continuous at* $x = 0$*, and satisfying*

$$\lim_{x\to 0}\frac{f(2x) - f(x)}{x} = m.$$

Prove that $f'(0) = m$.

Proof. The proof is very similar to that of Problem 4.11, we omit the detail here. □

Problem 7.4. *Suppose that* $f(x) = \begin{cases} e^{-\frac{1}{x^2}}, & x \neq 0, \\ 0, & x = 0. \end{cases}$ *Prove that* $f^{(n)}(0) = 0$, $\forall\, n \in \mathbb{N}_+$.

Proof. We firstly calculate $f'(0)$. By definition, there is

$$f'(0) = \lim_{x\to 0}\frac{1}{x}\cdot e^{-\frac{1}{x^2}} = \lim_{y\to\infty} ye^{-y^2} = 0.$$

Next, a direct calculation yields the formula of $f'(x)$ when $x \neq 0$

$$f'(x) = \frac{2}{x^3}e^{-\frac{1}{x^2}}.$$

By definition, again, we have

$$f''(0) = \lim_{x\to 0}\frac{f'(x) - f'(0)}{x} = \lim_{x\to 0}\frac{2}{x^4}\cdot e^{-\frac{1}{x^2}} = 0.$$

Then, a direct calculation yields the formula of $f''(x)$ when $x \neq 0$

$$f''(x) = \left(\frac{4}{x^6} - \frac{6}{x^4}\right)e^{-\frac{1}{x^2}}.$$

Hence, we claim that the nth-order derivative of f when $x \neq 0$ is

$$f^{(n)}(x) = P_n\left(\frac{1}{x}\right) \cdot e^{-\frac{1}{x^2}}, \tag{7.1}$$

where $P_n(y)$ is a polynomial of degree n. We prove this claim by induction. We already know that (7.1) is true for $n = 1, 2$. We suppose that (7.1) is valid for $f^{(k)}(x)$, we calculate $f^{(k+1)}(x)$ in the following.

$$\begin{aligned} f^{(k+1)}(x) &= (f^{(k)}(x))' \\ &= \left[P_k\left(\frac{1}{x}\right) \cdot e^{-\frac{1}{x^2}}\right]' \\ &= P_k'\left(\frac{1}{x}\right)\left(-\frac{1}{x^2}\right) \cdot e^{-\frac{1}{x^2}} + P_k\left(\frac{1}{x}\right) \cdot \frac{2}{x^3} \cdot e^{-\frac{1}{x^2}} \\ &= \left[P_k\left(y\right)\left(-y^2\right) + P_k(y)(2y^3)\right]\Big|_{y=\frac{1}{x}} \cdot e^{-\frac{1}{x^2}}. \end{aligned}$$

Denote $P_{k+1}\left(\frac{1}{x}\right) := \left[P_k\left(y\right)\left(-y^2\right) + P_k(y)(2y^3)\right]\Big|_{y=\frac{1}{x}}$, then we obtain (7.1).

Finally, we prove $f^{(n)}(0) = 0$ by induction. It is true for $n = 1, 2$. We suppose that there is $f^{(k)}(0) = 0$, we prove that $f^{(k+1)}(0) = 0$. Indeed,

$$\begin{aligned} f^{(k+1)}(0) &= \lim_{x\to 0} \frac{f^{(k)}(x) - f^{(k)}(0)}{x} = \lim_{x\to 0} \frac{f^{(k)}(x)}{x} \\ &= \lim_{x\to 0} \frac{1}{x} \cdot P_k\left(\frac{1}{x}\right) \cdot e^{-\frac{1}{x^2}} = 0. \end{aligned}$$

The graph of $f(x)$ is as follows:

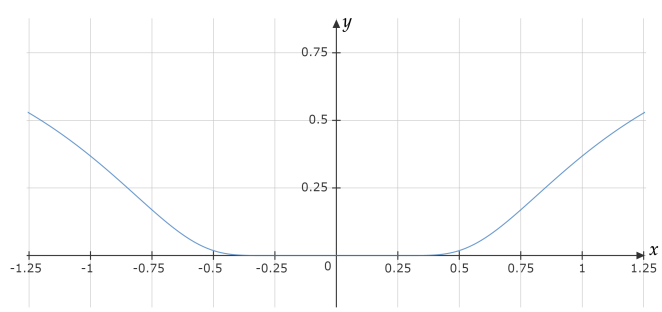

FIGURE 5. Graph of $f(x)$

□

Exercise 7.5. *Prove that there exists a smooth function* $f : \mathbb{R} \longrightarrow [0, 1]$ *such that* $f\big|_{(-\infty,0]} = 0$ *and* $f\big|_{[1,+\infty)} = 1$.

Hint: Firstly, define

$$g(x) = \begin{cases} e^{-\frac{1}{x^2}}, & x > 0, \\ 0, & x \le 0. \end{cases}$$

Next, let

$$f(x) = \frac{g(x)}{g(x) + g(1-x)}.$$

It's easy to verify that f satisfies the condition. □

Problem 7.6. *Suppose that f is a polynomial of degree 7. If $f(x)+1$ is divisible by $(x-1)^4$ and $f(x)-1$ is divisible by $(x+1)^4$. Find f by the method of derivatives.*

Proof. Note that

$$f(x)+1 = p(x)(x-1)^4 \Rightarrow f'(x) = p'(x)(x-1)^4 + 4p(x)(x-1)^3 \Rightarrow f'(1) = 0,$$
$$f(x)-1 = q(x)(x+1)^4 \Rightarrow f'(x) = q'(x)(x+1)^4 + 4q(x)(x+1)^3 \Rightarrow f'(-1) = 0.$$

Hence we have

$$f'(x) = a(x-1)^3(x+1)^3.$$

Then

$$f(x) = \frac{a}{7}x^7 - \frac{3a}{5}x^5 + ax^3 - ax + b.$$

Since $f(1) = -1, f(-1) = 1$, we have that

$$-\frac{16}{35}a + b = -1, \quad \frac{16}{35}a + b = 1,$$

i.e.

$$a = \frac{35}{16}, \quad b = 0.$$

Hence

$$f(x) = \frac{1}{16}x\left(5x^6 - 21x^4 + 35x^2 - 35\right).$$

□

Problem 7.7. *Suppose that $f(x) = ax^2 + bx + c$, and $|f(x)| \le 1$, when $|x| \le 1$. Prove that $|f'(x)| \le 4$, when $|x| \le 1$.*

Proof. By $|f(-1)| = |a-b+c| \le 1$, $|f(0)| = |c| \le 1$, $|f(1)| = |a+b+c| \le 1$, there is

$$\begin{aligned} |2a+b| &= \left|\frac{1}{2}(a-b+c) - 2c + \frac{3}{2}(a+b+c)\right| \\ &\le \frac{1}{2}|a-b+c| + 2|c| + \frac{3}{2}|a+b+c| \\ &\le 4. \end{aligned}$$

Similarly, we have

$$|-2a+b| = \left|-\frac{3}{2}(a-b+c) + 2c - \frac{1}{2}(a+b+c)\right| \le 4.$$

Since the maximum of linear functions is achieved at endpoints, we know that

$$|f'(x)| = |2ax + b| \leq \max\{|2a + b|, |-2a + b|\} \leq 4, \quad \forall\, x \in [-1, 1].$$

□

Problem 7.8. *Prove that for every $n \in \mathbb{N}_+$ there is*

$$\sum_{k=0}^{n} (-1)^k C_n^k k^m = \begin{cases} 0, & 0 \leq m \leq n-1, \\ (-1)^n n!, & m = n. \end{cases}$$

Proof. Let $S_n^m = \sum_{k=0}^{n} (-1)^k C_n^k k^m$. We show that $S_n^m = 0$ if $0 \leq m \leq n-1$. Firstly, we have

$$S_n^0 = \sum_{k=0}^{n} (-1)^k C_n^k = (1-1)^n = 0.$$

In particular, $S_1^0 = 0$. By

$$kC_n^k = k\frac{n!}{k!(n-k)!} = nC_{n-1}^{k-1},$$

we know that when $1 \leq m \leq n-1$, there is

$$\begin{aligned} S_n^m &= n\sum_{k=1}^{n} (-1)^k C_{n-1}^{k-1} k^{m-1} \\ &= -n\sum_{k=0}^{n-1} (-1)^k C_{n-1}^k (k+1)^{m-1} \\ &= -n\sum_{k=0}^{n-1}\sum_{l=0}^{m-1} (-1)^k C_{n-1}^k C_{m-1}^l k^l \\ &= -n\sum_{l=0}^{m-1}\sum_{k=0}^{n-1} (-1)^k C_{n-1}^k C_{m-1}^l k^l \\ &= -n\sum_{l=0}^{m-1} C_{m-1}^l S_{n-1}^l. \end{aligned} \tag{7.2}$$

Hence, by induction, we have that $S_n^m = 0$ for $0 \leq m \leq n-1$. What's more, by (7.2), we have

$$S_n^n = -n\sum_{l=0}^{n-1} C_{n-1}^l S_{n-1}^l = -nS_{n-1}^{n-1} = (-1)^n n!.$$

Then the result follows. □

Exercise 7.9. *Given a positive integer n. Find*

$$S = \sum_{k=0}^{n} (-1)^k \binom{n}{k} k^{n+2},$$

and

$$T = \sum_{k=0}^{n} (-1)^k \binom{n}{k} k^{n+3}.$$

Hint: Using (7.2) and induction, we have

$$S = \frac{(-1)^n n(3n+1)(n+2)!}{24},$$

and

$$T = \frac{(-1)^n n^2(n+1)(n+3)!}{48}.$$

□

Problem 7.10. *Suppose that $f(x) = x^n \ln x$, $n \in \mathbb{N}_+$. Calculate* $\lim\limits_{n\to\infty} \dfrac{f^{(n)}(1/n)}{n!}$.

Proof. Denote that $f_n(x) = x^n \ln x$. Then

$$f_n'(x) = nx^{n-1} \ln x + x^{n-1} = nf_{n-1}(x) + x^{n-1}.$$

Hence, we have

$$f_n^{(n)}(x) = nf_{n-1}^{(n-1)}(x) + (n-1)!,$$

which gives us

$$\frac{f_n^{(n)}(x)}{n!} = \frac{f_{n-1}^{(n-1)}(x)}{(n-1)!} + \frac{1}{n} = \cdots = \ln x + 1 + \frac{1}{2} + \cdots + \frac{1}{n}.$$

Then take $x = \frac{1}{n}$, ther is

$$\lim_{n\to\infty} \frac{f_n^{(n)}(1/n)}{n!} = \lim_{n\to\infty} \left(1 + \frac{1}{2} + \cdots + \frac{1}{n} - \ln n\right) = c,$$

where c is the Euler constant. □

Problem 7.11 (5.1)**.** *Prove the generalized Rolle's theorem, i.e. suppose that $f(x)$ is differentiable on (a, b), and $f(a+0) = f(b-0) = A$. Then there exists $\xi \in (a, b)$, such that $f'(\xi) = 0$, where a can be $-\infty$, b can be $+\infty$, A can be $+\infty$ or $-\infty$.*

Proof. We only prove the case that a, b and A are finite, others are similar and we leave them to the reader. The conclusion is clear if $f(x) = A$, $\forall\, x \in (a, b)$. Hence, without loss of generality, we may assume that there is at least a $x_0 \in (a, b)$ such that $f(x_0) > A$. By the definition of limits, we have there is a small $\delta > 0$ such that

$$f(x) < f(x_0), \quad \forall\, x \in (a, a+\delta) \cup (b-\delta, b).$$

Hence, we know that the maximum of $f(x)$ is achieved on $[a+\delta, b-\delta]$, thus there exists $\xi \in (a,b)$, such that $f'(\xi) = 0$. $\square$

Problem 7.12 (5.5)**.** *Prove that the Chebyshev-Laguerre polynomial*

$$L_n(x) = e^x \frac{d^n}{dx^n}(x^n e^{-x})$$

has n different zero points.

Proof. By Leibniz formula, it's easy to see that $L_n(x)$ is a polynomial of degree n. Hence, it has at most n zero points. The conclusion is clear for $n = 0, 1$, we show it's true for $n \geq 2$ in the following. Denote $g(x) = x^n e^{-x}$, then $L_n(x) = e^x g^{(n)}(x)$. It suffices to find all zero points of $g^{(n)}(x)$. Note that

$$g^{(l)}(x) = \sum_{k=0}^{l} C_l^k (x^n)^{(k)} (e^{-x})^{(l-k)} = \sum_{k=0}^{l} C_l^k n(n-1)\cdots(n-k+1)x^{n-k}(-1)^{l-k}e^{-x}.$$

Hence for $l < n$, there are always $g^{(l)}(0) = 0$ and $\lim\limits_{x\to+\infty} g^{(l)}(x) = 0$. Hence, by Rolle's theorem (Problem 7.11) and induction, we know that there is at least $n-1$ zero points of $g^{(l)}(x)$ between $(0, +\infty)$.By Rolle's theorem again, we have that $g^{(n)}(x)$ has at least n zero points in $(0, +\infty)$. Therefore, we know that $L_n(x)$ has n different zero points. $\square$

Problem 7.13. *Suppose that f is continuous on $[x_1, x_2]$, differentiable on (x_1, x_2). Show that there exists $\xi \in (x_1, x_2)$, such that* $\dfrac{1}{x_1 - x_2}\begin{vmatrix} x_1 & x_2 \\ f(x_1) & f(x_2) \end{vmatrix} = f(\xi) - \xi f'(\xi)$.

Proof. Note that

$$\frac{1}{x_1 - x_2}\begin{vmatrix} x_1 & x_2 \\ f(x_1) & f(x_2) \end{vmatrix} = \frac{x_1 f(x_2) - x_2 f(x_1)}{x_1 - x_2} = \frac{\frac{f(x_2)}{x_2} - \frac{f(x_1)}{x_1}}{\frac{1}{x_2} - \frac{1}{x_1}}.$$

By the Cauchy mean value theorem, we know that there exists $\xi \in (x_1, x_2)$, such that

$$\frac{\frac{f(x_2)}{x_2} - \frac{f(x_1)}{x_1}}{\frac{1}{x_2} - \frac{1}{x_1}} = \frac{\frac{\xi f'(\xi) - f()\xi}{\xi^2}}{-\frac{1}{\xi^2}} = f(\xi) - \xi f'(\xi).$$

Then the result follows. $\square$

Problem 7.14. *Suppose that $f(x)$ is differentiable on (a,b), $b < +\infty$, and $\lim\limits_{x\to b-0} f(x) = +\infty$. Prove that $\overline{\lim\limits_{x\to b-0}} f'(x) = +\infty$.*

Proof. Prove by contradiction. Assume that $\overline{\lim\limits_{x\to b-0}} f'(x) < +\infty$. Then there exist $M \in \mathbb{R}$, $\delta > 0$, such that $\forall\, x \in (b-\delta, b)$, there is $f'(x) \leq M$. Hence by the Lagrange mean value theorem, we know that $\forall\, x, y \in (b-\delta, b)$, $x > y$, there exist $\xi \in (b-\delta, b)$ such that

$$f(x) - f(y) = f'(\xi)(x-y) \leq M(x-y).$$

Let $x \to b-0$, and by $\lim\limits_{x\to b-0} f(x) = +\infty$, we have that

$$+\infty \le M(b-y),$$

contradiction. $\square$

Problem 7.15. *Suppose that $f(x)$ is continuous on $[a,b]$, differentiable on (a,b), and f is not a linear function. Prove that there exists $\xi \in (a,b)$, such that $f'(\xi) > \dfrac{f(b)-f(a)}{b-a}$.*

Proof. Prove by contradiction. Assume that $f'(x) \le \dfrac{f(b)-f(a)}{b-a}$, $\forall\, x \in (a,b)$. Define

$$F(x) = f(x) - f(a) - \frac{f(b)-f(a)}{b-a}(x-a).$$

Note that $F(a) = F(b) = 0$, and

$$F'(x) = f'(x) - \frac{f(b)-f(a)}{b-a} \le 0.$$

Hence

$$0 = F(a) \ge F(x) \ge F(b) = 0, \quad \forall\, x \in (a,b),$$

i.e. $F(x) = 0$, $\forall\, x \in (a,b)$. Therefore, we have

$$f(x) = f(a) + \frac{f(b)-f(a)}{b-a}(x-a), \quad \forall\, x \in (a,b),$$

which implies that f is a linear function, contradiction. $\square$

Exercise 7.16. *Suppose that $f(x)$ is continuous on $[0,1]$, differentiable on $(0,1)$, and f is not a constant function. If $f(0) = 0$, prove that there exists $\xi \in (0,1)$ such that $f(\xi)f'(\xi) > 0$.*

Hint: Consider the function $F(x) := f^2(x)$. $\square$

Problem 7.17. *Suppose that $f(x)$ is differentiable on $[0,1]$, $f(0) = 0$, $f(1) = 1$, $k_1, \cdots, k_n$ are positive numbers. Prove that there are $x_1, \cdots, x_n \in [0,1]$, $x_i \ne x_j$, such that*

$$\sum_{i=1}^{n} \frac{k_i}{f'(x_i)} = \sum_{i=1}^{n} k_i.$$

Proof. Denote that $m = \sum\limits_{i=1}^{n} k_i$, $\lambda_i = \dfrac{k_i}{m}$. Then $0 < \lambda_i < 1$, $\lambda_1 + \cdots + \lambda_n = 1$. Since $f(0) = 0$, $f(1) = 1$ and $f(x)$ is continuous on $[0,1]$, we know that there exists $c_1 \in (0,1)$ such that $f(c_1) = \lambda_1$. Again, we know that there exists $c_2 \in (c_1, 1)$ such that $f(c_2) = \lambda_1 + \lambda_2$ since $\lambda_1 < \lambda_1 + \lambda_2 < 1$. Proceeding like this, we can find

$$0 < c_1 < c_2 < \cdots < c_n = 1,$$

such that

$$f(c_i) = \sum_{k=1}^{i} \lambda_k \quad (i = 1, 2, \cdots, n).$$

By the Lagrange mean value theorem, we have $x_i \in (c_{i-1}, c_i)$ $(c_0 = 0)$, such that

$$f'(x_i) = \frac{f(c_i) - f(c_{i-1})}{c_i - c_{i-1}} = \frac{\lambda_i}{c_i - c_{i-1}},$$

i.e.

$$\frac{\lambda_i}{f'(x_i)} = c_i - c_{i-1} \quad (i = 1, 2, \cdots, n).$$

Hence, we have

$$\sum_{i=1}^{n} \frac{\lambda_i}{f'(x_i)} = \sum_{i=1}^{n} (c_i - c_{i-1}) = c_n - c_0 = 1.$$

Recall that $\lambda_i = \dfrac{k_i}{m}$, we obtain

$$\sum_{i=1}^{n} \frac{k_i}{f'(x_i)} = \sum_{i=1}^{n} k_i.$$

□

Problem 7.18. *Suppose that $f(x)$ is differentiable on (a, b). Prove that the points in (a, b) are either the continuous point of $f'(x)$, or the discontinuous point of secong kind. i.e. $f'(x)$ has no discontinous points of first kind.*

Proof. Since $f(x)$ is differentiable on (a, b), we know that $\forall\, x_0 \in (a, b)$, there is

$$\begin{aligned} f'(x_0) = f'_+(x_0) &= \lim_{x \to x_0 + 0} \frac{f(x) - f(x_0)}{x - x_0} \\ &= \lim_{x \to x_0 + 0} \frac{f'(\xi)(x - x_0)}{x - x_0} \\ &= \lim_{x \to x_0 + 0} f'(\xi) \quad (x_0 < \xi < x). \end{aligned}$$

Hence, if $\lim\limits_{x \to x_0 + 0} f'(x)$ exists, there must be

$$f'(x_0) = \lim_{\xi \to x_0 + 0} f'(\xi) = f'(x_0 + 0).$$

Similarly, if $\lim\limits_{x \to x_0 - 0} f'(x)$ exists, there must be

$$f'(x_0) = f'(x_0 - 0).$$

Therefore $f'(x)$ is continuous at $x = x_0$ unless at least one of $\lim\limits_{x \to x_0 + 0} f'(x)$, $\lim\limits_{x \to x_0 - 0} f'(x)$ does not exist. □

Exercise 7.19 (Darboux Theorem)**.** *Suppose that $f(x)$ is differentiable on $[a, b]$, and $f'(a) < f'(b)$. Then $\forall\, c : f'(a) < c < f'(b)$, there exists $\xi \in (a, b)$, such that $f'(\xi) = c$.*

Hint: Define

$$g(x) = f(x) - cx, \quad \forall\, x \in [a, b].$$

Hence, $g(x)$ is differentiable on $[a, b]$. What's more, there are $g'(a) = f'(a) - c < 0$, $g'(b) = f'(b) - c > 0$. Note that

$$\lim_{x \to a+0} \frac{g(x) - g(a)}{x - a} = g'(a) < 0,$$

which implies that there is $\delta > 0$, such that $\forall\, x \in (a, a+\delta)$, there is $g(x) < g(a)$. Similarly, there is $\delta > 0$, such that $\forall\, x \in (b - \delta, b)$, there is $g(x) < g(b)$. Hence, the minimum of $g(x)$ is achieved on $[a + \delta, b - \delta]$, which yields that there is $\xi \in (a, b)$, such that $g'(\xi) = 0$, i.e. $f'(\xi) = c$. □

Exercise 7.20. *Suppuse that $f(x)$ is continuous on $[a, b]$, and differentiable on (a, c), (c, b). Prove that there exists $\xi \in (a, b)$ such that $|f(b) - f(a)| \leq |f'(\xi)||b - a|$.*

Hint: Using the Lagrange mean value theorem on (a, c), (c, b), respectively, and taking the maxinmum of intermediate points. □

Exercise 7.21. *Suppuse that $f(x)$ is continuous on $[0, 1]$, differentiable on $(0, 1)$, and $|f'(x)| < 1$. If $f(0) = f(1)$, prove that for any $x_1, x_2 \in (0, 1)$, there is*

$$|f(x_1) - f(x_2)| < \frac{1}{2}.$$

Hint: Consider $|x_1 - x_2| < \dfrac{1}{2}$ and $|x_1 - x_2| \geq \dfrac{1}{2}$, respectively. □

Exercise 7.22 (Challenge!)**.** *Suppose that $f(x)$, $f'_+(x) \in C(\mathbb{R})$. Prove that $f(x)$ is differentiable on $\mathbb{R}$.*

Hint: Firstly, prove the two lemmas:

Suppose that $f(x) \in C[a, b]$, $f(a) = f(b)$, $f'_+(x)$ exists on $[a, b)$. Prove that there exist $c, d \in [a, b)$ such that $f'_+(c) \leq 0$, $f'_+(d) \geq 0$.

Suppose that $f(x) \in C[a, b]$, $f'_+(x)$ exists on $[a, b)$. Prove that there exist $c, d \in [a, b)$ such that $f'_+(c) \leq \dfrac{f(b) - f(a)}{b - a} \leq f'_+(d)$.

Finally, using the continuity of $f'_+(x)$ to prove that $\forall\, x_0 \in \mathbb{R}$, there is

$$f'_-(x_0) = \lim_{x \to x_0 - 0} \frac{f(x) - f(x_0)}{x - x_0} = f'_+(x_0).$$

□

8. Week 11 (11.14)

Problem 8.1. *Calculate the following limitations.*

(1) $\lim\limits_{x\to 0+0} x^x$;

(2) $\lim\limits_{x\to 0+0} x^{x^x-1}$;

(3) $\lim\limits_{x\to +\infty} \left(\sqrt[3]{x^3-3x}-\sqrt{x^2-2x}\right)$;

(4) $\lim\limits_{x\to 0+0} \dfrac{x^x-(\sin x)^x}{x^2\ln(1+x)}$.

Solution. (1) By L'Hospital's rule, we have

$$\begin{aligned}\lim_{x\to 0+0} x^x &= \lim_{x\to 0+0} e^{x\ln x}\\ &= \exp\left\{\lim_{x\to 0+0}\frac{\ln x}{\frac{1}{x}}\right\}\\ &= \exp\left\{\lim_{x\to 0+0}\frac{\frac{1}{x}}{-\frac{1}{x^2}}\right\}\\ &= \exp\left\{\lim_{x\to 0+0} -x\right\}\\ &= 1.\end{aligned}$$

(2) By Taylor's formula, we have

$$\begin{aligned}\lim_{x\to 0+0} x^{x^x-1} &= e^{\lim\limits_{x\to 0+0}(x^x-1)\ln x}\\ &= e^{\lim\limits_{x\to 0+0} x(\ln x)^2} = 1.\end{aligned}$$

(3) By Taylor's formula, we have

$$\begin{aligned}\lim_{x\to +\infty}\left(\sqrt[3]{x^3-3x}-\sqrt{x^2-2x}\right) &= \lim_{x\to +\infty} x\left(\sqrt[3]{1-3x^{-2}}-\sqrt{1-2x^{-1}}\right)\\ &= \lim_{x\to +\infty} x(1-x^{-2}-1+x^{-1})\\ &= 1.\end{aligned}$$

(4) By Taylor's formula, we have

$$\begin{aligned}\lim_{x\to 0+0}\frac{x^x-(\sin x)^x}{x^2\ln(1+x)} &= \lim_{x\to 0+0}\frac{e^{x\ln x}-e^{x\ln\sin x}}{x^3}\\ &= \lim_{x\to 0+0} e^{x\ln x}\frac{1-e^{x\ln\frac{\sin x}{x}}}{x^3}\\ &= \lim_{x\to 0+0}\frac{1-e^{-\frac{1}{6}x^3}}{x^3}\end{aligned}$$

$$= \frac{1}{6}.$$

□

Exercise 8.2. *Prove that*

$$\left(1+\frac{1}{n}\right)^n = e - \frac{e}{2n} + \frac{11e}{24n^2} + o\left(\frac{1}{n^2}\right), \quad (n \to \infty).$$

Hint: By Taylor's formula, we have

$$\begin{aligned}
\left(1+\frac{1}{n}\right)^n &= e^{n\ln\left(1+\frac{1}{n}\right)} \\
&= e^{1-\frac{1}{2n}+\frac{1}{3n^2}+o\left(\frac{1}{n^2}\right)} \\
&= e\left(1-\frac{1}{2n}+\frac{1}{3n^2}+\frac{1}{2}\left(-\frac{1}{2n}+\frac{1}{3n^2}\right)^2\right)+o\left(\frac{1}{n^2}\right) \\
&= e - \frac{e}{2n} + \frac{11e}{24n^2} + o\left(\frac{1}{n^2}\right), \quad (n \to \infty).
\end{aligned}$$

□

Remark 8.3. *Similarly, for* $(1+x)^{\frac{1}{x}}$*, there is*

$$(1+x)^{\frac{1}{x}} = e - \frac{e}{2}x + \frac{11e}{24}x^2 + o\left(x^2\right), \quad (x \to 0).$$

Problem 8.4. *Suppose that* $f(x)$ *is twice differentiable on* $[0,1]$*,* $f(0) = f(1) = 0$*,* $\max\limits_{x\in[0,1]} f(x) = 2$*. Prove that* $\inf\limits_{x\in[0,1]} f''(x) \le -16$*.*

Proof. Since $f(x)$ is continuous on $[0,1]$, $f(0) = f(1) = 0$, and $\max\limits_{x\in[0,1]} f(x) = 2$, we know that there is $x_0 \in (0,1)$ such that

$$f(x_0) = \max_{x\in[0,1]} f(x).$$

Hence, by Fermat's theorem, there is

$$f'(x_0) = 0.$$

By Taylor's formula at $x = x_0$, we have

$$\begin{aligned}
0 = f(0) &= f(x_0) + \frac{1}{2}f''(\xi)(0-x_0)^2 = 2 + \frac{1}{2}f''(\xi)x_0^2, \\
0 = f(1) &= f(x_0) + \frac{1}{2}f''(\eta)(1-x_0)^2 = 2 + \frac{1}{2}f''(\eta)(1-x_0)^2.
\end{aligned}$$

Hence, we know

$$\inf_{x\in[0,1]} f''(x) \le \min\left\{f''(\xi), f''(\eta)\right\} = \min\left\{-\frac{4}{x_0^2}, -\frac{4}{(1-x_0)^2}\right\}.$$

Note that

$$\min\left\{-\frac{4}{x_0^2}, -\frac{4}{(1-x_0)^2}\right\} = -\frac{4}{(1-x_0)^2} \le -16, \quad x_0 \in \left[\frac{1}{2}, 1\right],$$
$$\min\left\{-\frac{4}{x_0^2}, -\frac{4}{(1-x_0)^2}\right\} = -\frac{4}{x_0^2} \le -16, \quad x_0 \in \left[0, \frac{1}{2}\right].$$

Therefore, we obtain that

$$\inf_{x\in[0,1]} f''(x) \le -16.$$

□

Remark 8.5. *Use the same method, we can prove that if $f(x)$ is twice differentiable on $[0,1]$, $f(0) = f(1) = 0$, $\min\limits_{x\in[0,1]} f(x) = -1$, then $\inf\limits_{x\in[0,1]} f''(x) \le \frac{1}{8}$, $\sup\limits_{x\in[0,1]} f''(x) \ge 8$.*

Problem 8.6 (5.16)**.** *Suppose that $f(x)$ is three times differentiable on $[a,b]$. Prove that there is $\xi \in (a,b)$, such that*

$$f(b) = f(a) + \frac{1}{2}(b-a)(f'(a) + f'(b)) - \frac{1}{12}(b-a)^3 f^{(3)}(\xi).$$

Proof. Firstly, we choose M such that

$$f(b) = f(a) + \frac{1}{2}(b-a)(f'(a) + f'(b)) - \frac{1}{12}(b-a)^3 M.$$

Define

$$F(x) = f(x) - f(a) - \frac{1}{2}(x-a)(f'(x) + f'(a)) + \frac{1}{12}(x-a)^3 M.$$

Hence, there is $F(a) = F(b) = 0$. By Rolle's theorem, we know that there exists $\eta \in (a,b)$ such that $F'(\eta) = 0$. Note that

$$F'(x) = f'(x) - \frac{1}{2}(f'(x) + f'(a)) - \frac{1}{2}(x-a)f''(x) + \frac{1}{4}(x-a)^2 M.$$

Hence, there is $F'(a) = F'(\eta) = 0$. By Rolle's theorem again, we know that there exists $\xi \in (a,\eta)$ such that $F''(\xi) = 0$. Note that

$$F''(x) = -\frac{1}{2}(x-a)f^{(3)}(x) + \frac{1}{2}(x-a)M.$$

Therefore, we have $M = f^{(3)}(\xi)$, i.e.

$$f(b) = f(a) + \frac{1}{2}(b-a)(f'(a) + f'(b)) - \frac{1}{12}(b-a)^3 f^{(3)}(\xi).$$

□

Problem 8.7. *Suppose that $f(x)$ is twice differentiable on $[-1,1]$, $f(0) = f'(0) = 0$. Assume that $|f''(x)| \le |f(x)| + |f'(x)|$, $\forall\, x \in [-1,1]$. Prove that there exists $\delta > 0$, such that $f(x) = 0$, $\forall\, x \in (-\delta, \delta)$.*

Proof. Choose $\delta = 1/4$. Since $|f(x)| + |f'(x)|$ is continuous on $[-1/4, 1/4]$, we know that there is $x_0 \in [-1/4, 1/4]$ such that

$$|f(x_0)| + |f'(x_0)| = \max_{x\in[-1/4,1/4]} |f(x)| + |f'(x)| =: M.$$

By Taylor's formula, we have

$$\begin{aligned} f(x_0) &= f(0) + f'(0)x_0 + \frac{f''(\xi)}{2}x_0^2 = \frac{f''(\xi)}{2}x_0^2, \\ f'(x_0) &= f''(\eta)x_0. \end{aligned}$$

Hence, there is

$$\begin{aligned} |f(x_0)| + |f'(x_0)| &= \left|\frac{f''(\xi)}{2}x_0^2\right| + |f''(\eta)x_0| \\ &\le \frac{1}{4}(|f(\xi)| + |f'(\xi)|) + \frac{1}{4}(|f(\eta)| + |f'(\eta)|) \\ &\le \frac{1}{2}M, \end{aligned}$$

which implies that $M = 0$, i.e. $f(x) = 0$, $\forall\, x \in [-1/4, 1/4]$. □

Problem 8.8. *Suppose that $f(x)$ is twice differentiable on $\mathbb{R}$, and $f(x)$ is also a bounded function. Prove that there is $\xi \in \mathbb{R}$ such that $f''(\xi) = 0$.*

Proof. Assume that $f''(x) \neq 0$, $\forall\, x \in \mathbb{R}$. By Darboux's theorem (Exercise 7.19), we know that $f''(x)$ does not change sign. Without loss of generality, we may assume that $f''(x) > 0$, $\forall\, x \in \mathbb{R}$. Choosing $x_0 \in \mathbb{R}$ such that $f'(x_0) \neq 0$. By Taylor' formula, we have

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(\xi)}{2}(x - x_0)^2 \ge f(x_0) + f'(x_0)(x - x_0),$$

which contradicts with f is bounded on $\mathbb{R}$. □

Problem 8.9. *Suppose that $f \in C^n(\mathbb{R})$, and there exist constants M_0, M_1 such that $|f(x)| \le M_0$, $|f^{(n)}(x)| \le M_1$, $\forall\, x \in \mathbb{R}$. Prove that there is $M > 0$ such that $|f^{(j)}(x)| \le M$, $j = 1, 2, \cdots, n-1$, $\forall\, x \in \mathbb{R}$.*

Proof. By Taylor's formula, we have

$$f(x+m) = f(x) + mf'(x) + \frac{m^2}{2!}f''(x) + \cdots + \frac{m^{n-1}}{(n-1)!}f^{(n-1)}(x) + \frac{m^n}{n!}f^{(n)}(\xi_m),$$

where $x < \xi_m < x+m$, $m = 1, 2, \cdots, n$. This is a linear system of equations about $f'(x)$, $f''(x)$, $\cdots$, $f^{(n-1)}(x)$, and the determinant of its coefficients is

$$\begin{vmatrix} 1 & 1 & \frac{1}{2!} & \cdots & \frac{1}{(n-1)!} \\ 1 & 2 & \frac{2^2}{2!} & \cdots & \frac{2^{n-1}}{(n-1)!} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & n & \frac{n^2}{2!} & \cdots & \frac{n^{n-1}}{(n-1)!} \end{vmatrix} = \frac{1}{1!2!\cdots(n-1)!} \begin{vmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & 2 & 2^2 & \cdots & 2^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & n & n^2 & \cdots & n^{n-1} \end{vmatrix} = 1.$$

Hence, we know that $f'(x)$, $f''(x)$, $\cdots$, $f^{(n-1)}(x)$ can be written as linear combinations of $f(x+m)$ and $f^{(n)}(\xi_m)$, $m = 1, 2, \cdots, n$. Since $|f(x)| \le M_0$, $|f^{(n)}(x)| \le M_1$, $\forall\, x \in \mathbb{R}$, we have that there exists $M > 0$ such that $|f^{(j)}(x)| \le M$, $j = 1, 2, \cdots, n-1$, $\forall\, x \in \mathbb{R}$. □

Problem 8.10. *Suppose that $f(x)$ is bounded on $\mathbb{R}$ and $f'(x)$ is uniformly continuous on $\mathbb{R}$. Prove that $f'(x)$ is also bounded.*

Proof. Prove by contradiction. Without loss of generality, we may assume that $f'(x)$ has no upper bound. Hence, we know that $\forall\, n \in \mathbb{N}$, there exists $x_n \in \mathbb{R}$, such that $f'(x_n) > n$. Since $f'(x)$ is uniformly continuous on $\mathbb{R}$, we have that there exists $\delta > 0$ such that $\forall\, x, y : |x - y| < \delta$, there is

$$|f(x) - f(y)| < 1.$$

Then, there is $f(x) > f(x_n) - 1 > n - 1$, $\forall\, x \in (x_n, x_n + \delta)$. By Taylor's formula, we have

$$2 \sup_{x \in \mathbb{R}} |f(x)| \ge |f(x_n + \delta) - f(x_n)| = |f'(\xi_n)\delta| > (n-1)\delta \to +\infty, \quad n \to \infty,$$

which contradicts with $f(x)$ is bounded on $\mathbb{R}$. □

Remark 8.11. *Note that $\lim\limits_{x \to +\infty} f'(x)$ may not exist, for example, consider $f(x) = \sin x$. But if $\lim\limits_{x \to +\infty} f'(x)$ exists, there must be $\lim\limits_{x \to +\infty} f'(x) = 0$.*

Problem 8.12. *Suppose that $f \in C^3(\mathbb{R})$, and there exists $\theta \in (0, 1)$ such that*

$$f(x+h) = f(x) + hf'(x + \theta h), \quad \forall\, h \in \mathbb{R}. \tag{8.1}$$

Prove that f is a linear function or a quadratic function.

Proof. Differentiating (8.1) respect to h, we have

$$f'(x+h) = f'(x + \theta h) + \theta h f''(x + \theta h). \tag{8.2}$$

Hence, there is

$$\frac{f'(x+h) - f'(x) + f'(x) - f'(x + \theta h)}{h} = \theta f''(x + \theta h).$$

Letting $h \to 0$, we have

$$f''(x) - \theta f''(x) = \theta f''(x),$$

i.e.

$$f''(x) = 2\theta f''(x).$$

If $\theta \neq \dfrac{1}{2}$, we know that $f''(x) = 0$, $\forall\, x \in \mathbb{R}$, thus $f(x)$ is a linear function. If $\theta = \dfrac{1}{2}$, (8.2) yields

$$f'(x+h) = f'\left(x + \frac{1}{2}h\right) + \frac{1}{2}hf''\left(x + \frac{1}{2}h\right).$$

Differentiating the above formula respect to h yields

$$f''(x+h) = f''\left(x + \frac{1}{2}h\right) + \frac{1}{4}hf'''\left(x + \frac{1}{2}h\right).$$

Hence

$$\frac{f''(x+h) - f''\left(x + \frac{1}{2}h\right)}{\frac{1}{2}h} = \frac{1}{2}hf'''\left(x + \frac{1}{2}h\right).$$

Letting $h \to 0$, we have

$$f'''(x) = \frac{1}{2}f'''(x),$$

i.e. $f'''(x) = 0$, $\forall\, x \in \mathbb{R}$, thus $f(x)$ is a quadratic function. □

Problem 8.13. *If f is defined on $(0, +\infty)$ and f', f'' exists , with $\lim\limits_{x\to+\infty} f(x)$ exists and f'' bounded, prove that $\lim\limits_{x\to+\infty} f'(x) = 0$.*

Proof. Without loss of generality, we assume that $\lim\limits_{x\to+\infty} f(x) = 0$ (otherwise, replace $f(x)$ by $f(x) - \lim\limits_{x\to+\infty} f(x)$, and those conditions are still satisfied). Then $\forall\, \varepsilon > 0$, $\exists\, a \in \mathbb{R}$ such that $\sup\limits_{x\in(a,\infty)} |f(x)| < \varepsilon$.

Let

$$\sup_{x\in(0,\infty)} |f''(x)| = M_2,$$

(exists finitely in $\mathbb{R}$ as f'' is bounded.) So, for a defined above, there is

$$\sup_{x\in(a,\infty)} |f''(x)| \leq M_2.$$

Taking $h > 0$, by Taylor's theorem we have

$$f'(x) = \frac{1}{2h}[f(x+2h) - f(x)] - hf''(\xi)$$

for some $\xi \in (x, x+2h)$. Hence

$$|f'(x)| \leq \frac{\varepsilon}{h} + hM_2.$$

Then we obtain

$$h^2M_2 - h\,|f'(x)| + \varepsilon \geq 0, \quad \forall\, x \in (a, \infty),$$

which is a quadratic in h, and since $M_2 > 0$, we have

$$|f'(x)|^2 \leq 4M_2\varepsilon, \quad \forall\, x \in (a, \infty).$$

Hence

$$\lim_{x\to\infty} |f'(x)|^2 = 0 \Longrightarrow \lim_{x\to\infty} |f'(x)| = 0 \Longrightarrow \lim_{x\to\infty} f'(x) = 0.$$

□

Problem 8.14. *Suppose that $f \in C^\infty(a, b)$, and $f^{(n)}(x) \geq 0$ for all $n \in \mathbb{N}_+$. If $|f(x)| \leq M$, prove that for every $x \in (a, b)$, $r > 0$, $x + r \in (a, b)$, there is*

$$f^{(n)}(x) \leq \frac{2Mn!}{r^n}, \quad \forall\, n \in \mathbb{N}_+.$$

Proof. By Taylor's formula, we have

$$\begin{aligned} f(x+r) &= f(x) + f'(x)r + \cdots + \frac{f^{(n)}(x)}{n!}r^n + \frac{f^{(n+1)}(\xi)}{(n+1)!}r^{n+1} \\ &\geq f(x) + \frac{f^{(n)}(x)}{n!}r^n, \end{aligned}$$

which implies

$$f^{(n)}(x) \leq \frac{(f(x+r) - f(x))n!}{r^n} \leq \frac{2Mn!}{r^n}.$$

□

Problem 8.15 (Bernstein Theorem)**.** *Suppose that $f \in C^\infty(a, b)$, and $f^{(n)}(x) \geq 0$ for all $n \in \mathbb{N}_+$. Prove that for every $x_0 \in (a, b)$, there exists $r > 0$, such that $\forall\, x \in [x_0 - r, x_0 + r] \subset (a, b)$, there is*

$$f(x) = \lim_{n\to\infty} \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k.$$

Proof. Choosing $r > 0$ small enough, such that $[x_0 - 3r, x_0 + 3r] \subset (a, b)$. Denote

$$M = \sup_{x\in[x_0-2r, x_0+2r]} |f(x)|.$$

By Problem 8.14, we have for every $x \in [x_0 - 2r, x_0 + 2r]$, there is $f^{(n)}(x) \leq \dfrac{2Mn!}{(2r)^n}$. Then, we have

$$\begin{aligned} \left| f(x) - \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k \right| &= \frac{f^{(n+1)}(\xi)}{(n+1)!}|x - x_0|^{n+1} \\ &\leq \frac{f^{(n+1)}(\xi)}{(n+1)!}r^{n+1} \leq \frac{M}{2^n}. \end{aligned}$$

Hence, there is

$$f(x) = \lim_{n\to\infty} \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k.$$

$\square$

Exercise 8.16 (5.12)**.** *Suppose that $f(x)$ is differentiable on (a,b), and $f'(x)$ is monotonic. Prove that $f'(x)$ is continuous on (a,b).*

Hint: Since $f'(x)$ is monotonic, we know that for any $x_0 \in (a,b)$, $\lim\limits_{x\to x_0+0} f'(x)$ and $\lim\limits_{x\to x_0-0} f'(x)$ exist. Then by Problem 7.18, we have that $f'(x)$ is continuous on (a,b). $\square$

Exercise 8.17. *Suppose that $f(x)$ is continuous on $[a,b]$, and twice differentiable on (a,b). If $|f''(x)| \geq m > 0$, and $f(a) = f(b) = 0$. Prove that $\max\limits_{x\in[a,b]} |f(x)| \geq \dfrac{m}{8}(b-a)^2$.*

Hint: Denote $|f(x_0)| = \max\limits_{x\in[a,b]} |f(x)|$. It's easy to see that $f'(x_0) = 0$. Then

$$f(a) = f(x_0) + f'(x_0)(a - x_0) + \frac{f''(\xi)}{2}(x_0 - a)^2 = f(x_0) + \frac{f''(\xi)}{2}(x_0 - a)^2,$$
$$f(b) = f(x_0) + f'(x_0)(b - x_0) + \frac{f''(\eta)}{2}(x_0 - b)^2 = f(x_0) + \frac{f''(\eta)}{2}(x_0 - b)^2.$$

Hence

$$|f(x_0)| \geq \frac{m}{2}(x_0 - a)^2 \geq \frac{m}{8}(b-a)^2, \quad x_0 \in \left[\frac{a+b}{2}, b\right],$$
$$|f(x_0)| \geq \frac{m}{2}(x_0 - b)^2 \geq \frac{m}{8}(b-a)^2, \quad x_0 \in \left[a, \frac{a+b}{2}\right].$$

$\square$

9. Week 12 (11.21)

Problem 9.1 (Schwarz Theorem)**.** *Define the generalized second order derivative as follows*

$$f^{[2]}(x) = \lim_{h\to 0+0} \frac{f(x+h)+f(x-h)-2f(x)}{h^2}.$$

If $f \in C[a,b]$, and $f^{[2]}(x) = 0$ on (a,b), prove that f is a linear function.

Proof. For any $x \in [a,b]$ and $\forall\, \varepsilon > 0$, define

$$f_\varepsilon(x) = f(x) - f(a) - \frac{f(b)-f(a)}{b-a}(x-a) + \varepsilon(x-a)(x-b).$$

Then $f_\varepsilon(a) = f_\varepsilon(b) = 0$. Note that

$$\begin{aligned}
f_\varepsilon^{[2]}(x) &= \lim_{h\to 0+0} \frac{f_\varepsilon(x+h)+f_\varepsilon(x-h)-2f_\varepsilon(x)}{h^2} \\
&= \lim_{h\to 0+0} \frac{1}{h^2}\Bigg(f(x+h) - f(a) - \frac{f(b)-f(a)}{b-a}(x+h-a) + \varepsilon(x+h-a)(x+h-b) \\
&\quad + f(x-h) - f(a) - \frac{f(b)-f(a)}{b-a}(x-h-a) + \varepsilon(x-h-a)(x-h-b) \\
&\quad - 2\left(f(x) - f(a) - \frac{f(b)-f(a)}{b-a}(x-a) + \varepsilon(x-a)(x-b)\right) \\
&= \lim_{h\to 0+0} \frac{1}{h^2}\left(f(x+h)+f(x-h)-2f(x)+\varepsilon\cdot 2h^2\right) \\
&= f^{[2]}(x) + 2\varepsilon = 2\varepsilon.
\end{aligned}$$

We claim that $f_\varepsilon \le 0$, when $\varepsilon > 0$. Indeed, if there is $x_0 \in (a,b)$ such that $f_\varepsilon(x_0) > 0$, by $f_\varepsilon(a) = f_\varepsilon(b) = 0$ and the continuity of f_ε, we know that the maximum of f_ε is achieved in (a,b), say $f_\varepsilon(x^*) = \max\limits_{x\in[a,b]} f_\varepsilon(x)$. Hence, there is

$$f_\varepsilon(x^*+h) + f_\varepsilon(x^*-h) \le 2f_\varepsilon(x^*).$$

Since $f_\varepsilon^{[2]}(x^*) = 2\varepsilon > 0$, we know that there is $h > 0$ such that

$$f_\varepsilon(x^*+h) + f_\varepsilon(x^*-h) - 2f_\varepsilon(x^*) > 0.$$

Therefore, there is

$$f_\varepsilon(x^*) < \frac{1}{2}(f_\varepsilon(x^*+h) + f_\varepsilon(x^*-h)) \le f_\varepsilon(x^*),$$

contradiction. Similarly, we can prove that $f_\varepsilon \ge 0$, when $\varepsilon < 0$. By the continuity of f_ε respect to ε , we know that

$$f(x) - f(a) - \frac{f(b)-f(a)}{b-a}(x-a) = f_0(x) = 0,$$

i.e. f is a linear function. $\square$

Problem 9.2. *If $f'(x_0) > (<) 0$, can we say that f is monotonic on a small enough neighborhood of $x = x_0$?*

Solution. No. For example (6.59 in book), let

$$f(x) = \begin{cases} x + 2x^2 \sin \dfrac{1}{x}, & x \neq 0, \\ 0, & x = 0. \end{cases}$$

It's easy to see that $f'(0) = 1 > 0$. However, by

$$f'(x) = 1 + 4x \sin \frac{1}{x} - 2 \cos \frac{1}{x}, \quad x \neq 0,$$

we have

$$f'\left(\frac{1}{n\pi}\right) = 1 - 2(-1)^n,$$

which means that $f'(x)$ can not be always positve or negative on any neighboehoods of $x = 0$, hence f is not monotonic on any neighboehoods of $x = 0$. The graph of $f(x)$ is as follows:

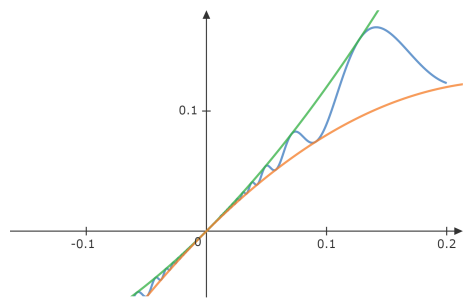

FIGURE 6. Graph of $f(x)$

□

Problem 9.3 (5.37)**.** *Suppose that $f(x)$ has nth-order derivative on $(a, +\infty)$. If $\lim\limits_{x \to +\infty} f(x) = A$, $\lim\limits_{x \to +\infty} f^{(n)}(x) = B$. Prove that $B = 0$.*

Proof. By L'Hospital's rule, we have

$$\begin{aligned} 0 &= \lim_{x \to +\infty} \frac{f(x)}{x^n} \\ &= \lim_{x \to +\infty} \frac{f'(x)}{nx^{n-1}} \\ &\vdots \end{aligned}$$

$$
\begin{aligned}
&= \lim_{x\to+\infty} \frac{f^{(n)}(x)}{n!} \\
&= \frac{B}{n!},
\end{aligned}
$$

which gives us $B = 0$. $\square$

Problem 9.4 (5.48)**.** *Suppose that $f(x)$ is defined on $[a, b]$ satisfying*

$$|f(x) - f(y)| \le k|x - y|^{1+\alpha}, \quad \forall\, x, y \in [a, b],\ \alpha > 0. \tag{9.1}$$

Prove that $f(x)$ is constant.

Proof. For $\forall\, x_0 \in [a, b]$, by (9.1), we have

$$\left|\frac{f(x) - f(x_0)}{x - x_0}\right| \le k|x - x_0|^{\alpha}, \quad \forall\, x \ne x_0.$$

Hence, letting $x \to x_0$, we know

$$f'(x_0) = 0, \quad \forall\, x_0 \in [a, b],$$

which means that $f(x)$ is constant. $\square$

Problem 9.5. *Find the extreme value of the following functions.*

(50(3)) $f(x) = \dfrac{(\ln x)^2}{x}$;

(50(4)) $f(x) = |x(x^2 - 1)|$;

(51(2)) $f(x) = 2\tan x - \tan^2 x$, $x \in [0, \pi/2)$;

(51(3)) $f(x) = \sqrt{x}\ln x$, $x \in (0, +\infty)$.

Solution. (50(3)) The derivative of $f(x)$ is

$$f'(x) = \frac{\ln x(2 - \ln x)}{x^2}.$$

Hence the local minimum is $f(1) = 0$ and the local maximum is $f(e^2) = 4/e^2$. The graph of $f(x)$ is as follows:

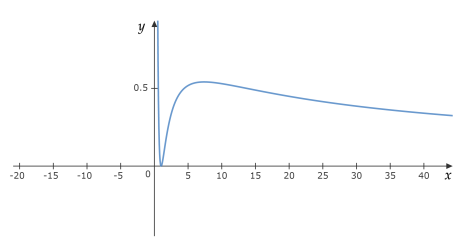

FIGURE 7. Graph of $f(x)$

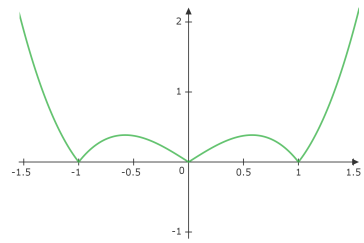

FIGURE 8. Graph of $f(x)$

(50(4))The graph of $f(x)$ is as follows: When $x = 0$, $x = \pm 1$, $f(x)$ has local minimun 0. When $x = \pm\frac{\sqrt{3}}{3}$, $f(x)$ has local maximun $\frac{2\sqrt{3}}{9}$.

(51(2)) The graph of $f(x)$ is as follows:

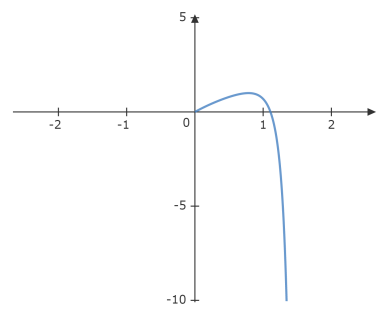

FIGURE 9. Graph of $f(x)$

(51(3)) The graph of $f(x)$ is as follows:

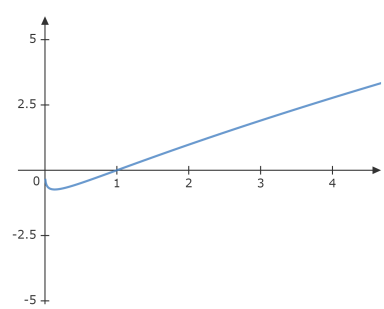

FIGURE 10. Graph of $f(x)$

□

Problem 9.6 (5.54(2)). *Suppose that $f(x)$ and $g(x)$ are differentiable on (a,b), and denote*

$$F(x) = f(x)g'(x) - f'(x)g(x), \quad x \in (a,b).$$

If $F(x) > 0$, $x \in (a,b)$. Prove that there must be zero points of $g(x)$ between the two zero points of $f(x)$.

Proof. Assume that x_1, $x_2 \in (a,b)$ are two zero points of $f(x)$. If $\forall\, x \in (x_1, x_2)$, there is $g(x) \neq 0$. Define

$$G(x) = \frac{f(x)}{g(x)}, \quad \forall\, x \in (x_1, x_2).$$

Hence, we have

$$G'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{g^2(x)} < 0,$$

which gives us that $G(x)$ is monotonic decreasing on $[x_1, x_2]$. Then there is

$$0 = G(x_1) \geq G(x) \geq G(x_2) = 0,$$

i.e. $G(x) \equiv 0$, which contradicts with $F(x) > 0$. Hence we know that there must be zero points of $g(x)$ between x_1 and x_2. □

Problem 9.7. *Suppose that $f(x)$ is twice differentiable on $[a,b]$ satisfying*

$$f''(x) + b(x)f'(x) + c(x)f(x) = 0, \quad \forall\, x \in [a,b], \tag{9.2}$$

where $c(x) < 0$. Prove that

(1) *$f(x)$ can not admit a positive maximum or negative minimum in (a,b);*
(2) *If $f(a) = f(b) = 0$, then $f(x) \equiv 0$.*

Proof. (1) Prove by contradiction. Assume that $f(x)$ has a positive maximum in (a,b), i.e. there is $x_0 \in (a,b)$ such that $f(x_0) = \max\limits_{x\in[a,b]} f(x) > 0$. By the necessary condition for maximum value, we know that $f'(x_0) = 0$, $f''(x_0) \leq 0$. Then combining with $c(x_0) < 0$, we have

$$f''(x_0) + b(x_0)f'(x_0) + c(x_0)f(x_0) \leq c(x_0)f(x_0) < 0,$$

which contradicts with (9.2). Hence, we have that $f(x)$ can not admit a positive maximum in (a,b). Using the same method, we can prove that $f(x)$ can not admit a negative minimum in (a,b).

(2) By (1), we know that $\max\limits_{x\in[a,b]} f(x) \leq 0$ and $\min\limits_{x\in[a,b]} f(x) \geq 0$. Hence

$$0 \leq \min_{x\in[a,b]} f(x) \leq \max_{x\in[a,b]} f(x) \leq 0,$$

i.e. $f(x) \equiv 0$. □

Exercise 9.8 (5.67). *Suppose that the tangent line of the elliptic $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$ intersects the x-axis and y-axis at A and B, respectively.*

(1) *Find the minimum length of AB;*

(2) *Find the minimum area of the triangle formed by AB and the coordinate axis.*

Hint: (1) The tangent line at (x_0, y_0) is

$$\frac{x_0 x}{a^2} + \frac{y_0 y}{b^2} = 1.$$

Hence we know that $A\left(\frac{a^2}{x_0}, 0\right)$, $B\left(\frac{b^2}{y_0}, 0\right)$. Then

$$\begin{aligned} |AB| &= \sqrt{\frac{a^4}{x_0^2} + \frac{b^4}{y_0^2}} \\ &= \sqrt{\frac{a^2}{\frac{x_0^2}{a^2}} + \frac{b^2}{\frac{y_0^2}{b^2}}} \\ &\geq \sqrt{\frac{(a+b)^2}{\frac{x_0^2}{a^2} + \frac{y_0^2}{b^2}}} \\ &= a + b. \end{aligned}$$

(2) By (1), we know

$$S_{\Delta AOB} = \frac{1}{2}\frac{a^2 b^2}{|x_0 y_0|} \geq \frac{ab}{\frac{x_0^2}{a^2} + \frac{y_0^2}{b^2}} = ab.$$

The graph of the elliptic when $a = 2$, $b = \sqrt{3}$ is as follows:

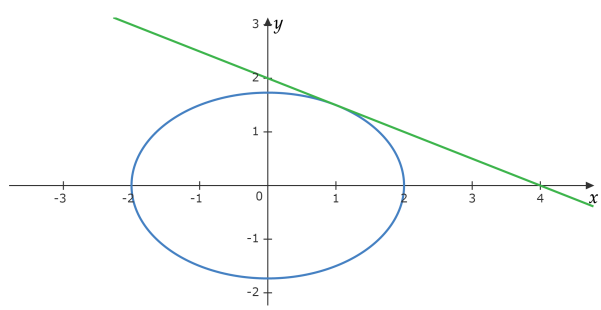

FIGURE 11. Graph of $\frac{x^2}{4} + \frac{y^2}{3} = 1$

□

Problem 9.9. *Suppose that $P(x)$ is a polynomial function. Prove that*

(1) *If $P'(x) + P(x) \geq 0$, then $P(x) \geq 0$;*

(2) *If $P(x) - P'(x) \geq 0$, then $P(x) \geq 0$;*

(3) *If* $P'''(x) - P''(x) - P'(x) + P(x) \geq 0$, *then* $P(x) \geq 0$.

Proof. (1) Let

$$F(x) = P(x)e^x.$$

Then there is

$$F'(x) = (P'(x) + P(x))e^x \geq 0.$$

Since $\lim\limits_{x\to-\infty} F(x) = 0$, we know that $F(x) \geq 0$, $\forall\, x \in \mathbb{R}$. Note that $e^x > 0$, $\forall\, x \in \mathbb{R}$, we have $P(x) \geq 0$.

(2) Let

$$G(x) = P(x)e^{-x}.$$

Then there is

$$G'(x) = (P'(x) - P(x))e^{-x} \leq 0.$$

Since $\lim\limits_{x\to+\infty} G(x) = 0$, we know that $G(x) \geq 0$, $\forall\, x \in \mathbb{R}$. Note that $e^{-x} > 0$, $\forall\, x \in \mathbb{R}$, we have $P(x) \geq 0$.

(3) Denote

$$Q(x) = P''(x) - P(x).$$

We have

$$Q'(x) - Q(x) \geq 0.$$

Hence by (2), we know that $Q(x) \leq 0$, i.e. $P(x) - P'(x) \geq 0$. Then by (2) again, we have that $P(x) \geq 0$. □

Problem 9.10. *Suppose that* $f(x)$ *is a bounded convex function on* (a, b). *Prove that* $\lim\limits_{x\to a+0} f(x)$ *and* $\lim\limits_{x\to b-0} f(x)$ *exist.*

Proof. Since $f(x)$ is bounded, we may assume that $|f(x)| \leq M$, $\forall\, x \in (a, b)$. Let $x > x_1 > x_0$ be any points in (a, b). By the convexity of $f(x)$, we have $\dfrac{f(x) - f(x_0)}{x - x_0}$ is monotonic inreasing respect to x. Since

$$\frac{f(x) - f(x_0)}{x - x_0} \leq \frac{M - f(x_0)}{x_1 - x_0}, \quad \forall\, x > x_1 > x_0,$$

we have

$$\lim_{x\to b-0} \frac{f(x) - f(x_0)}{x - x_0} = A$$

by the monotone bounded convergence theorem. Then

$$\begin{aligned}\lim_{x\to b-0} f(x) &= \lim_{x\to b-0} \left[(x - x_0) \cdot \frac{f(x) - f(x_0)}{x - x_0} + f(x_0)\right] \\ &= A(b - x_0) + f(x_0),\end{aligned}$$

which is $\lim\limits_{x\to b-0} f(x)$ exists. Similarly, we can prove that $\lim\limits_{x\to a+0} f(x)$ exists. □

Problem 9.11. *Suppose that* $f(x)$ *is convex on* (a, b). *Prove that* $\forall\, [c, d] \subset (a, b)$, $f(x)$ *is Lipschitz continuous on* $[c, d]$.

Proof. Since $[c,d] \subset (a,b)$, we can choose $h > 0$ small enough such that

$$[c-h, d+h] \subset (a,b).$$

Indeed, it suffices to choose $0 < h < \min\{c-a, b-d\}$. Then $\forall\, x_1, x_2 \in [c,d]$, if $x_1 < x_2$, we take $x_3 = x_2 + h$. By the convexity of $f(x)$, we have

$$\frac{f(x_2)-f(x_1)}{x_2-x_1} \le \frac{f(x_3)-f(x_2)}{x_3-x_2} \le \frac{M-m}{h},$$

where $M = \sup\limits_{x\in[c-h,d+h]} f(x)$, $m = \inf\limits_{x\in[c-h,d+h]} f(x)$. If $x_2 < x_1$, we take $x_3 = x_2 - h$. By the convexity of $f(x)$, we have

$$\frac{f(x_2)-f(x_3)}{x_2-x_3} \le \frac{f(x_1)-f(x_2)}{x_1-x_2}.$$

Then

$$\frac{f(x_2)-f(x_1)}{x_1-x_2} \le \frac{f(x_3)-f(x_2)}{x_2-x_3} \le \frac{M-m}{h}.$$

Hence, we have

$$f(x_2)-f(x_1) \le \frac{M-m}{h}|x_1-x_2|.$$

Switching x_1 and x_2, we know the above inequality is still valid. Hence, we know

$$|f(x_1)-f(x_2)| \le \frac{M-m}{h}|x_1-x_2|, \quad \forall\, x_1, x_2 \in [c,d],$$

i.e. $f(x)$ is Lipschitz continuous on $[c,d]$. □

Problem 9.12. *Prove that the non-differentiable points of a convex function are countable.*

Proof. Let $f(x)$ be a convex function. For $\forall\, x < x_0 < y$, by the convexity of f, we have

$$\frac{f(x)-f(x_0)}{x-x_0} \le \frac{f(y)-f(x_0)}{y-x_0}.$$

It's easy to see that $\dfrac{f(x)-f(x_0)}{x-x_0}$ is monotonic inreasing respect to x. Hence by the monotone bounded convergence theorem, we have

$$f'_-(x_0) \le f'_+(x_0).$$

If $f(x)$ is non-differentiable at $x = x_0$, there is $f'_-(x_0) < f'_+(x_0)$. Then every non-differentiable point x_0 is corresponding to a rational number in $\big(f'_-(x_0), f'_+(x_0)\big)$, and $\big\{\big(f'_-(x_0), f'_+(x_0)\big)\big\}$ is disjoint pairwise, thus the non-differentiable points of $f(x)$ are countable. □

Exercise 9.13. *Suppose that $f(x)$ is a bounded convex function on $\mathbb{R}$. Prove that f is a constant function.*

Hint: Prove by contradiction. Suppose that f is not a constant function. By Problem 9.12, we know that $f'_-(x)$ and $f'_+(x)$ exist. Then there is at least a x_0 such that $f'_-(x_0) \neq 0$ or $f'_+(x_0) \neq 0$. Without loss of generality, we assume that $f'_+(x_0) \neq 0$. Then by the convexity of f, we have

$$f(x) \geq f(x_0) + f'_+(x_0)(x - x_0),$$

which contradicts with f is bounded. □

Problem 9.14. *Prove that $f(x)$ is a convex function on (a, b) if and only if $\forall\, x_1, x_2 \in (a, b)$, there is $\varphi(\lambda) = f(\lambda x_1 + (1-\lambda)x_2)$ is a convex function on $[0, 1]$.*

Proof. "$\Rightarrow$" For $\forall\, \lambda_1, \lambda_2 \in [0, 1]$, $\alpha \in [0, 1]$, there is

$$\begin{aligned}\varphi(\alpha\lambda_1 + (1-\alpha)\lambda_2) &= f((\alpha\lambda_1 + (1-\alpha)\lambda_2)x_1 + (1 - (\alpha\lambda_1 + (1-\alpha)\lambda_2))x_2) \\ &= f(\alpha(\lambda_1 x_1 + (1-\lambda_1)x_2) + (1-\alpha)(\lambda_2 x_1 + (1-\lambda_2)x_2)) \\ &\leq \alpha\varphi(\lambda_1) + (1-\alpha)\varphi(\lambda_2),\end{aligned}$$

where we used the convexity of f in the last inequality. Hence $\varphi(\lambda)$ is a convex function on $[0, 1]$.
"$\Leftarrow$" It is easy to see that

$$\begin{aligned}f(\lambda x_1 + (1-\lambda)x_2) = \varphi(\lambda) &= \varphi(\lambda \cdot 1 + (1-\lambda)\cdot 0) \\ &\leq \lambda\varphi(1) + (1-\lambda)\varphi(0) \\ &= \lambda f(x_1) + (1-\lambda)f(x_2),\end{aligned}$$

i.e. $f(x)$ is a convex function on (a, b). □

Problem 9.15. *Suppose that $f(x)$ is a strict convex function on I. Prove that if $f(x)$ has minimum $f(x_0)$, then $f(x_0)$ is unique, i.e. $\forall\, x \in I\backslash\{x_0\}$, there is $f(x) > f(x_0)$.*

Proof. Prove by contradiction. If there exists $x_1 \in I\backslash\{x_0\}$, such that $f(x_1) \leq f(x_0)$. Then $\forall\, \lambda \in (0, 1)$, there is

$$f(\lambda x_0 + (1-\lambda)x_1) < \lambda f(x_0) + (1-\lambda)f(x_1) \leq f(x_0).$$

Then for any neighborhood of x_0, say $U(x_0, \delta)$ $(0 < \delta < |x_1 - x_0|)$, in I, we know that if we let $\lambda : 1 - \lambda < \dfrac{\delta}{|x_1 - x_0|}$, and take $x = \lambda x_0 + (1-\lambda)x_1$, then there is $x \in U(x_0, \delta)$ and

$$f(x) = f(\lambda x_0 + (1-\lambda)x_1) < f(x_0),$$

which contradicts with $f(x_0)$ is the minimum of $f(x)$. □

Exercise 9.16. *Suppose that $f \in C^2[0, 1]$ is a nonegative function. Assume that $f(0) = 0$, $f'(0) = 1$, $f''(0) = -1$, and $\forall\, x \in (0, 1]$, $f(x) \neq x$. For any given $x_0 \in (0, 1]$, define $x_{n+1} = f(x_n)$. Calculate the limitation $\lim\limits_{n\to\infty} nx_n$.*

Hint: By Taylor's formula, we have

$$f(x) = x - \frac{1}{2}x^2 + o(x^2).$$

Hence, for $x > 0$ small enough, we have $f(x) < x$. By $\forall\, x \in (0,1]$, $f(x) \neq x$ and the continuity of f, we know that $f(x) \leq x$. Hence, $\lim\limits_{n\to\infty} x_n$ exists and $\lim\limits_{n\to\infty} x_n = 0$. By Stolz theorem and Taylor's formula, we have

$$\begin{aligned}\lim_{n\to\infty} nx_n &= \lim_{n\to\infty} \frac{n}{\frac{1}{x_n}} = \lim_{n\to\infty} \frac{x_n x_{n+1}}{x_n - x_{n+1}} \\ &= \lim_{n\to\infty} \frac{x_n f(x_n)}{x_n - f(x_n)} = \lim_{n\to\infty} \frac{x_n^2}{\frac{1}{2}x_n^2} = 2.\end{aligned}$$

□

Exercise 9.17. *Suppose that f has nth-order derivatives at $x = x_0$. Prove that*

$$f^{(n)}(x_0) = \lim_{h\to 0} \frac{1}{h^n} \sum_{k=0}^{n} (-1)^{n-k} C_n^k f(x_0 + kh).$$

Hint:

$$\begin{aligned}&\lim_{h\to 0} \frac{\sum_{k=0}^{n} (-1)^{n-k} \mathrm{C}_n^k f\left(x_0 + kh\right)}{h^n} \\ &= \lim_{h\to 0} \frac{\sum_{k=0}^{n} (-1)^{n-k} \mathrm{C}_n^k \left(\sum_{m=0}^{n} \frac{f^{(m)}(x_0)}{m!} k^m h^m + o\left(h^n\right)\right)}{h^n} \\ &= \lim_{h\to 0} \frac{\sum_{k=0}^{n} (-1)^{n-k} C_n^k \sum_{m=0}^{n} \frac{f^{(m)}(x_0)}{m!} k^m h^m}{h^n} \\ &= \lim_{h\to 0} \frac{\sum_{k=0}^{n} \sum_{m=0}^{n} (-1)^{n-k} C_n^k \frac{f^{(m)}(x_0)}{m!} k^m h^m}{h^n} \\ &= \lim_{h\to 0} \frac{\sum_{m=0}^{n} \sum_{k=0}^{n} (-1)^{n-k} C_n^k \frac{f^{(m)}(x_0)}{m!} k^m h^m}{h^n} \\ &= \lim_{h\to 0} \frac{\sum_{m=0}^{n} \frac{f^{(m)}(x_0)}{m!} h^m \sum_{k=0}^{n} (-1)^{n-k} C_n^k k^m}{h^n} \\ &= \lim_{h\to 0} \frac{(-1)^n \cdot (-1)^n n! \frac{f^{(n)}(x_0)}{n!} h^n}{h^n} \\ &= f^{(n)}\left(x_0\right),\end{aligned}$$

where we used Problem 7.8 in the last equality. □

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